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Foreword:

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This Vocoder Standard (102BABA), titled Vocoder Description, has been developed with inputs from the APCO Project 25 Interface Committee (APIC), the APIC Vocoder Task Group, and TIA Industry Members. This Standard will be maintained by TR-8.4.

This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December, 1993. That MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard. The symbol "P25" or the wording "Project 25" appearing on the cover sheet of this document when published or in the title of any TIA-102 document or Telecommunications Service Bulletin (TSB) referenced herein, that indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted the document as part of the Standard. The appearance of the abbreviation "P25" or the wording "Project 25" on the cover sheet of this document or in the title of any document referenced herein does not limit the applicability of the information contained in this document to "P25" or "Project 25" implementations exclusively. This document is being published to provide technical information on the emerging digital techniques for Land Mobile Radio Service. This document defines vocoders for land mobile radios capable of 12.5 kHz efficiency and/or 6.25 kHz efficiency.

The reader's attention is called to the possibility that compliance with this Standard may require the use of one or more inventions covered by patent rights.

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Vocoder Description

Digital Voice Systems Inc. (DVS) claims certain rights, including patent rights, in the Improved Multi-Band Excitation (IMBE) voice coding algorithm described in this document and elsewhere. Any use of this technology requires a separate written license from DVS.

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1 Scope

This document serves as the interoperability specification for the Dual-Rate Vocoder employed within the TIA-102 series of documents. This document merges two previous vocoder standard documents, one published as TIA-102.BABA to describe a Full-Rate Vocoder at a gross bit rate of 7200 bps, and one published as TIA-102.BABA-1 to describe a Half-Rate Vocoder at a gross bit rate of 3600 bps. This document now describes the functional requirements for the transmission and reception of voice information using digital communication media described in the TIA-102 series of documents at either bit rate. This document is specifically intended to define the conversion of voice from an analog representation to a Full-Rate or a Half-Rate digital representation. The Full-Rate digital representation consists of a net bit rate of 4400 bps for voice information, and a gross bit rate of 7200 bps after error control coding. The Half-Rate digital representation consists of a net bit rate of 2450 bps for voice information, and a gross bit rate of 3600 bps after error control coding.

The voice coder (or vocoder) presented in this document is intended to be used in any equipment that requires an analog-to-digital or digital-to-analog voice interface. Specifically, mobile and portable radios as well as console equipment and gateways to voice networks may contain the vocoder described in this document. The reader is referred to the TIA-102 series of documents for additional information on the integration of the vocoder into the overall communication system.

The Full-Rate Vocoder is intended for use over a radio air interface providing 7200 bps for voice data plus forward error correction (FEC). The reader is referred to the FDMA Common Air Interface standard, TIA-102.BAAA, and other related standards for more information pertaining to the radio FDMA air interface.

The Half-Rate Vocoder is intended for use over a radio air interface providing 3600 bps for voice data plus FEC. The reader is referred to the Phase 2 Two-Slot TDMA Media Access Control standard, TIA-102.BBAC, and other related standards for more information pertaining to the radio TDMA air interface.

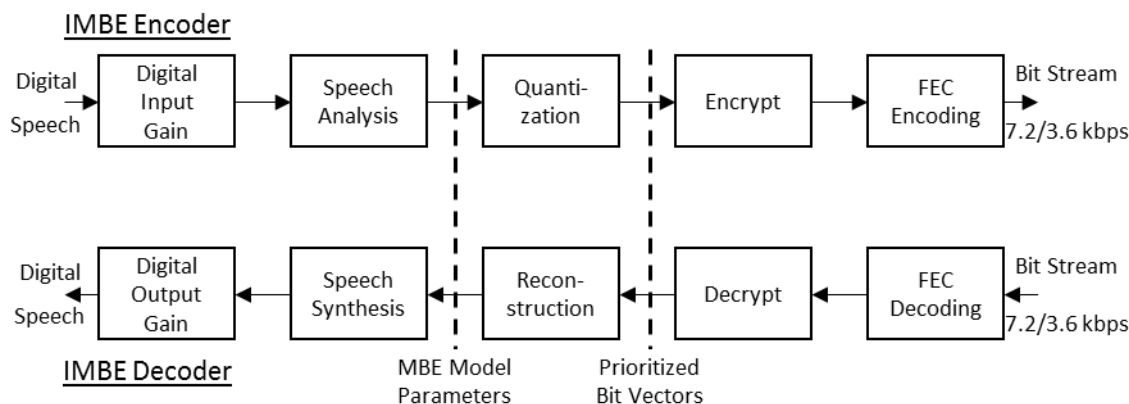


Figure 1 Improved Multi-Band Excitation Speech Coder

2 Introduction

The Dual-Rate Vocoder is described in this document within the context of the Improved Multi-Band Excitation (IMBE) model of a voice coding algorithm. The IMBE algorithm is then, in its turn, described in the context of the Multi-Band Excitation (MBE) speech model.

This document provides a functional description of the Improved Multi-Band Excitation (IMBE) voice coding algorithm. This document describes the essential operations that are necessary and sufficient to implement this voice coding algorithm. However, it is highly recommended that the references be studied prior to the implementation of this algorithm. It is also recommended that implementations begin with a high-level language simulation of the algorithm, and then proceed to a real-time implementation using a digital signal processor. High performance real-time implementations have been demonstrated using both floating-point and fixed-point processors. The reader is cautioned that this document does not attempt to describe the most efficient means of implementing the IMBE vocoder. The reader should consult one or more references on efficient real-time programming for more information on this subject. Additionally this document does not address vocoder testing and verification. These subjects are addressed in separate documents in the TIA-102 series of documents.

The IMBE speech coder is based on a robust speech model which is referred to as the Multi-Band Excitation (MBE) speech model [3]. The basic methodology of the coder is to divide a digital speech input signal into overlapping speech segments (or frames) using a window such as a Kaiser window. Each speech frame is then compared with the underlying speech model, and a set of model parameters are estimated for that particular frame. The encoder quantizes these model parameters and transmits a bit stream at 7.2 kbps or 3.6 kbps for the Full-Rate or Half-Rate modes, respectively. The decoder receives this bit stream, reconstructs the model parameters, and uses these model parameters to generate a synthetic speech signal. This synthesized speech signal is the output of the IMBE speech coder as shown in Figure 1. One should note that the IMBE speech coder shown in this figure and defined by this document is a digital-to-digital function.

The IMBE speech coder is a model-based speech coder, or vocoder, which does not try to reproduce the input speech signal on a sample by sample basis. Instead the IMBE speech coder constructs a synthetic speech signal that contains the same perceptual information as the original speech signal.

Many previous vocoders (such as LPC vocoders, homomorphic vocoders, and channel vocoders) have not been successful in producing high quality synthetic speech. The IMBE speech coder has two primary advantages over these vocoders. First, the IMBE speech coder is based on the MBE speech model which is a more robust model than the traditional speech models used in previous vocoders. Second, the IMBE speech coder uses more sophisticated algorithms to estimate the speech model parameters, and to synthesize the speech signal from these model parameters.

This document is organized as follows. Sections 1 and 2 give the scope and introduction to the vocoder, respectively. In Section 3 the MBE speech model is briefly reviewed. This section presents background material which is useful in understanding the operation of the IMBE speech coder. Section 4 describes the basic speech input/output requirements. Section 5 examines the methods used to estimate the speech model parameters. The first five sections are common for both the Full-Rate and the Half-Rate modes of the vocoder.

Section 6 examines the quantization and reconstruction of the MBE model parameters for the Full-Rate mode. The error correction coding and the format of the 7.2 kbps bit stream is discussed in Section 7 for the Full-Rate mode. This is followed by Section 8 which describes the enhancement of the spectral amplitudes, and Section 9 which describes the adaptive smoothing method used to reduce the effect of uncorrectable bit errors. Section 10 then demonstrates the encoding of a typical set of model parameters for the Full-Rate mode. Section 11 discusses the synthesis of speech from the MBE model parameters. Sections 9 and 11 are used for both the Full-Rate and Half-Rate modes. A few additional comments on the algorithm and this document are provided in Section 12.

Section 13 describes the quantization and reconstruction of the MBE model parameters for the Half-Rate mode. The error correction coding and the format of the 3.6 kbps bit stream is described in Section 14. This is followed by Section 15 which describes the enhancement of the spectral amplitudes. The operations of adaptive smoothing and speech synthesis previously described in sections 9 and 11 are then also used for the Half-Rate mode.

Section 16 describes tone frames to encode and decode tone transmissions. This section specifically defines Half-Rate frame formats for tones. There is no special Full-Rate frame format for tones, although the Full-Rate vocoder can transmit tones without any special formats. Section 17 describes the conversion of MBE model parameters between the Full-Rate and Half-Rate modes.

Other information such as bit allocation tables, quantization levels and initialization vectors are contained in the attached annexes. Annexes A through J are used for the Full-Rate mode. Annex K contains a set of flow charts describing certain elements of this vocoder. Annexes L through T are used for the Half-Rate mode. Note that the flow charts in Annex K have been designed to help clarify the various algorithmic steps and do not necessarily describe the best or most efficient method of implementing the vocoder. A list of informative references is given after the annexes.

The vocoder described in this document is compatible with many other signal processing functions which may be combined with the vocoder to improve performance in certain circumstances. Examples of optional processing functions include echo cancellation [13], noise removal [12], level adjustment, and others. These functions are outside the scope of this document, however additional information can be found in the aforementioned references and elsewhere in the open literature.

For the purposes of this Standard, the following definitions apply.

<i>Abbreviation</i>	<i>Definition</i>
BER	Bit Error Rate
bps	Bits per Second
dB	Decibel
DCT	Discrete Cosine Transform
DTMF	Dual Tone Multi-Frequency
FDMA	Frequency Division Multiple Access
FEC	Forward Error Correction
HOC	Higher Order Coefficient
Hz	Hertz
IMBE	Improved Multi-Band Excitation
KNOX	Dual Tone
LPC	Linear Prediction Coding
LSB	Least Significant Bit
MBE	Multi-Band Excitation
MSB	Most Significant Bit
PRBA	Predictive Residual Block
TDMA	Time Division Multiple Access
TIA	Telecommunications Industry Association
V/UV	Voiced Unvoiced

3 Multi-Band Excitation Speech Model

Let $s(n)$ denote a discrete speech signal obtained by sampling an analog speech signal. In order to focus attention on a short segment of speech over which the model parameters are assumed to be constant, a window $w(n)$ is applied to the speech signal $s(n)$. The windowed speech signals $s_w(n)$ is defined by

$$s_w(n) = s(n) w(n) \quad (1)$$

The sequence $s_w(n)$ is referred to as a speech segment or a speech frame. The IMBE analysis algorithm actually uses two different windows, $w_R(n)$ and $w_I(n)$, each of which is applied separately to the speech signal via Equation (1). This will be explained in more detail in Section 5 of this document. The speech signal $s(n)$ is shifted in time to select any desired segment. For notational convenience $s_w(n)$ refers to the current speech frame. The next speech frame is obtained by shifting $s(n)$ by 20 ms.

A speech segment $s_w(n)$ is modeled as the response of a linear filter $h_w(n)$ to some excitation signal $e_w(n)$. Therefore, $S_w(\omega)$, the Fourier Transform of $s_w(n)$, can be expressed as

$$S_w(\omega) = H_w(\omega) E_w(\omega) \quad (2)$$

where $H_w(\omega)$ and $E_w(\omega)$ are the Fourier Transforms of $h_w(n)$ and $e_w(n)$, respectively.

In traditional speech models, speech is divided into two classes depending upon the nature of the excitation signal. For voiced speech the excitation signal is a periodic impulse sequence, where the distance between impulses is the pitch period P_0 . For unvoiced speech the excitation signal is a white noise sequence. One of the primary distinctions between traditional vocoders is the method in which they model the linear filter $h_w(n)$. The frequency response of this filter is generally referred to as the spectral envelope of the speech signal. In a LPC vocoder, for example, the spectral envelope is modeled with a low order all-pole model. Similarly, in a homomorphic vocoder, the spectral envelope is modelled with a small number of cepstral coefficients.

A primary difference between traditional speech models and the MBE speech model is the excitation signal. In conventional speech models a single voiced/unvoiced (V/UV) decision is used for each speech segment. In contrast the MBE speech model divides the excitation spectrum into a number of non-overlapping frequency bands and makes a V/UV decision for each frequency band. This allows the excitation signal for a particular speech segment to be a mixture of periodic (voiced) energy and noise-like (unvoiced) energy. This added degree of freedom in the modelling of the excitation allows the MBE speech model to generate higher quality speech than conventional speech models. In addition it allows the MBE speech model to be robust to the presence of background noise.

In the MBE speech model the excitation spectrum is obtained from the pitch period (or the fundamental frequency) and the V/UV decisions. A periodic spectrum is used in the frequency bands declared voiced, while a random noise spectrum is used in the frequency bands declared unvoiced. The periodic spectrum is generated from a windowed periodic impulse train which is completely determined by the window and the pitch period. The random noise spectrum is generated from a windowed random noise sequence.

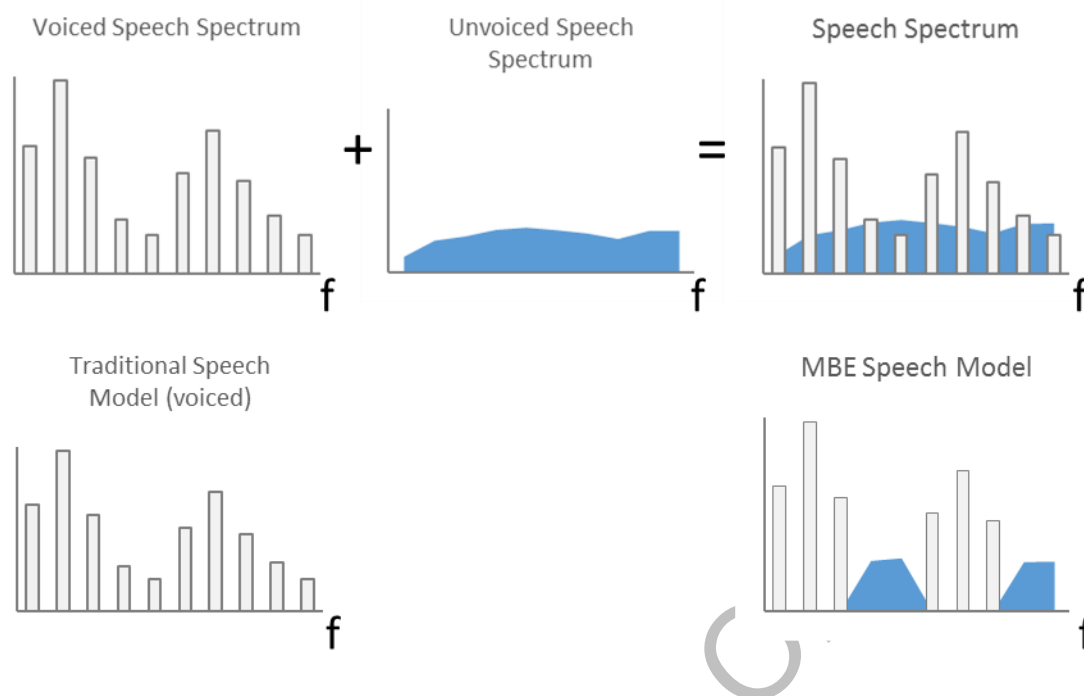


Figure 2 Comparison of Traditional and MBE Speech Models

A comparison of a traditional speech model and the MBE speech model is shown in Figure 2. In this example the traditional model has classified the speech segment as voiced, and consequently the traditional speech model is composed completely of periodic energy. The MBE model has divided the spectrum into 10 frequency bands in this example. The fourth, fifth, ninth and tenth bands have been declared unvoiced while the remaining bands have been declared voiced. The excitation in the MBE model is comprised of periodic energy only in the frequency bands declared voiced, while the remaining bands are comprised of noise-like energy. This example shows an important feature of the MBE speech model. Namely, the V/UV determination is performed such that frequency bands where the ratio of periodic energy to noise-like energy is high are declared voiced, while frequency bands where this ratio is low are declared unvoiced. The details of this procedure are discussed in Section 5.2.

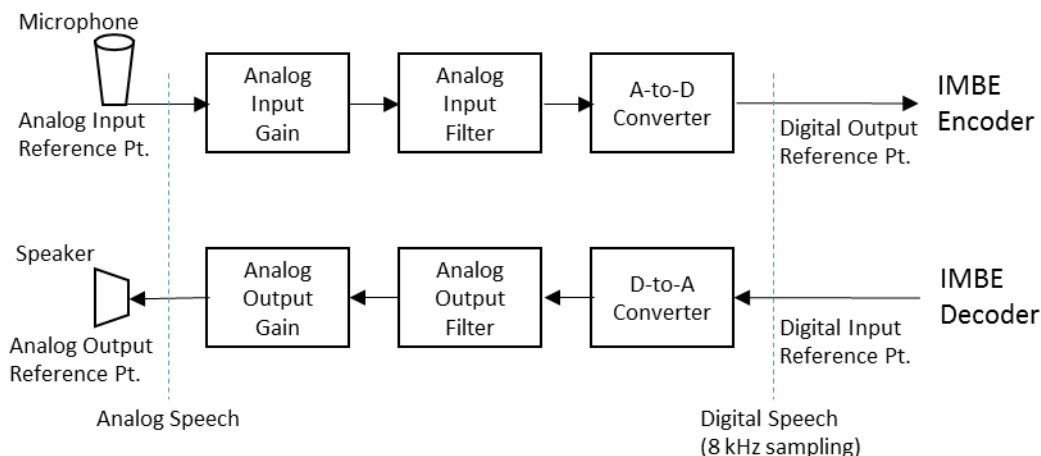


Figure 3 Analog Front End

4 Speech Input / Output Requirements

This section presents a number of performance recommendations for the analog front end of a voice codec, including the gain, filtering, and conversion elements as depicted in Figure 3. The objective is to establish a set of input/output requirements that will ensure that the voice codec operates at its maximum capability. The reader should note that Figure 3 shows four reference points (analog input, analog output, digital input and digital output) which are used in this document and will be used in future documents describing the test and verification procedure used with this vocoder.

The voice encoder and decoder defined in the remainder of this document operates with unity (i.e. 0 dB) gain. Consequently the analog input and output gain elements shown in Figure 3 are only used to match the sensitivity of the microphone and speaker with the A-to-D converters and D-to-A converters, respectively. It is recommended that the analog input gain be set such that the RMS speech level under nominal input conditions is 25 dB below the saturation point of the A-to-D converter. This level (-22 dBm0) is designed to provide sufficient margin to prevent the peaks of the speech waveform from being clipped by the A-to-D converter.

The voice coder defined in this document requires the A-to-D and D-to-A converters to operate at an 8 kHz sampling rate (i.e. a sampling period of 125 microseconds) at the digital input/output reference points. This requirement necessitates the use of analog filters at both the input and output to eliminate any frequency components above the Nyquist frequency (4 kHz). The recommended input and output filter masks are shown in Figure 4. For proper operation, the frequency response of the analog filters should be bounded by the shaded zone depicted in this figure.

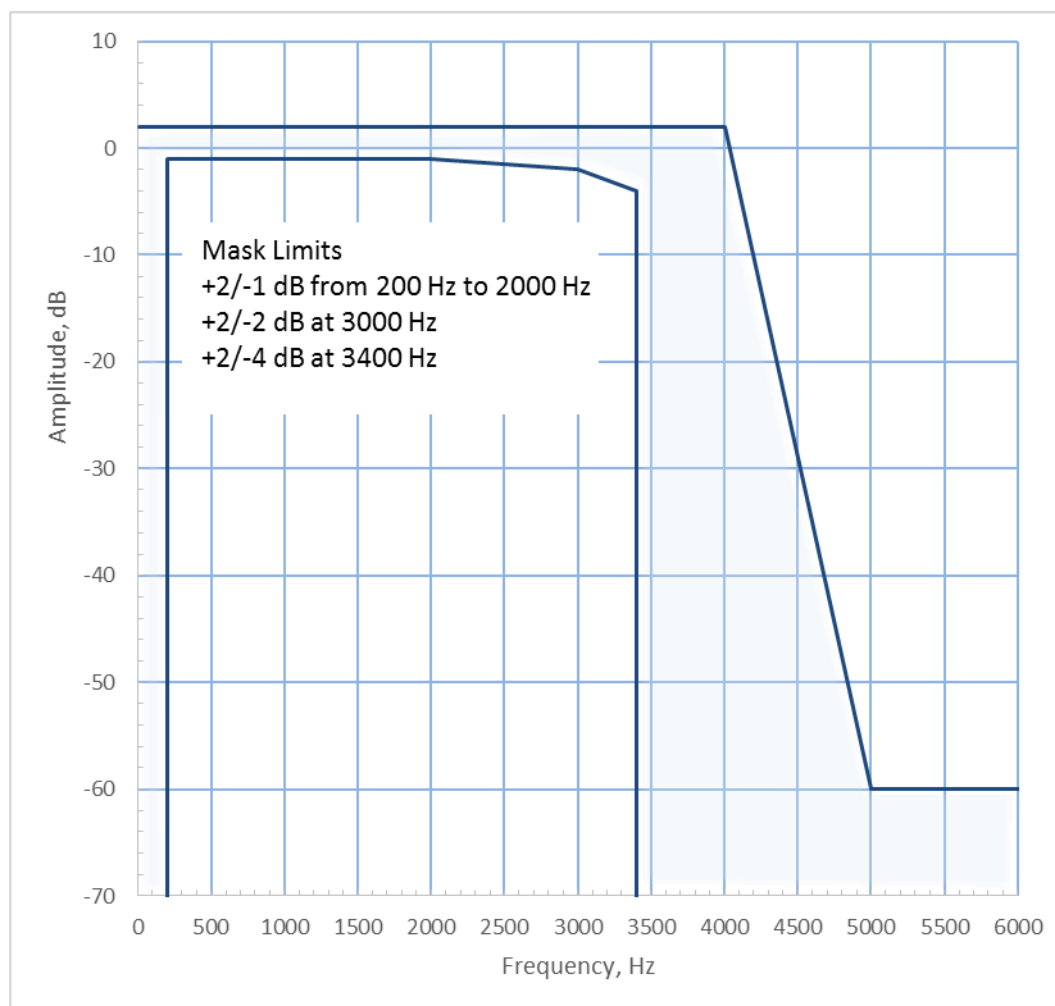


Figure 4 Analog Input / Output Filter Mask

This vocoder description assumes that the A-to-D converter produces digital speech which is confined to the range $[-32768, 32767]$, and similarly that the D-to-A converter accepts digital speech within this same range. If a converter is used which does not meet these assumptions then the digital gain elements shown in Figure 1 should be adjusted appropriately. Note that these assumptions are automatically satisfied if 16 bit linear A-to-D and D-to-A converters are used, in which case the digital gain elements should be set to unity gain. Also note that the vocoder requires that any companding which is applied by the A-to-D converter (i.e. A-law or μ -law) should be removed prior to speech encoding. Similarly any companding used by the D-to-A converter should be applied after speech decoding.

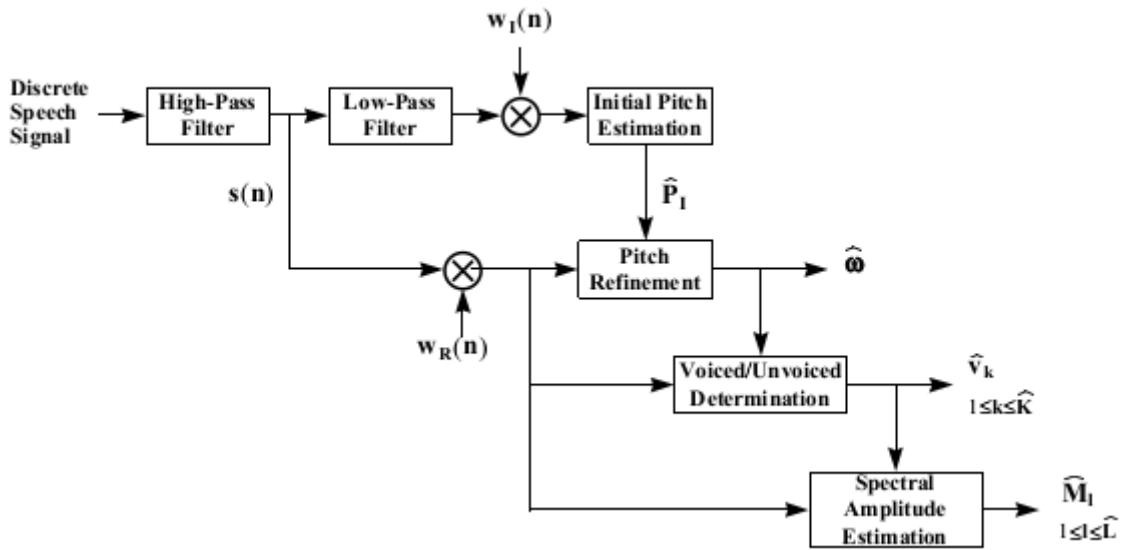


Figure 5 IMBE Speech Analysis Algorithm

5 Speech Analysis

This section presents the methods used to estimate the MBE speech model parameters. To develop a high quality vocoder it is essential that robust and accurate algorithms are used to estimate the model parameters. The approach which is presented here differs from conventional approaches in a fundamental way. Typically algorithms for the estimation of the excitation parameters and algorithms for the estimation of the spectral envelope parameters operate independently. These parameters are usually estimated based on some reasonable but heuristic criterion without explicit consideration of how close the synthesized speech will be to the original speech. This can result in a synthetic spectrum quite different from the original spectrum. In the approach used in the IMBE speech coder the excitation and spectral envelope parameters are estimated simultaneously, so that the synthesized spectrum is closest in a least squares sense to the original speech spectrum. This approach can be viewed as an “analysis-by-synthesis” method. The theoretical derivation and justification of this approach is presented in references [3, 4, 6].

A block diagram of the analysis algorithm is shown in Figure 5. The MBE speech model parameters which are estimated for each speech frame are the pitch period (or equivalently the fundamental frequency), the V/UV decisions, and the spectral amplitudes which characterize the spectral envelope. The organization of this section is as follows. First, the pitch estimation method is presented in Section 5.1. The V/UV determination is discussed in Section 5.2, and finally Section 5.3 discusses the estimation of the spectral amplitudes.

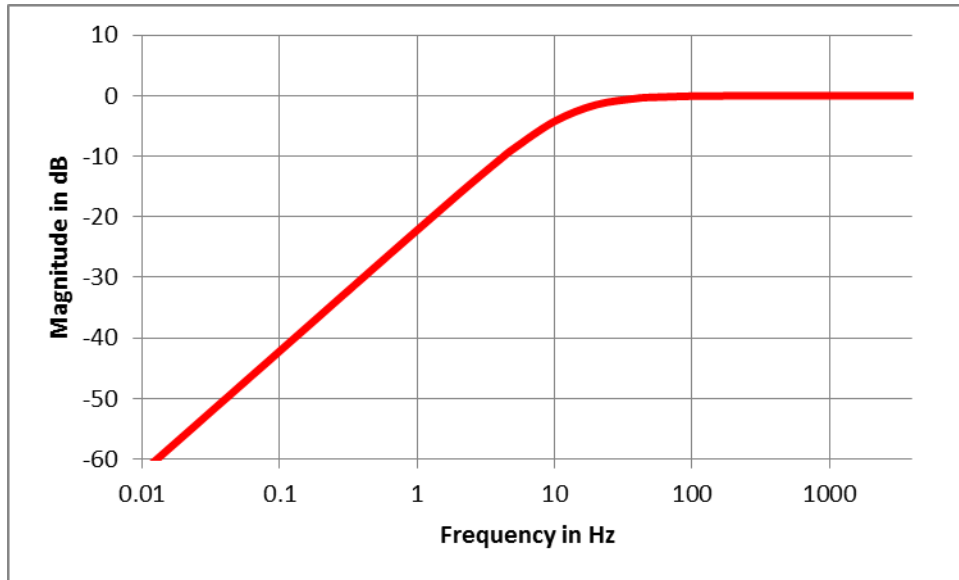


Figure 6 High Pass Filter Frequency Response at 8 kHz Sampling Rate

The input to the speech analyzer, and, consequently, the encoder, is a discrete speech signal generated using an A-to-D converter as described in Section 4. This speech signal shall first be digitally filtered to remove any residual energy at D.C. This is accomplished by passing the input signal through a discrete high-pass filter with the following transfer function:

$$H(z) = \frac{1-z^{-1}}{1-0.99z^{-1}} \quad (3)$$

The resulting high-pass filtered signal is denoted by $s(n)$ throughout the remainder of this section. Figure 6 shows the frequency response of the filter specified in equation (3) using the convention that the Nyquist frequency (4 kHz) is mapped to a discrete frequency of π radians. For more information on this frequency convention, which is used throughout this document, the reader is referred to reference [11].

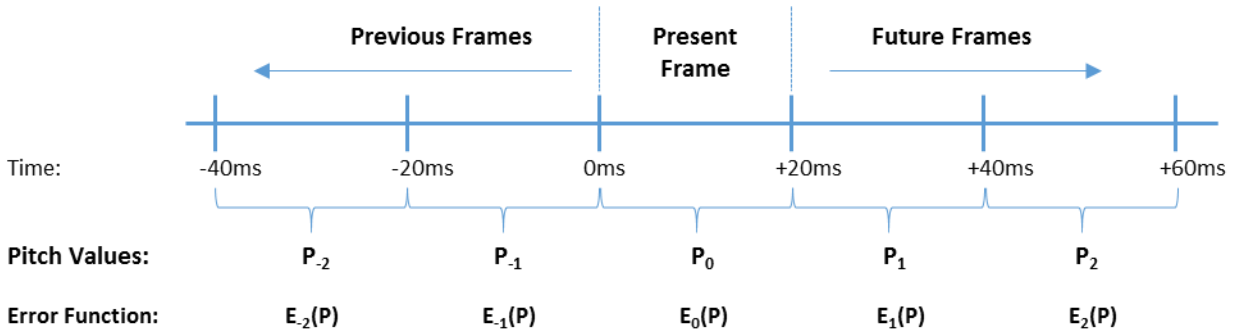


Figure 7 Relationship between Speech Frames

5.1 Pitch Estimation

The objective in pitch estimation is to determine the pitch P_0 corresponding to the “current” speech frame $s_w(n)$. P_0 is related to the fundamental frequency ω_0 by

$$P_0 = \frac{2\pi}{\omega_0} \quad (4)$$

where P_0 is measured in samples (at 8 kHz) and ω_0 is measured in radians.

The pitch estimation algorithm attempts to preserve some continuity of the pitch between neighboring speech frames. A pitch tracking algorithm considers the pitch from previous and future frames, when determining the pitch of the current frame. Previous and future speech frames are obtained by shifting the speech signal in 160 sample (20 ms) time increments prior to the application of the window in Equation (1). The pitches corresponding to the two future speech frames are denoted by P_1 and P_2 . Similarly, the pitches of the two previous speech frames are denoted by P_{-1} and P_{-2} . These relationships are shown in Figure 7.

The pitch is estimated using a two-step procedure. First an initial pitch estimate, denoted by $\hat{P}_l$, is obtained. The initial pitch estimate is restricted to be a member of the set $\{21, 21.5, \dots, 121.5, 122\}$. It is then refined to obtain the final estimate of the fundamental frequency $\hat{\omega}_0$, which has one-quarter-sample accuracy. This two-part procedure is used in part to reduce the computational complexity, and in part to improve the robustness of the pitch estimate.

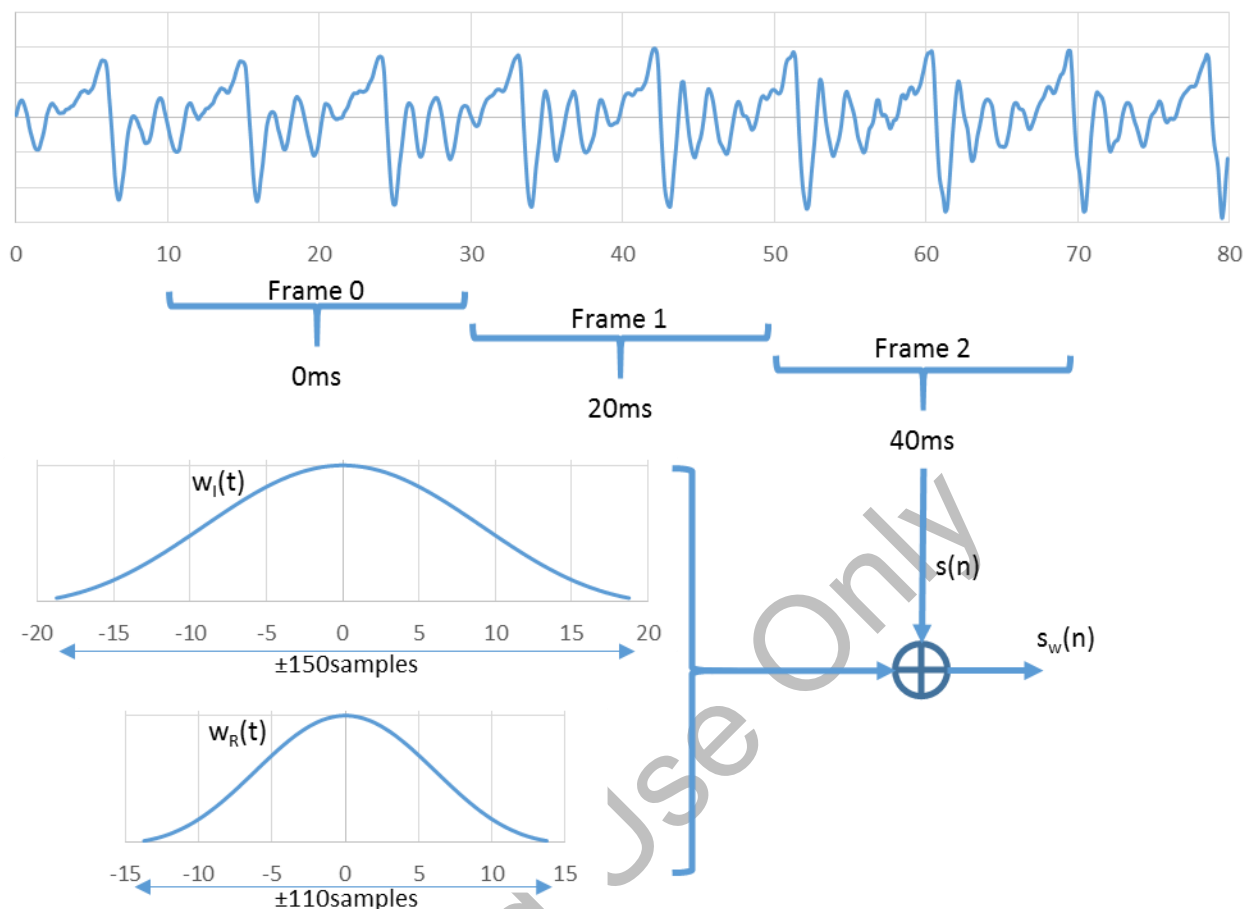


Figure 8 Window Alignment

One important feature of the pitch estimation algorithm is that the initial pitch estimation algorithm uses a different window than the pitch refinement algorithm. The window used for initial pitch estimation, $w_l(n)$, is 301 samples long and is given in Annex B. The window used for pitch refinement (and also for spectral amplitude estimation and V/UV determination), $w_r(n)$, is 221 samples long and is given in Annex C. Throughout this document the window functions are assumed to be equal to zero outside the range given in the Annexes. The center point of the two windows shall coincide, therefore the first non-zero point of $w_r(n)$ shall begin 40 samples after the first non-zero point of $w_l(n)$. This constraint is typically met by adopting the convention that $w_r(n) = w_r(-n)$ and $w_l(n) = w_l(-n)$, as shown in Figure 8. The amount of overlap between neighboring speech segments is a function of the window length. Specifically the overlap is equal to the window length minus the distance between frames (160 samples). Therefore the overlap when using $w_r(n)$ is equal to 61 samples and the overlap when using $w_l(n)$ is equal to 141 samples.

5.1.1 Determination of $E(P)$

To obtain the initial pitch estimate an error function, $E(P)$, is evaluated for every P in the set $\{21, 21.5, \dots, 121.5, 122\}$. Pitch tracking is then used to compare the evaluations of $E(P)$, and the best candidate from this set is chosen as $\hat{P}_l$. This procedure is shown in Figure 9. The function $E(P)$ is defined by

$$E(P) = \frac{\sum_{j=-150}^{150} s_{LPF}^2(j) w_I^2(j) - P \cdot \sum_{n=-\lfloor 150/P \rfloor}^{\lfloor 150/P \rfloor} r(n \cdot P)}{\left[\sum_{j=-150}^{150} s_{LPF}^2(j) w_I^2(j) \right] \left[1 - P \cdot \sum_{j=-150}^{150} w_I^4(j) \right]} \quad (5)$$

where $w_I(n)$ is normalized to meet the constraint

$$\sum_{j=-150}^{150} w_I^2(j) = 1.0 \quad (6)$$

This constraint is satisfied for $w_I(n)$ listed in Annex B. The function $r(t)$ is defined for integer values of t by

$$r(t) = \sum_{j=-150}^{150} s_{LPF}(j) w_I^2(j) s_{LPF}(j+t) w_I^2(j+t) \quad (7)$$

The function $r(t)$ is evaluated at non-integer values of t through linear interpolation:

$$r(t) = (1 + [t] - t) \cdot r([t]) + (t - [t]) \cdot r([t] + 1) \quad (8)$$

where $[x]$ is equal to the largest integer less than or equal to x (i.e. truncating values of x). The low-pass filtered speech signal is given by

$$s_{LPF}(n) = \sum_{j=-10}^{10} s(n-j) h_{LPF}(j) \quad (9)$$

where $h_{LPF}(n)$ is the 21 point FIR filter given in Annex D.

The theoretical justification for the error function $E(P)$ is presented in [3, 6]. The initial pitch estimate $\hat{P}_I$ is chosen such that $E(\hat{P}_I)$ is small; however, $\hat{P}_I$ is not chosen simply to minimize $E(P)$. Instead pitch tracking shall be used to account for pitch continuity between neighboring speech frames.

5.1.2 Pitch Tracking

Pitch tracking is used to improve the pitch estimate by attempting to limit the pitch deviation between consecutive frames. If the pitch estimate is chosen to strictly minimize $E(P)$, then the pitch estimate may change abruptly between succeeding frames. This abrupt change in the pitch can cause degradation in the synthesized speech. In addition, pitch typically changes slowly; therefore, the pitch estimates from neighboring frames can aid in estimating the pitch of the current frames.

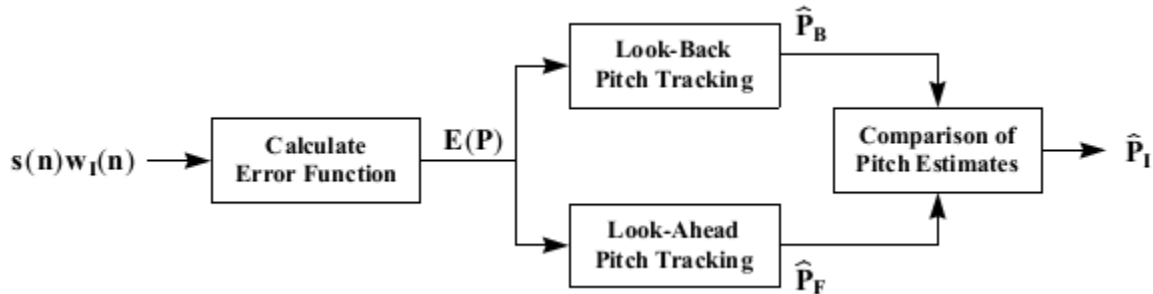


Figure 9 Initial Pitch Estimation

For each speech frame two different pitch estimates are computed. The first, $\hat{P}_B$, is a backward estimate which maintains pitch continuity with previous speech frames. The second, $\hat{P}_F$, is a forward estimate which maintains pitch continuity with future speech frames. The backward pitch

estimate is calculated with the look-back pitch tracking algorithm, while the forward pitch estimate is calculated with the look-ahead pitch tracking algorithm. These two estimates are compared with a set of decision rules defined below, and either the backward or forward estimate is chosen as the initial pitch estimate, $\hat{P}_I$.

5.1.3 Look-Back Pitch Tracking

Let $\hat{P}_{-1}$ and $\hat{P}_{-2}$ denote the initial pitch estimates which are calculated during the analysis of the previous two speech frames. Let $E_1(P)$ and $E_2(P)$ denote the error functions of Equation (5) obtained from the analysis of these previous two frames as shown in Figure 7. Then $E_1(\hat{P}_{-1})$ and $E_2(\hat{P}_{-2})$ will have some specific values. Upon initialization the error functions $E_1(P)$ and $E_2(P)$ are assumed to be equal to zero, and $\hat{P}_{-1}$ and $\hat{P}_{-2}$ are assumed to be equal to 100.

Since pitch continuity with previous frames is desired, the pitch for the current speech frame is considered in a range near $\hat{P}_{-1}$. First, the error function $E(P)$ is evaluated at each value of P which satisfies constraints (10) and (11).

$$0.8 \hat{P}_{-1} \leq P \leq 1.2 \hat{P}_{-1} \quad (10)$$

$$P \in \{21, 21.5, \dots, 121.5, 122\} \quad (11)$$

These values of $E(P)$ are compared and $\hat{P}_B$ is defined as the value of P which satisfies these constraints and which minimizes $E(P)$. The backward cumulative error $CE_B(\hat{P}_B)$ is then computed using the following formula:

$$CE_B(\hat{P}_B) = E(\hat{P}_B) + E_1(\hat{P}_{-1}) + E_2(\hat{P}_{-2}) \quad (12)$$

The backward cumulative error provides a confidence measure for the backward pitch estimate. It is compared against the forward cumulative error using a set of heuristics defined in Section 5.1.4. This comparison determines whether the forward pitch estimate or the backward pitch estimate is selected as the initial pitch estimate for the current frame.

5.1.4 Look-Ahead Pitch Tracking

Look-ahead tracking attempts to preserve pitch continuity between future speech frames. Let $E_1(P)$ and $E_2(P)$ denote the error functions of Equation (5) obtained from the two future speech frames as shown in Figure 7. Since the pitch has not been determined for these future frames, the look-ahead pitch tracking algorithm shall select the pitch of these future frames. This is done in the following manner. First, P_0 is assumed to be fixed. Then the P_1 and P_2 are found which jointly minimize $E_1(P_1) + E_2(P_2)$, subject to constraints (13) through (16).

$$P_1 \in \{21, 21.5, \dots, 121.5, 122\} \quad (13)$$

$$0.8 \hat{P}_0 \leq P_1 \leq 1.2 \hat{P}_0 \quad (14)$$

$$P_2 \in \{21, 21.5, \dots, 121.5, 122\} \quad (15)$$

$$0.8 \hat{P}_1 \leq P_2 \leq 1.2 \hat{P}_1 \quad (16)$$

The values of P_1 and P_2 which jointly minimize $E_1(P_1) + E_2(P_2)$ subject to these constraints are denoted by $\hat{P}_1$ and $\hat{P}_2$, respectively. Once $\hat{P}_1$ and $\hat{P}_2$ have been computed the forward cumulative error function $CE_F(P_0)$ is computed according to:

$$CE_F(\hat{P}_0) = E(\hat{P}_0) + E_1(\hat{P}_1) + E_2(\hat{P}_2) \quad (17)$$

This process is repeated for each P_0 in the set $\{21, 21.5, \dots, 121.5, 122\}$. The corresponding values of $CE_F(P_0)$ are compared and $\hat{P}_0$ is defined as the value of P_0 in this set which results in the minimum value of $CE_F(P_0)$. Note that references [3, 6] can be consulted for more information on the theory and implementation of the look-ahead pitch tracking algorithm.

Once $\hat{P}_0$ has been found, the integer sub-multiples of $\hat{P}_0$ (i.e. $\frac{\hat{P}_0}{2}, \frac{\hat{P}_0}{3}, \dots, \frac{\hat{P}_0}{n}$) shall be considered. Every sub-multiple which is greater than or equal to 21 is computed and replaced with the closest member of the set $\{21, 21.5, \dots, 121.5, 122\}$ (where closeness is measured with mean-square error). Sub-multiples which are less than 21 are disregarded.

The smallest of these sub-multiples is checked against constraints (18), (19) and (20). If this sub-multiple satisfies any of these constraints then it is selected as the forward pitch estimate, $\hat{P}_F$. Otherwise the next largest sub-multiple is checked against these constraints, and it is selected as the forward pitch estimate if it satisfies any of these constraints. This process continues until all pitch sub-multiples have been tested against these constraints. If no pitch sub-multiple satisfies any of these constraints then $\hat{P}_F = \hat{P}_0$. Note that this procedure will always select the smallest sub-multiple which satisfies any of these constraints as the forward pitch estimate.

$$CE_F\left(\frac{\hat{P}_0}{n}\right) \leq 0.85 \quad \text{and} \quad \frac{CE_F(\hat{P}_0/n)}{CE_F(\hat{P}_0)} \leq 1.7 \quad (18)$$

$$CE_F\left(\frac{\hat{P}_0}{n}\right) \leq 0.4 \quad \text{and} \quad \frac{CE_F(\hat{P}_0/n)}{CE_F(\hat{P}_0)} \leq 3.5 \quad (19)$$

$$CE_F(\hat{P}_0/n) \leq 0.05 \quad (20)$$

Once the forward pitch estimate and the backward pitch estimate have both been computed the forward cumulative error and the backward cumulative error are compared. Depending on the result of this comparison either $\hat{P}_F$ or $\hat{P}_B$ will be selected as the initial pitch estimate $\hat{P}_I$. The following set of decision rules is used to select the initial pitch estimate from among these two candidates:

$$\text{If} \quad CE_B(\hat{P}_B) \leq 0.48 \quad \text{then} \quad \hat{P}_I = \hat{P}_B \quad (21)$$

$$\text{Else if} \quad CE_B(\hat{P}_B) \leq CE_F(\hat{P}_F) \quad \text{then} \quad \hat{P}_I = \hat{P}_B \quad (22)$$

$$\text{Else} \quad \hat{P}_I = \hat{P}_F \quad (23)$$

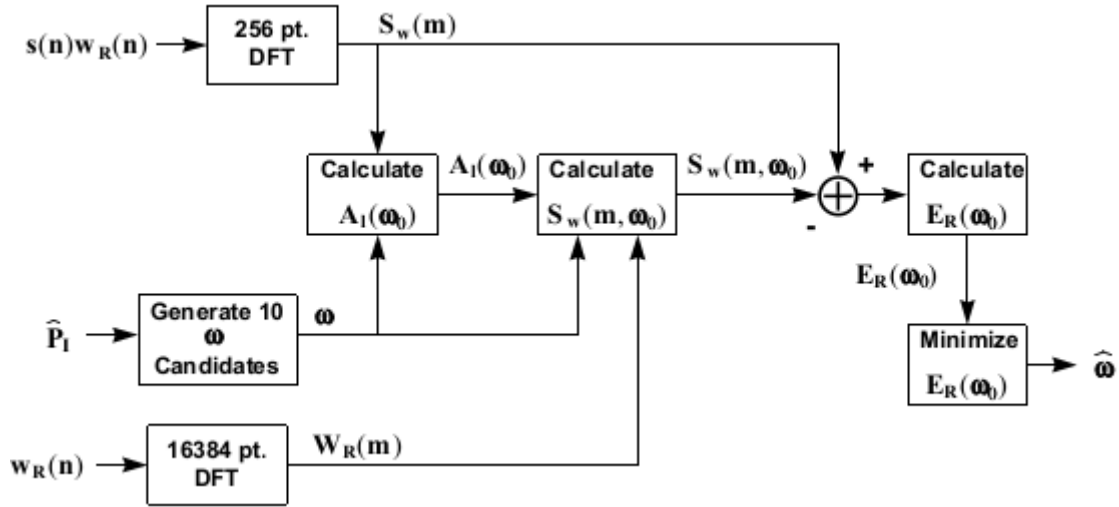


Figure 10 Pitch Refinement

The flowcharts in Annex K should be examined for more information on initial pitch estimation. Note that the initial pitch estimate, $\hat{P}_I$, is a member of the set $\{21, 21.5, \dots, 121.5, 122\}$, and therefore it has half-sample accuracy.

5.1.5 Pitch Refinement

The pitch refinement algorithm improves the resolution of the pitch estimate from one half sample to one quarter sample. Ten candidate pitches are formed from the initial pitch estimate. These are $\hat{P}_I - \frac{9}{8}$, $\hat{P}_I - \frac{7}{8}$, ..., $\hat{P}_I + \frac{7}{8}$, and $\hat{P}_I + \frac{9}{8}$. These candidates are converted to their equivalent fundamental frequency using Equation (4). The error function $E_R(\omega_0)$, defined in Equation (24), is evaluated for each candidate fundamental frequency ω_0 . The candidate fundamental frequency which results in the minimum value of $E_R(\omega_0)$ is selected as the refined fundamental frequency $\hat{\omega}_0$. A block diagram of this process is shown in Figure 10.

$$E_R(\omega_0) = \sum_{m=50}^{\lfloor 0.9254\pi/\omega_0 - 0.5 \rfloor 256 \omega_0/(2\pi)} |S_w(m) - S_w(m, \omega_0)|^2 \quad (24)$$

The synthetic spectrum $S_w(m, \omega_0)$ is given by,

$$S_w(m, \omega_0) = \begin{cases} A_0(\omega_0)W_R(64m) & \text{for } [a_0] \leq m < [b_0] \\ A_1(\omega_0)W_R\left(\left\lfloor 64m - \frac{16384}{2\pi}\omega_0 + 0.5 \right\rfloor\right) & \text{for } [a_1] \leq m < [b_1] \\ A_l(\omega_0)W_R\left(\left\lfloor 64m - \frac{16384}{2\pi}l\omega_0 + 0.5 \right\rfloor\right) & \text{for } [a_l] \leq m < [b_l] \end{cases} \quad (25)$$

where a_l , b_l and A_l are defined in equations (26) thru (28), respectively. The notation $[x]$ denotes the smallest integer greater than or equal to x .

$$a_l = \frac{256}{2\pi}(l - 0.5)\omega_0 \quad (26)$$

$$b_l = \frac{256}{2\pi}(l + 0.5)\omega_0 \quad (27)$$

$$A_l(\omega_0) = \frac{\sum_{m=[a_l]}^{[b_l]-1} S_w(m) W_R^* \left(\left\lfloor 64 m - \frac{16384}{2\pi} l \omega_0 + 0.5 \right\rfloor \right)}{\sum_{m=[a_l]}^{[b_l]-1} |W_R \left(\left\lfloor 64 m - \frac{16384}{2\pi} l \omega_0 + 0.5 \right\rfloor \right)|^2} \quad (28)$$

The function $S_w(m)$ refers to the 256 point Discrete Fourier Transform of $s(n) \cdot w_R(n)$, and $W_R(m)$ refers to the 16384 point Discrete Fourier Transform of $w_R(n)$. These relationships are expressed below. Reference [11] can be consulted for more information on the DFT.

$$S_w(m) = \sum_{n=-110}^{110} s(n) w_R(n) e^{-j2\pi m n / 256} \quad \text{for } -127 \leq m \leq 128 \quad (29)$$

$$W_R(m) = \sum_{n=-110}^{110} w_R(n) e^{-j2\pi m n / 16384} \quad \text{for } -8191 \leq m \leq 8192 \quad (30)$$

The notation $W_R^*(m)$ refers to the complex conjugate of $W_R(m)$. However, since $w_R(n)$ is a real symmetric sequence, $W_R^*(m) = W_R(m)$.

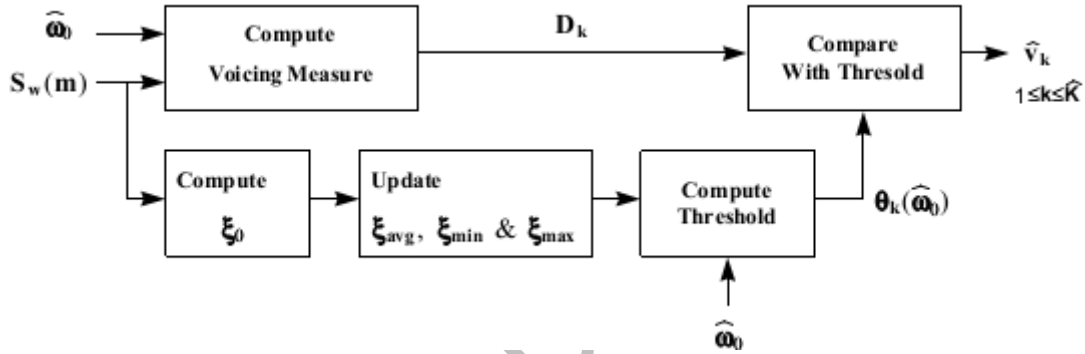


Figure 11 IMBE Voiced/Unvoiced Determination

Once the refined fundamental frequency has been selected from among the ten candidates, it is used to compute the number of harmonics in the current segment, $\hat{L}$, according to the relationship:

$$\hat{L} = \left\lfloor 0.9254 * \left\lfloor \frac{\pi}{\hat{\omega}_0} + 0.25 \right\rfloor \right\rfloor \quad (31)$$

Due to the limits on $\hat{\omega}_0$, equation (31) confines $\hat{L}$ to the range $9 \leq \hat{L} \leq 56$. Once this equation has been computed, the parameters $\hat{a}_l$ and $\hat{b}_l$ for $1 \leq l \leq \hat{L}$ are computed from $\hat{\omega}_0$ according to equations (32) and (33), respectively.

$$\hat{a}_l = \frac{256}{2\pi} (l - 0.5) \hat{\omega}_0 \quad (32)$$

$$\hat{b}_l = \frac{256}{2\pi} (l + 0.5) \hat{\omega}_0 \quad (33)$$

5.2 Voiced / Unvoiced Determination

The voiced / unvoiced (V/UV) decisions, $\hat{v}_k$ for $1 \leq k \leq \hat{K}$, are found by dividing the spectrum into $\hat{K}$ frequency bands and evaluating a voicing measure, D_k , for each band. The number of frequency bands is a function of $\hat{L}$ and is given by:

$$\hat{K} = \begin{cases} \lfloor (\hat{L} + 2)/3 \rfloor & \text{if } \hat{L} \leq 36 \\ 12 & \text{otherwise} \end{cases} \quad (34)$$

The voicing measure for $1 \leq k \leq \hat{K} - 1$ is given by

$$D_k = \frac{\sum_{m=\lfloor \hat{a}_{3k-2} \rfloor}^{\lfloor \hat{b}_{3k} \rfloor - 1} |S_w(m) - S_w(m, \hat{\omega}_0)|^2}{\sum_{m=\lfloor \hat{a}_{3k-2} \rfloor}^{\lfloor \hat{b}_{3k} \rfloor - 1} |S_w(m)|^2} \quad (35)$$

where $\hat{\omega}_0$ is the refined fundamental frequency, and $\hat{a}_l$, $\hat{b}_l$, $S_w(m)$, and $S_w(m, \omega_0)$ are defined in section 5.1.5. Similarly, the voicing measure for the highest frequency band is given by

$$D_{\hat{K}} = \frac{\sum_{m=\lfloor \hat{a}_{3\hat{K}-2} \rfloor}^{\lfloor \hat{b}_{\hat{L}} \rfloor - 1} |S_w(m) - S_w(m, \hat{\omega}_0)|^2}{\sum_{m=\lfloor \hat{a}_{3\hat{K}-2} \rfloor}^{\lfloor \hat{b}_{\hat{L}} \rfloor - 1} |S_w(m)|^2} \quad (36)$$

The parameters D_k for $1 \leq k \leq \hat{K}$ are compared with the threshold function $\Theta_{\xi}(k, \hat{\omega}_0)$ given by:

$$\Theta_{\xi}(k, \hat{\omega}_0) = \begin{cases} 0 & \text{if } E(\hat{P}_l) > 0.5 \text{ and } k \geq 2 \\ 0.5625 [1.0 - 0.3096 (k - 1) \hat{\omega}_0] \cdot M(\xi) & \text{else if } \hat{v}_k(-1) = 1 \\ 0.45 [1.0 - 0.3096 (k - 1) \hat{\omega}_0] \cdot M(\xi) & \text{otherwise} \end{cases} \quad (37)$$

where $M(\xi)$ is an energy dependent function which is computed from a set of local energy parameters and $\hat{v}_k(-1)$ is the value of the k -th V/UV decision for the previous frame. Evaluation of this threshold function requires the parameters ξ_{LF} , ξ_{HF} , and ξ_0 to be computed for the current segment in the following manner, where the value $W_R(0)$ is found by evaluating equation (30) at $m=0$.

$$\xi_{LF} = \sum_{m=0}^{63} \frac{|S_w(m)|^2}{|W_R(0)|^2} \quad (38)$$

$$\xi_{HF} = \sum_{m=64}^{128} \frac{|S_w(m)|^2}{|W_R(0)|^2} \quad (39)$$

$$\xi_0 = \xi_{LF} + \xi_{HF} \quad (40)$$

These parameters are then used to update the parameter ξ_{max} according to the rules presented below. Throughout this section the notation $\xi_{max}(0)$ or ξ_{max} is used to refer to the value of the parameter in the current frame, while the notation $\xi_{max}(-1)$ is used to refer to the value of the parameter in the previous frame.

$$\xi_{max}(0) = \begin{cases} 0.5 \xi_{max}(-1) + 0.5 \xi_0 & \text{if } \xi_0 > \xi_{max}(-1) \\ 0.99 \xi_{max}(-1) + 0.01 \xi_0 & \text{else if } 0.99 \xi_{max}(-1) + 0.01 \xi_0 > 20000 \\ 20000 & \text{otherwise} \end{cases} \quad (41)$$

The completed set of energy parameters for the current frame is used to calculate the function $M(\xi)$ as shown below.

$$M(\xi) = \begin{cases} \left(\frac{0.0025 \xi_{max} + \xi_0}{0.01 \xi_{max} + \xi_0} \right) & \text{if } \xi_{LF} \geq 5 \xi_{HF} \\ \left(\frac{0.0025 \xi_{max} + \xi_0}{0.01 \xi_{max} + \xi_0} \right) \left(\frac{\xi_{LF}}{5 \xi_{HF}} \right)^{1/2} & \text{otherwise} \end{cases} \quad (42)$$

This function is then used in Equation (37) to calculate the V/UV threshold function. If D_k is less than the threshold function then the frequency band $\hat{a}_{3k-2} \leq \omega < \hat{b}_{3k}$ is declared voiced; otherwise this frequency band is declared unvoiced. A block diagram of this procedure is shown in Figure 11. The adopted convention is that if the frequency band $\hat{a}_{3k-2} \leq \omega < \hat{b}_{3k}$ is declared voiced, then $\hat{v}_k = 1$. Alternatively, if the frequency band $\hat{a}_{3k-2} \leq \omega < \hat{b}_{3k}$ is declared unvoiced, then $\hat{v}_k = 0$.

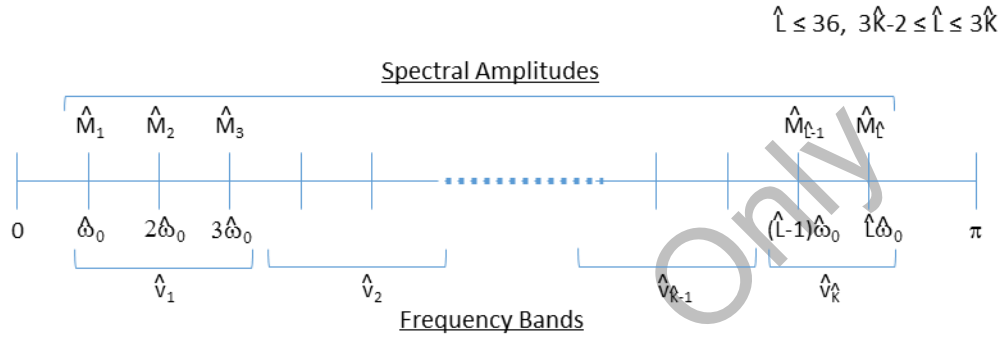


Figure 12 IMBE Frequency Band Structure

With the exception of the highest frequency band, the width of each frequency band is equal to $3\hat{\omega}_0$. Therefore all but the highest frequency band contain three harmonics of the refined fundamental frequency. The highest frequency band (as defined by Equation (34)) may contain more or less than three harmonics of the fundamental frequency. If a particular frequency band is declared voiced, then all of the harmonics within that band are defined to be voiced harmonics. Similarly, if a particular frequency band is declared unvoiced, then all of the harmonics within that band are defined to be unvoiced harmonics.

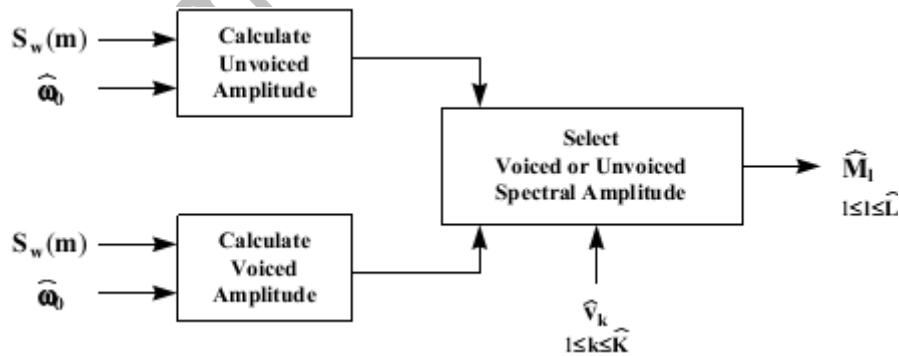


Figure 13 IMBE Spectral Amplitude Estimation

5.3 Estimation of the Spectral Amplitudes

Once the V/UV decisions have been determined the spectral envelope can be estimated as shown in Figure 13. In the IMBE speech coder the spectral envelope in the frequency band $\hat{a}_{3k-2} \leq \omega < \hat{b}_{3k}$ is

specified by 3 spectral amplitudes, which are denoted by $\hat{M}_{3k-2}$, $\hat{M}_{3k-1}$, and $\hat{M}_{3k}$ are estimated by the following equation for l in the range $3k-2 \leq l \leq 3k$.

$$\hat{M}_l = \left[\frac{\sum_{m=\lceil \hat{a}_l \rceil}^{\lceil \hat{b}_l \rceil - 1} |S_w(m)|^2}{\sum_{m=\lceil a_l \rceil}^{\lceil b_l \rceil - 1} |W_R\left(\left[64 m - \frac{16384}{2\pi} l \hat{\omega}_0 + 0.5\right]\right)|^2} \right]^{\frac{1}{2}} \quad (43)$$

Alternatively, if the frequency band for $\hat{a}_{3k-2} \leq \omega < \hat{b}_{3k}$ is declared unvoiced, then $\hat{M}_{3k-2}$, $\hat{M}_{3k-1}$, and $\hat{M}_{3k}$ are estimated according to the following equation for l in the range $3k-2 \leq l \leq 3k$.

$$\hat{M}_l = \left[\frac{1}{\sum_{n=-110}^{110} w_R(n)} \right] \cdot \left[\frac{\sum_{m=\lceil \hat{a}_l \rceil}^{\lceil \hat{b}_l \rceil - 1} |S_w(m)|^2}{(\lceil \hat{b}_l \rceil - \lceil \hat{a}_l \rceil)} \right]^{\frac{1}{2}} \quad (44)$$

This procedure shall be modified slightly for the highest frequency band which covers the frequency interval $\hat{a}_{3\hat{K}-2} \leq \omega < \hat{b}_{\hat{L}}$. The spectral envelope in this frequency band is represented by $\hat{L} - 3\hat{K} + 3$ spectral amplitudes, denoted $\hat{M}_{3\hat{K}-2}$, $\hat{M}_{3\hat{K}-1}$, ... $\hat{M}_{\hat{L}}$. If this frequency band is declared voiced then these spectral amplitudes are estimated using equation (43) for $3\hat{K}-2 \leq l \leq \hat{L}$. Alternatively, if this frequency band is declared unvoiced then these spectral amplitudes are estimated using equation (44) for $3\hat{K}-2 \leq l \leq \hat{L}$.

As described above, the spectral amplitudes $\hat{M}_l$ are estimated in the range $1 \leq l \leq \hat{L}$, where $\hat{L}$ is given in Equation (31). Note that the lowest frequency band, $a_1 \leq \omega < b_3$, is specified by $\hat{M}_1$, $\hat{M}_2$, and $\hat{M}_3$. The D.C. spectral amplitude, $\hat{M}_0$, is ignored in the IMBE speech coder and can be assumed to be zero.

<i>Parameter</i>	<i>Number of Bits</i>
Fundamental Frequency	8
Voiced / Unvoiced Decisions	$\hat{K}$
Spectral Amplitudes	$79 - \hat{K}$
Synchronization	1

Table 1 Bit Allocation Among Model Parameters

6 Full-Rate Parameter Encoding and Decoding

The analysis of each speech frame generates a set of model parameters consisting of the fundamental frequency, $\hat{\omega}_0$, the V/UV decisions, $\hat{v}_k$ for $1 \leq k \leq \hat{K}$, and the spectral amplitudes, $\hat{M}_l$ for $1 \leq l \leq \hat{L}$. Since the speech coder is designed to operate at 7.2 kbps with a 20 ms frame length, 144 bits per frame are available for encoding the model parameters. Of these 144 bits, 56 are reserved for error control as is discussed in Section 7 of this document, and the remaining 88 bits are divided among the model parameters as shown in Table 1. This section describes the manner in which these bits are used to quantize, encode, decode and reconstruct the model parameters. In Section 6.1 the encoding and decoding of the fundamental frequency is discussed, while Section 6.2 discusses the encoding and decoding of the V/UV decisions. Section 6.3 discusses the quantization and encoding of the spectral amplitudes, and Section 6.4 discusses the decoding and reconstruction of the spectral amplitudes. Reference [7] provides general information on many of the techniques used in this section.

6.1 Fundamental Frequency Encoding and Decoding

The fundamental frequency is estimated with one-quarter sample resolution in the interval $\frac{2\pi}{123.125} \leq \hat{\omega}_0 \leq \frac{2\pi}{19.875}$; however, it is only encoded at half-sample resolution. This is accomplished by finding the value of $\hat{b}_0$ which satisfies:

$$\hat{b}_0 = \left\lfloor \frac{4\pi}{\hat{\omega}_0} - 39 \right\rfloor \quad (45)$$

Value	bits
0	0000 0000
1	0000 0001
2	0000 0010
.	.
.	.
.	.
255	1111 1111

Table 2 Eight Bit Binary Representation

The quantizer value $\hat{b}_0$ is represented with 8 bits using the unsigned binary representation shown in Table 2. This representation is used throughout the IMBE speech coder to convert quantized values into a specific bit pattern.

The fundamental frequency is decoded and reconstructed at the receiver by using Equation (46) to convert $\hat{b}_0$ to the received fundamental frequency $\tilde{\omega}_0$. In addition $\hat{b}_0$ is used to calculate $\tilde{K}$ and $\tilde{L}$,

the number of V/UV decisions and the number of spectral amplitudes, respectively. These relationships are given in Equations (47) and (48).

$$\tilde{\omega}_0 = \frac{4\pi}{\tilde{b}_0 + 39.5} \quad (46)$$

$$\tilde{L} = \left\lceil 0.9254 \left\lfloor \frac{\pi}{\tilde{\omega}_0} + 0.25 \right\rfloor \right\rceil \quad (47)$$

$$\tilde{K} = \begin{cases} \left\lfloor \frac{\tilde{L}+2}{3} \right\rfloor & \text{if } \tilde{L} \leq 36 \\ 12 & \text{otherwise} \end{cases} \quad (48)$$

Since $\tilde{K}$ and $\tilde{L}$ control subsequent bit allocation by the receiver, it is important that they equal $\hat{K}$ and $\hat{L}$, respectively. This occurs if there are no uncorrectable bit errors in the six most significant bits (MSB) of $\tilde{b}_0$. For this reason these six bits are well protected by the error correction scheme discussed in Section 7. A block diagram of the fundamental frequency encoding and decoding process is shown in Figure 14.

Since the pitch estimation algorithm described in Section 5.1 restricts the range of $\hat{\omega}_0$ to $\frac{2\pi}{123.125} \leq \hat{\omega}_0 \leq \frac{2\pi}{19.875}$, the value of $\hat{b}_0$ computed according to Equation (45) is limited to the range $0 \leq \hat{b}_0 \leq 207$. The use of 8 bits to represent $\hat{b}_0$ leaves 48 values of $\hat{b}_0$ (i.e. $208 \leq \hat{b}_0 \leq 255$) which are outside the valid range of pitch values. These 48 values are reserved for future use.

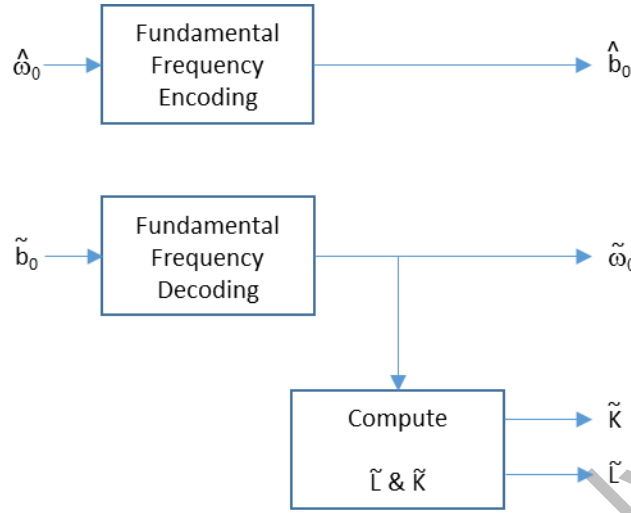


Figure 14 Fundamental Frequency Encoding and Decoding

6.2 Voiced / Unvoiced Decision Encoding and Decoding

The V/UV decisions $\hat{v}_k$, for $1 \leq k \leq \hat{K}$, are binary values which classify each frequency band as either voiced or unvoiced. These values are encoded using

$$\hat{b}_1 = \sum_{k=1}^{\hat{K}} \hat{v}_k 2^{\hat{K}-k} \quad (49)$$

The quantizer value $\hat{b}_1$ is represented with $\hat{K}$ bits using an unsigned binary representation which is analogous to that shown in Table 2.

At the receiver the $\tilde{K}$ bits correspond to $\tilde{b}_1$ are decoded into the V/UV decisions $\tilde{v}_l$ for $1 \leq l \leq \tilde{L}$. Note that this is a departure from the V/UV convention used by the encoder, which used a single V/UV decision to represent an entire frequency band. Instead the decoder uses a separate V/UV decision for each spectral amplitude. The decoder performs this conversion by using $\tilde{b}_1$ to determine which frequency bands are voiced or unvoiced. The state of $\tilde{v}_l$ is then set depending upon whether the frequency $\omega = l \tilde{\omega}_0$ is within a voiced or unvoiced frequency band. This can be expressed mathematically as shown in the following two equations.

$$\kappa_l = \begin{cases} \left\lfloor \frac{l+2}{3} \right\rfloor & \text{if } l \leq 36 \\ 12 & \text{otherwise} \end{cases} \quad (50)$$

$$\tilde{v}_l = \left\lfloor \frac{\tilde{b}_1}{2^{\tilde{K}-\kappa_l}} \right\rfloor - 2 \left\lfloor \frac{\tilde{b}_1}{2^{\tilde{K}+1-\kappa_l}} \right\rfloor \quad \text{for } 1 \leq l \leq \tilde{L} \quad (51)$$

Figure 15 shows a block diagram of the V/UV decision encoding and decoding process.

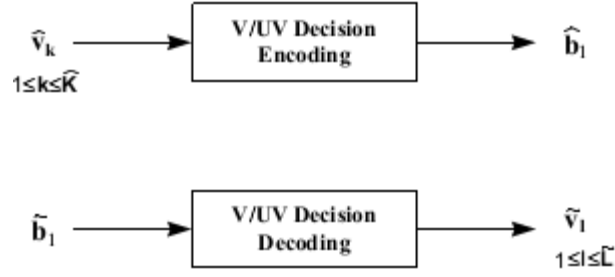


Figure 15 V/UV Decision Encoding and Decoding

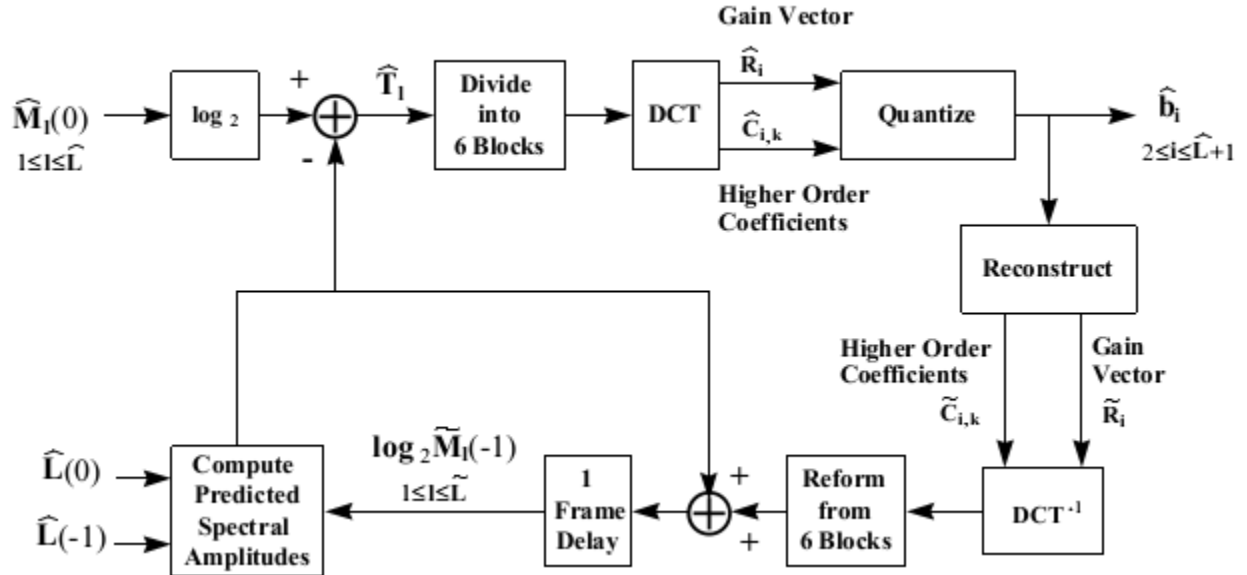


Figure 16 Encoding of the Spectral Amplitudes

6.3 Spectral Amplitudes Encoding

The spectral amplitudes $\hat{M}_l$ for $1 \leq l \leq \hat{L}$, are real values which shall be quantized prior to encoding. This is accomplished as shown in Figure 16, by forming the spectral amplitude prediction residuals $\hat{T}_l$ for $1 \leq l \leq \hat{L}$, according to Equations (52) through (57). The reader is referred to [6] for more information on this topic.

For the purpose of this discussion $\hat{L}(0)$ or $\hat{L}$ refer to the number of harmonics in the current frame, while $\hat{L}(-1)$ refers to the number of harmonics in the previous frame. Similarly, $\hat{M}_l(0)$ for $1 \leq l \leq \hat{L}$, refers to the unquantized spectral amplitudes of the current frame, while $\hat{M}_l(-1)$ for $1 \leq l \leq \hat{L}$ refers to the quantized spectral amplitudes of the previous frame.

$$\hat{k}_l = \frac{\hat{L}(-1)}{\hat{L}(0)} \cdot l \quad (52)$$

$$\hat{\delta}_l = \hat{k}_l - \lfloor \hat{k}_l \rfloor \quad (53)$$

$$\begin{aligned}\hat{T}_l = & \log_2 \hat{M}_l(0) - \rho (1 - \hat{\delta}_l) \log_2 \tilde{M}_{[\hat{k}_l]}(-1) - \rho \hat{\delta}_l \log_2 \tilde{M}_{[\hat{k}_l]+1}(-1) \\ & + \frac{\rho}{\hat{L}(0)} \sum_{\lambda=1}^{\hat{L}(0)} \left[(1 - \hat{\delta}_\lambda) \log_2 \tilde{M}_{[\hat{k}_\lambda]}(-1) + \hat{\delta}_\lambda \log_2 \tilde{M}_{[\hat{k}_\lambda]+1}(-1) \right]\end{aligned}\quad (54)$$

The prediction coefficient, ρ , is adjusted for each frame according to the following rule:

$$\rho = \begin{cases} 0.4 & \text{if } \hat{L}(0) \leq 15 \\ 0.03 \hat{L}(0) - 0.05 & \text{if } 15 < \hat{L}(0) \leq 24 \\ 0.7 & \text{otherwise} \end{cases} \quad (55)$$

In order to form $\hat{T}_l$ using equations (52) through (55), the following assumptions are made:

$$\tilde{M}_0(-1) = 1.0 \quad (56)$$

$$\tilde{M}_l(-1) = \tilde{M}_{\tilde{L}(-1)}(1) \quad \text{for } l > \tilde{L}(-1) \quad (57)$$

Also upon initialization $\tilde{M}_l(-1)$ should be set equal to 1.0 for all l , and $\hat{L}(-1) = 30$.

The $\hat{L}$ prediction residuals are then divided into 6 blocks. The length of each block, denoted $\hat{j}_i$ for $1 \leq i \leq 6$, is adjusted such that the following constraints are satisfied:

$$\sum_{i=1}^6 \hat{j}_i = \hat{L} \quad (58)$$

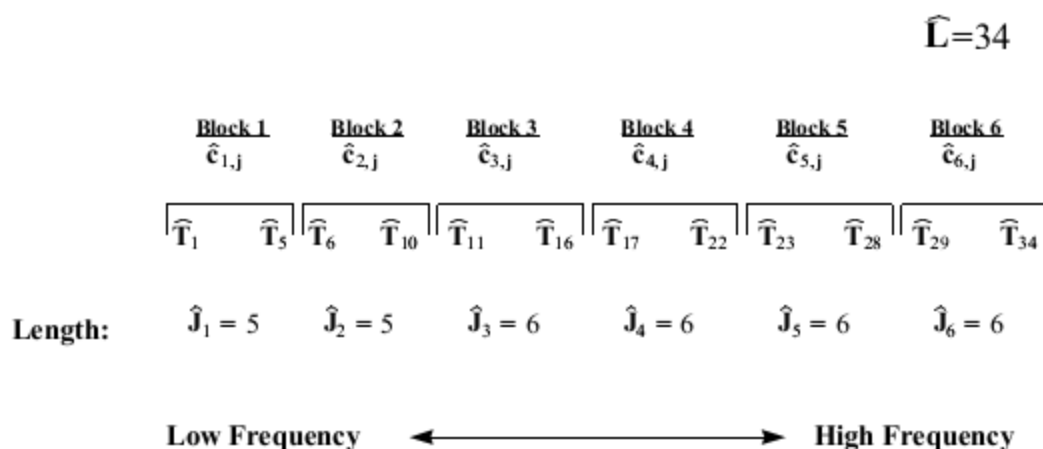
$$\left\lfloor \frac{\hat{L}}{6} \right\rfloor \leq \hat{j}_i \leq \hat{j}_{i+1} \leq \left\lceil \frac{\hat{L}}{6} \right\rceil \quad \text{for } 1 \leq i \leq 5 \quad (59)$$

The table shown in Annex J lists the six block lengths for all possible values of $\hat{L}$. The first or lowest frequency block is denoted by $\hat{c}_{1,j}$ for $1 \leq j \leq \hat{j}_1$, and it consists of the first $\hat{j}_1$ consecutive elements of $\hat{T}_l$, (i.e. $1 \leq l \leq \hat{j}_1$). The second block is denoted by $\hat{c}_{2,j}$ for $1 \leq j \leq \hat{j}_2$, and it consists of the first $\hat{j}_2$ consecutive elements of $\hat{T}_l$, (i.e. $\hat{j}_1 + 1 \leq l \leq \hat{j}_1 + \hat{j}_2$). This continues through the sixth or highest block which is denoted by $\hat{c}_{6,j}$ for $1 \leq j \leq \hat{j}_6$. It consists of the last $\hat{j}_6$ consecutive elements of $\hat{T}_l$, (i.e. $\hat{L} + 1 - \hat{j}_6 \leq l \leq \hat{L}$). An example of this process is shown in Figure 17 for $\hat{L} = 34$.

Each of the six blocks is transformed using a Discrete Cosine Transform (DCT), which is discussed in [7]. The length of the DCT for the i -th block is equal to $\hat{j}_i$. The DCT coefficients are denoted by $\hat{C}_{i,k}$, where $1 \leq i \leq 6$ refers to the block number, and $1 \leq k \leq \hat{j}_i$ refers to the particular coefficient within each block. The formula for the computation of these DCT coefficients is as follows:

$$\hat{C}_{i,k} = \frac{1}{\hat{j}_i} \sum_{j=1}^{\hat{j}_i} \hat{c}_{i,j} \cos \left[\frac{\pi (k-1)(j-1/2)}{\hat{j}_i} \right] \quad \text{for } 1 \leq k \leq \hat{j}_i \quad (60)$$

The DCT coefficients from each of the six blocks are then divided into two groups. The first group consists of the first DCT coefficient from each of the six blocks. These coefficients are used to form a six element vector, $\hat{R}_i$ for $1 \leq i \leq 6$, where $\hat{R}_i = \hat{C}_{i,1}$. The vector $\hat{R}_i$ is referred to as the gain vector, and its construction is shown in Figure 18. The quantization of the gain vector is discussed in section 6.3.1.

Figure 17 Prediction Residual Blocks for $L = 34$

The second group consists of the remaining higher order DCT coefficients. These coefficients correspond to $\hat{c}_{i,j}$, where $1 \leq i \leq 6$, and $2 \leq j \leq \hat{J}_i$. Note that if $\hat{J}_i = 1$, then there are no higher order DCT coefficients in the i -th block. The quantization of the higher order DCT coefficients is discussed in section 6.3.2.

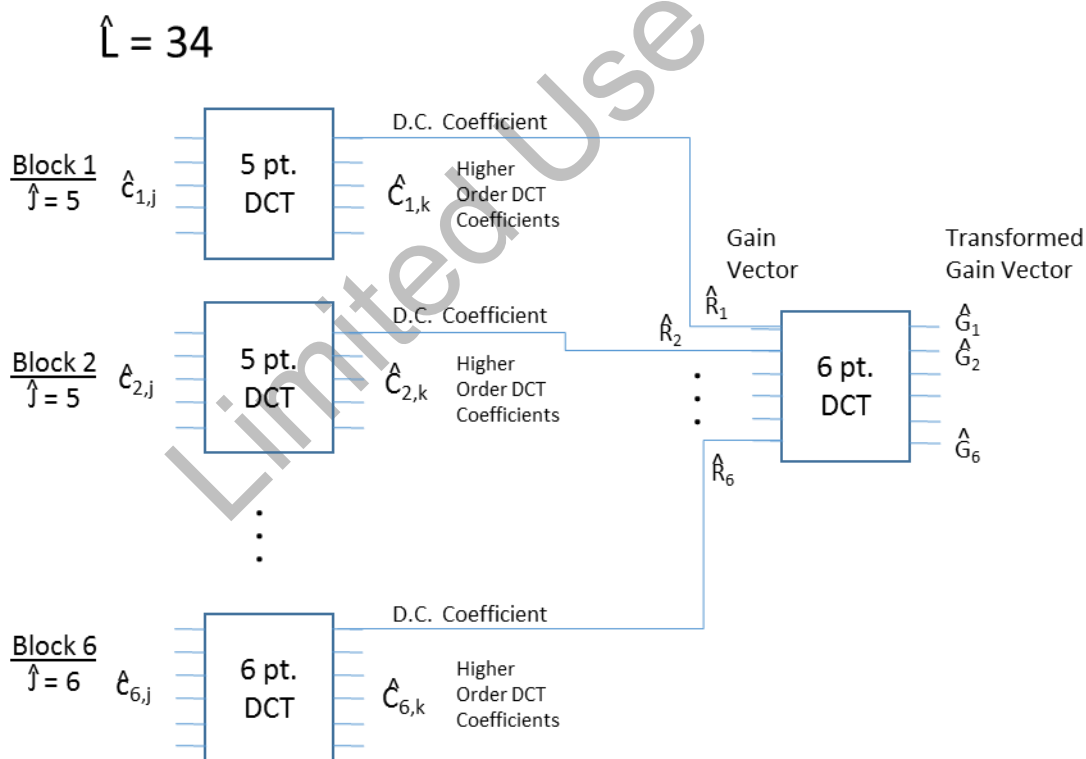


Figure 18 Formation of Gain Vector

One important feature of the spectral amplitude encoding algorithm is that the spectral amplitude information is transmitted differentially. Specifically, a prediction residual is transmitted which measures the change in the spectral envelope between the current frame and the previous frame. In order for a differential scheme of this type to work properly, the encoder needs to simulate the

operation of the decoder and use the reconstructed spectral amplitudes from the previous frame to predict the spectral amplitudes of the current frame. The IMBE spectral amplitude encoder simulates the spectral amplitude decoder by setting $\tilde{L} = \hat{L}$ and then reconstructing the spectral amplitudes as discussed above. This is shown as the feedback path in Figure 16.

6.3.1 Encoding the Gain Vector

The gain vector can be viewed as a coarse representation of the spectral envelope of the current segment of speech. The quantization of the gain vector begins with a six point DCT of $\hat{R}_i$ for $1 \leq i \leq 6$ as shown in the following equation.

$$\hat{G}_m = \frac{1}{6} \sum_{i=1}^6 \hat{R}_i \cos \left[\frac{\pi (m-1)(i-1/2)}{6} \right] \quad \text{for } 1 \leq m \leq 6 \quad (61)$$

The resulting vector, denoted by $\hat{G}_m$ for $1 \leq m \leq 6$, is quantized in two parts. The first element, $\hat{G}_1$, can be viewed as representing the overall gain or level of the speech segment. This element is quantized using the 6 bit non-uniform quantizer given in Annex E. The 6 bit value $\hat{b}_2$ is defined as the index of the quantizer value (as shown in Annex E) which is nearest to $\hat{G}_1$. The remaining five elements of $\hat{G}_m$ are quantized using uniform scalar quantizers where the five quantizer values $\hat{b}_3$ to $\hat{b}_7$ are computed from the vector elements as shown in Equation (62).

$$\hat{b}_m = \begin{cases} 0 & \text{if } \left\lfloor \frac{\hat{G}_{m-1}}{\hat{\Delta}_m} \right\rfloor < -2^{\hat{B}_m-1} \\ 2^{\hat{B}_m} - 1 & \text{if } \left\lfloor \frac{\hat{G}_{m-1}}{\hat{\Delta}_m} \right\rfloor \geq 2^{\hat{B}_m-1} \\ \left\lfloor \frac{\hat{G}_{m-1}}{\hat{\Delta}_m} \right\rfloor + 2^{\hat{B}_m-1} & \text{otherwise} \end{cases} \quad \text{for } 3 \leq m \leq 7 \quad (62)$$

The parameters $\hat{B}_m$ and $\hat{\Delta}_m$ in Equation (62) are the number of bits and the step sizes used to quantize each element. These values are dependent upon $\hat{L}$, which is the number of harmonics in the current frame. This dependence is tabulated in Annex F. Since $\hat{L}$ is known by the encoder, the correct values of $\hat{B}_m$ and $\hat{\Delta}_m$ are first obtained using Annex F and then the quantizer values $\hat{b}_m$ for $3 \leq m \leq 7$ are computed using Equation (62). The final step is to convert each quantizer value into an unsigned binary representation using the same method as shown in Table 2.

6.3.2 Encoding the Higher Order DCT Coefficients

Once the gain vector has been quantized, the remaining bits are used to encode the $\hat{L} - 6$ higher order DCT coefficients which complete the representation of the spectral amplitudes. Annex G shows the bit allocation as a function of $\hat{L}$ for these coefficients. For each value of $\hat{L}$ the $\hat{L} - 6$ entries, labeled $\hat{b}_8$ through $\hat{b}_{\hat{L}+1}$, provide the bit allocation for the higher order DCT coefficients. The adopted convention is that $[\hat{b}_8, \hat{b}_9, \dots, \hat{b}_{\hat{L}+1}]$ correspond to $[\hat{C}_{1,2}, \hat{C}_{1,3}, \dots, \hat{C}_{1,\hat{J}_1}, \dots, \hat{C}_{6,2}, \hat{C}_{6,3}, \dots, \hat{C}_{6,\hat{J}_6}]$, respectively.

Once the bit allocation for the higher order DCT coefficients has been obtained, these coefficients are quantized using uniform quantization. The step size used to quantize each coefficient shall be computed from the bit allocation and the standard deviation of the DCT coefficient using Tables 3 and 4. For example, if 4 bits are allocated for a particular coefficient, then from Table 3 the step size, $\hat{\Delta}$, equals 0.40σ . If this was the third DCT coefficient from any block (i.e. $\hat{C}_{i,3}$), then $\sigma = 0.241$ as

shown in Table 4. Performing this multiplication gives a step size of 0.0964. Once the bit allocation and the step sizes for the higher order DCT coefficients have been determined, then the bit encodings $\hat{b}_m$ for $8 \leq m \leq \hat{L} + 1$ are computed according to Equation (63).

<i>Number of Bits</i>	<i>Step Size</i>
1	1.20 σ
2	0.85 σ
3	0.65 σ
4	0.40 σ
5	0.28 σ
6	0.15 σ
7	0.08 σ
8	0.04 σ
9	0.02 σ
10	0.01 σ

Table 3 Uniform Quantizer Step Size for Higher Order DCT Coefficients

$$\hat{b}_m = \begin{cases} 0 & \text{if } \left\lfloor \frac{\hat{c}_{i,k}}{\hat{\Delta}_m} \right\rfloor < -2^{\hat{B}_m-1} \\ 2^{\hat{B}_m} - 1 & \text{if } \left\lfloor \frac{\hat{c}_{i,k}}{\hat{\Delta}_m} \right\rfloor \geq 2^{\hat{B}_m-1} \\ \left\lfloor \frac{\hat{c}_{i,k}}{\hat{\Delta}_m} \right\rfloor + 2^{\hat{B}_m-1} & \text{otherwise} \end{cases} \quad \text{for } 8 \leq m \leq \hat{L} + 1 \quad (63)$$

The parameters $\hat{b}_m$, $\hat{B}_m$, and $\hat{\Delta}_m$ in equation (63) refer to the quantizer value, the number of bits and the step size which has been computed for $\hat{c}_{i,k}$, respectively. Note that the relationship between m , i , and k in Equation (63) is known and can be expressed as:

$$m = 6 + k + \sum_{n=1}^{i-1} \hat{J}_n \quad (64)$$

Finally, each quantizer value is converted into the appropriate unsigned binary representation which is analogous to that shown in Table 2.

<i>DCT Coefficient</i>	σ
$C_{i,2}$	0.307
$C_{i,3}$	0.241
$C_{i,4}$	0.207
$C_{i,5}$	0.190
$C_{i,6}$	0.179
$C_{i,7}$	0.173
$C_{i,8}$	0.165
$C_{i,9}$	0.170
$C_{i,10}$	0.170

Table 4 Standard Deviation of Higher Order DCT Coefficients

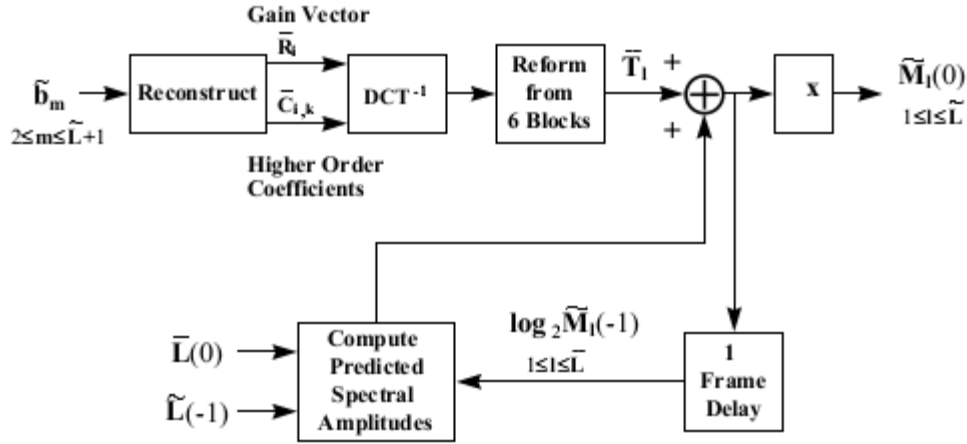


Figure 19 Decoding of the Spectral Amplitudes

6.4 Spectral Amplitudes Decoding

In order for the decoder to reconstruct the spectral amplitudes, the parameter $\tilde{L}$ shall first be computed from $\tilde{b}_0$ using Equations (46) and (47). Then the spectral amplitudes can be decoded and reconstructed by inverting the quantization and encoding procedure described above. A block diagram of the spectral amplitude decoder is shown in Figure 19.

The first step in the spectral amplitude reconstruction process is to divide the spectral amplitudes into six blocks. The length of each block, $\tilde{J}_i$ for $1 \leq i \leq 6$, is adjusted to meet the following constraints.

$$\sum_{i=1}^6 \tilde{J}_i = \tilde{L} \quad (65)$$

$$\left\lfloor \frac{\tilde{L}}{6} \right\rfloor \leq \tilde{J}_i \leq \tilde{J}_{i+1} \leq \left\lceil \frac{\tilde{L}}{6} \right\rceil \quad \text{for } 1 \leq i \leq 5 \quad (66)$$

The elements of these blocks are denoted by $\tilde{C}_{i,k}$ where $1 \leq i \leq 6$ denotes the block number and where $1 \leq k \leq \tilde{J}_i$ denotes the element within that block. The first element of each block is then set equal to the decoded gain vector, $\tilde{R}_i$, via equation (67). The formation of the decoded gain vector is discussed in Section 6.4.1.

$$\tilde{C}_{i,1} = \tilde{R}_i \quad \text{for } 1 \leq i \leq 6 \quad (67)$$

The remaining elements of each block correspond to the decoded higher order DCT coefficients which are discussed in Section 6.4.2.

6.4.1 Decoding the Gain Vector

The gain is decoded in two parts. First the six bit value $\tilde{b}_2$ is used to decode the first element of the transformed gain vector, denoted by $\tilde{G}_1$. This is done by using the 6 bit value $\tilde{b}_2$ as an index into the quantizer values listed in Annex E. Next the five quantizer values $\tilde{b}_3$ through $\tilde{b}_7$ are used to reconstruct the remaining five elements of the transformed gain vector, denoted by $\tilde{G}_2$ through $\tilde{G}_6$. This is done by using $\tilde{L}$, the number of harmonics in the current frame, in combination with the table in Annex F to establish the bit allocation and step size for each of these five elements. The

relationship between the quantizer values and the transformed gain vector elements is expressed in Equation (68), where $\tilde{\Delta}_m$ and $\tilde{B}_m$ are the step sizes and the number of bits found via Annex F.

$$\tilde{G}_{m-1} = \begin{cases} 0 & \text{if } \tilde{B}_m = 0 \\ \tilde{\Delta}_m (\tilde{b}_m - 2^{\tilde{B}_m-1} + 0.5) & \text{otherwise} \end{cases} \quad \text{for } 3 \leq m \leq 7 \quad (68)$$

Once the transformed gain vector has been reconstructed in this manner, the gain vector $\tilde{R}_i$ for $1 \leq i \leq 6$ shall be computed through an inverse DCT of $\tilde{G}_m$ in the following equations.

$$\tilde{R}_i = \sum_{m=1}^6 \alpha(m) \tilde{G}_m \cos \left[\frac{\pi (m-1)(i-1/2)}{j_i} \right] \quad \text{for } 1 \leq i \leq 6 \quad (69)$$

$$\alpha(m) = \begin{cases} 1 & \text{if } m = 1 \\ 2 & \text{otherwise} \end{cases} \quad (70)$$

6.4.2 Decoding the Higher Order DCT Coefficients

The higher order DCT coefficients, which are denoted by $\tilde{C}_{i,k}$ for $2 \leq i \leq 6$ and $1 \leq k \leq \tilde{J}_i$, are reconstructed from the quantizer values $\tilde{b}_8, \tilde{b}_9, \dots, \tilde{b}_{\tilde{L}+1}$. First the bit allocation table listed in Annex G is used to determine the appropriate bit allocation. The adopted convention is that $[\tilde{b}_8, \tilde{b}_9, \dots, \tilde{b}_{\tilde{L}+1}]$ correspond to $[\tilde{C}_{1,2}, \tilde{C}_{1,3}, \dots, \tilde{C}_{1,\tilde{J}_1}, \dots, \tilde{C}_{6,2}, \tilde{C}_{6,3}, \dots, \tilde{C}_{6,\tilde{J}_6}]$, respectively. Once the bit allocation has been determined the step sizes for each $\tilde{C}_{i,k}$ are computed using Tables 3 and 4. The determination of the bit allocation and the step sizes proceeds in the same manner as is discussed in Section 6.3.2. Using the notation $\tilde{B}_m$ and $\tilde{\Delta}_m$ to denote the number of bits and the step size, respectively, then each higher order DCT coefficient can be reconstructed according to the following formula:

$$\tilde{C}_{i,k} = \begin{cases} 0 & \text{if } \tilde{B}_m = 0 \\ \tilde{\Delta}_m (\tilde{b}_m - 2^{\tilde{B}_m-1} + 0.5) & \text{otherwise} \end{cases} \quad \text{for } 8 \leq m \leq \tilde{L}+1 \quad (71)$$

As in Equation (64), the following equation can be used to relate m , i , and k .

$$m = 6 + k + \sum_{n=1}^{i-1} \tilde{J}_n \quad (72)$$

Once the DCT coefficients $\tilde{C}_{i,k}$ have been reconstructed, an inverse DCT is computed on each of the six blocks to form the vectors $\tilde{c}_{i,j}$. This is done using the following equations for $1 \leq i \leq 6$.

$$\tilde{c}_{i,j} = \sum_{k=1}^{\tilde{J}_i} \alpha(k) \tilde{C}_{i,k} \cos \left[\frac{\pi (k-1)(j-1/2)}{\tilde{J}_i} \right] \quad \text{for } 1 \leq j \leq \tilde{J}_i \quad (73)$$

$$\alpha(k) = \begin{cases} 1 & \text{if } k = 1 \\ 2 & \text{otherwise} \end{cases} \quad (74)$$

The six transformed blocks $\tilde{c}_{i,j}$ are then joined to form a single vector of length $\tilde{L}$, which is denoted $\tilde{T}_l$ for $1 \leq l \leq \tilde{L}$. The vector $\tilde{T}_l$ corresponds to the reconstructed spectral amplitude prediction residuals. The adopted convention is that the first $\tilde{J}_1$ elements of $\tilde{T}_l$ are equal to $\tilde{c}_{1,j}$ for $1 \leq j \leq \tilde{J}_1$. The next $\tilde{J}_2$ elements of $\tilde{T}_l$ are equal to $\tilde{c}_{2,j}$ for $1 \leq j \leq \tilde{J}_2$. This continues until the last $\tilde{J}_6$ elements of $\tilde{T}_l$ are equal to $\tilde{c}_{6,j}$ for $1 \leq j \leq \tilde{J}_6$. Finally, the reconstructed log₂ spectral amplitudes for the current frame are computed using the following equations.

$$\tilde{k}_l = \frac{\tilde{L}(-1)}{\tilde{L}(0)} \cdot l \quad (75)$$

$$\tilde{\delta}_l = \tilde{k}_l - \lfloor \tilde{k}_l \rfloor \quad (76)$$

$$\begin{aligned} \log_2 \tilde{M}_l(0) &= \tilde{T}_l + \rho (1 - \tilde{\delta}_l) \log_2 \tilde{M}_{\lfloor \tilde{k}_l \rfloor}(-1) + \rho \tilde{\delta}_l \log_2 \tilde{M}_{\lfloor \tilde{k}_l \rfloor + 1}(-1) \\ &\quad - \frac{\rho}{\tilde{L}(0)} \sum_{\lambda=1}^{\tilde{L}(0)} [(1 - \tilde{\delta}_\lambda) \log_2 \tilde{M}_{\lfloor \tilde{k}_\lambda \rfloor}(-1) + \tilde{\delta}_\lambda \log_2 \tilde{M}_{\lfloor \tilde{k}_\lambda \rfloor + 1}(-1)] \end{aligned} \quad (77)$$

In order to reconstruct $\tilde{M}_l(0)$ using equations (75) through (77), the following assumptions are always made:

$$\tilde{M}_0(-1) = 1.0 \quad (78)$$

$$\tilde{M}_l(-1) = \tilde{M}_{\tilde{L}(-1)}(-1) \quad \text{for } l > \tilde{L}(-1) \quad (79)$$

In addition it is assumed that upon initialization $\tilde{M}_l(-1) = 1$ for all l , and $\tilde{L}(-1) = 30$. Note that later sections of the IMBE decoder require the spectral amplitudes, $\tilde{M}_l$ for $1 \leq l \leq \tilde{L}$, which shall be computed by applying the inverse $\log_2$ to each of the values computed with Equation (77).

One final note is that it should be clear that the IMBE speech coder uses adaptive bit allocation and quantization which is dependent upon the number of harmonics in each frame. At the encoder the value $\hat{L}$ is used to determine the bit allocation and quantizer step sizes, while at the decoder the value $\tilde{L}$ is used to determine the bit allocation and quantizer step sizes. In order to ensure proper operation it is necessary that these two values be equal (i.e. $\hat{L} = \tilde{L}$). The encoder and decoder are designed to ensure this property except in the presence of a very large number of bit errors. In addition, the decoder is designed to detect frames where a large number of bit errors may prevent the generation of the correct bit allocation and quantizer step sizes. In this case the decoder discards the bits for the current frame and repeats the parameters from the previous frame. This is discussed in more detail in latter sections of this document.

6.5 Synchronization Encoding and Decoding

A final one bit quantizer value is reserved in each speech frame for synchronization. This quantizer value, denoted by $\hat{b}_{\hat{L}+2}$ is set to an alternating sequence by the encoder. If this bit was set to 0 during the previous speech frame, then this bit should be set to a 1 for the current speech frame. Otherwise, if this bit was set to 1 during the previous speech frame, then this bit should be set to a 0 for the current speech frame. This is expressed in the following equation, where $\hat{b}_{\hat{L}+2}(0)$ refers to the value for the current frame, while $\hat{b}_{\hat{L}+2}(-1)$ refers to the value for the previous frame.

$$\hat{b}_{\hat{L}+2}(0) = \begin{cases} 0 & \text{if } \hat{b}_{\hat{L}+2}(-1) = 1 \\ 1 & \text{otherwise} \end{cases} \quad (80)$$

It is assumed that $\hat{b}_{\hat{L}+2}(0)$ should be set equal to 0 during the first frame following initialization.

The decoder may use this bit to establish synchronization. As presented in Section 7, this bit is not error correction encoded or modulated, and it is placed in a fixed offset relative to the beginning of each 144 bit frame of speech data. The decoder may check each possible offset in the received data stream and establish which offset is most likely to correspond to the synchronization bit. The beginning of each speech frame can then be established using the known distance between the beginning of each speech frame and the synchronization bit. Note that the number of received speech frames which is used to establish synchronization can be modified to trade off the probability of false synchronization, the synchronization delay, and the ability to acquire synchronization in the presence of bit errors. Also note that other synchronization fields may be provided outside the IMBE speech coder which may eliminate the need to use $\hat{b}_{\hat{L}+2}$ for synchronization.

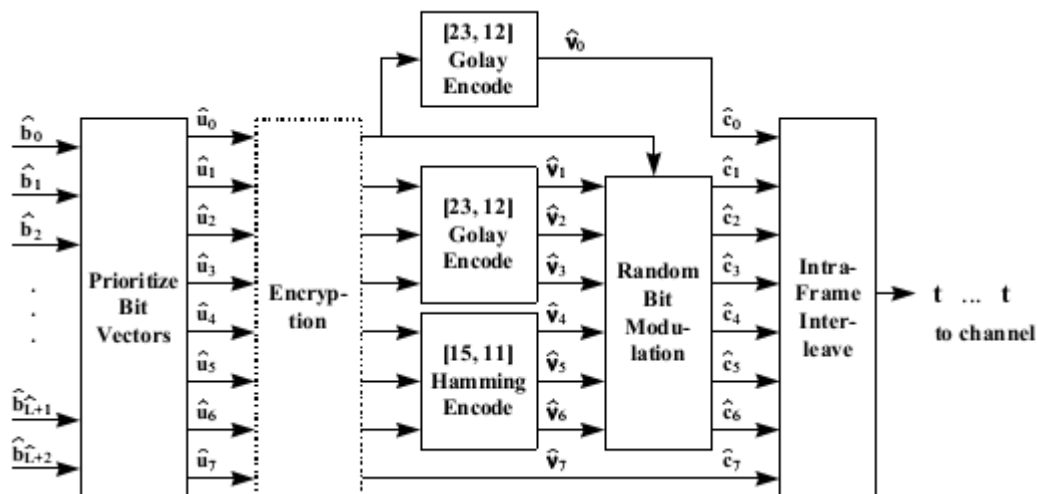


Figure 20 Encoder Bit Manipulations

7 Full-Rate Bit Manipulations

The IMBE speech coder uses a number of different bit manipulations in order to increase its robustness to channel degradations. The quantizer values, $\hat{b}_0, \dots, \hat{b}_{L+2}$, are first prioritized into a set of bit vectors, denoted by $\hat{u}_0, \dots, \hat{u}_7$. These vectors are optionally encrypted, and then they are protected with error control codes, including both [23, 12] Golay codes and [15, 11] Hamming codes, to produce a set of code vectors denoted by $\hat{v}_0, \dots, \hat{v}_7$. These code vectors are then modulated to produce a set of modulated code vectors denoted by $\hat{c}_0, \dots, \hat{c}_7$. Finally, intra-frame bit interleaving is used on the modulated code vectors in order to spread the effect of short burst errors. A block diagram of the bit manipulations performed by the encoder is shown in Figure 20.

The IMBE decoder reverses the bit manipulations performed by the encoder. First the decoder de-interleaves each frame of 144 bits to obtain the eight modulated code vectors $\tilde{c}_0, \dots, \tilde{c}_7$. The decoder then demodulates these vectors to produce the code vectors $\tilde{v}_0, \dots, \tilde{v}_7$ and then error control decodes these code vectors to produce the bit vectors $\tilde{u}_0, \dots, \tilde{u}_7$. In order to ensure sufficient performance it is necessary that the decoder decode all error control codes up to their maximum error correction capability (i.e. 3 errors for the Golay codes and 1 error for the Hamming codes). Optionally, soft-decision decoding can be used to further improve the robustness to bit errors. Note that at higher bit error rates (BER > 4%) soft-decision decoding can substantially improve performance. Next the decoder shall decrypt the bit vectors (if encryption is employed at the encoder), and then it shall rearrange the bit vectors to reconstruct the quantizer values, denoted by $\tilde{b}_0, \tilde{b}_1, \dots, \tilde{b}_{L+2}$. These values are further decoded using the techniques described in Section 6 and finally used to synthesize the current frame of speech. A block diagram of the bit manipulations performed by the decoder is shown in Figure 21.

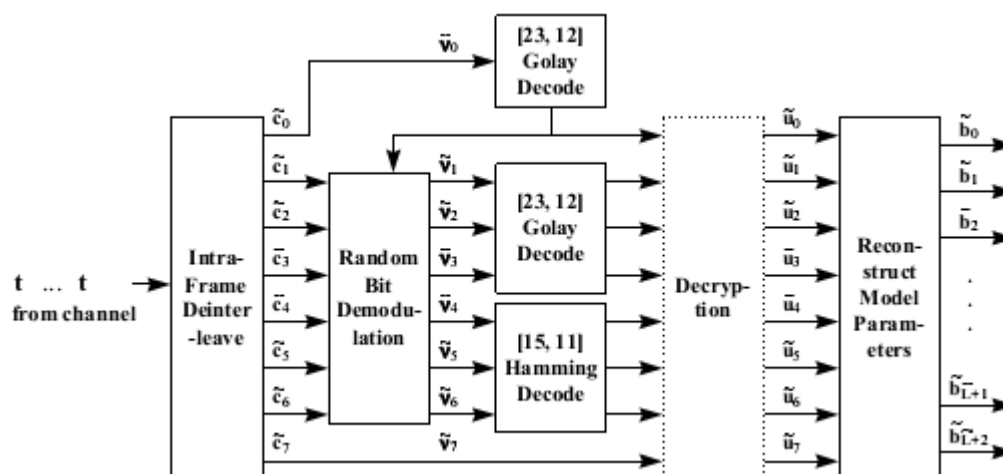


Figure 21 Decoder Bit Manipulations

One should note that the IMBE speech decoder employs a number of different mechanisms to improve performance in the presence of bit errors. These mechanisms consist first of error control codes, which are able to remove a significant number of errors. In addition, the IMBE speech coder uses bit modulation combined with frame repeats and frame mutes to detect and discard highly corrupted frames. Finally, the IMBE speech decoder uses adaptive smoothing to reduce the perceived effect of any remaining errors. These mechanisms are all discussed in the following sections of this description.

7.1 Bit Prioritization

The first bit manipulation performed by the IMBE encoder is a rearrangement of the quantizer values $\hat{b}_0, \hat{b}_1, \dots, \hat{b}_{\hat{L}+2}$ into a set of 8 prioritized bit vectors denoted by $\hat{u}_0, \hat{u}_1, \dots, \hat{u}_7$. The bit vectors $\hat{u}_0$, through $\hat{u}_3$ are 12 bits long, while the bit vectors $\hat{u}_4$ through $\hat{u}_6$ are 11 bits long, and the bit vector $\hat{u}_7$ is seven bits long. Throughout this section the convention has been adopted that bit $N-1$, where N is the vector length, is the MSB and bit 0 is the LSB.

The prioritization of the quantizer values into the set of bit vectors begins with $\hat{u}_0$. The six most significant bits of $\hat{u}_0$ (i.e. bits 11 through 6) are set equal to the six most significant bits of $\hat{b}_0$ (i.e. bits 7 through 2). The next three most significant bits of $\hat{u}_0$ (i.e. bits 5 through 3) are set equal to the three most significant bits of $\hat{b}_2$ (i.e. bits 5 through 3). The remaining three bits of $\hat{u}_0$ are generated from the spectral amplitude quantizer values $\hat{b}_0$ through $\hat{b}_{\hat{L}+1}$. Specifically, these quantizer values are arranged as shown in Figure 22. In this figure the shaded areas represent the number of bits which were allocated to each of these values assuming $\hat{L}=16$. Note that for other values of $\hat{L}$ this figure would change in accordance with the bit allocation information contained in Annex F and Annex G. The remaining three bits of $\hat{u}_0$ are then selected by beginning in the upper left hand corner of this figure (i.e. bit 5 of $\hat{b}_3$) and scanning left to right. When the end of any row is reached the scanning proceeds from left to right on the next lower row. Bit 2 of $\hat{u}_0$ is set equal to the bit corresponding to the first shaded block which is encountered using the prescribed scanning order. Similarly, bit 1 of $\hat{u}_0$ is set equal to the bit corresponding to the second shaded block which is encountered and bit 0 of $\hat{u}_0$ is set equal to the bit corresponding to the third shaded block which is encountered.

The scanning of the spectral amplitude quantizer values $\hat{b}_3$ through $\hat{b}_{\hat{L}+1}$ which is used to generate the last three bits of $\hat{u}_0$ is continued for the bit vectors $\hat{u}_1$ through $\hat{u}_3$. Each successive bit in these vectors is set equal to the bit corresponding to the next shaded block. This process begins with bit 11 of $\hat{u}_1$, proceeds through bit 0 of $\hat{u}_1$ followed by bit 11 of $\hat{u}_2$ and continues in this manner until finally reaching bit 0 of $\hat{u}_3$. At this point the 48 highest priority bits have been assigned to the bit vectors $\hat{u}_0$ through $\hat{u}_3$ as shown in Figure 23.

The next bits to be inserted into the bit vectors are all of the bits of $\hat{b}_1$ (starting with the MSB), followed by bit 2 and then bit 1 of $\hat{b}_2$, and then continuing with the scanning of $\hat{b}_3$ through $\hat{b}_{\hat{L}+1}$ as described above. These bits are inserted into the bit vectors beginning with bit 10 of $\hat{u}_4$, proceeding through bit 0 of $\hat{u}_4$ followed by bit 10 of $\hat{u}_5$, and continuing in this manner until finally reaching bit 4 of $\hat{u}_7$. The final four bits of $\hat{u}_7$, beginning with bit 3 and ending with bit 0, are set equal to bit 0 of $\hat{b}_2$, bit 1 of $\hat{b}_0$, bit 0 of $\hat{b}_0$, and then bit 0 of $\hat{b}_{\hat{L}+2}$, respectively. A block diagram of this procedure is shown in Figure 24 for $\hat{K} = 6$.

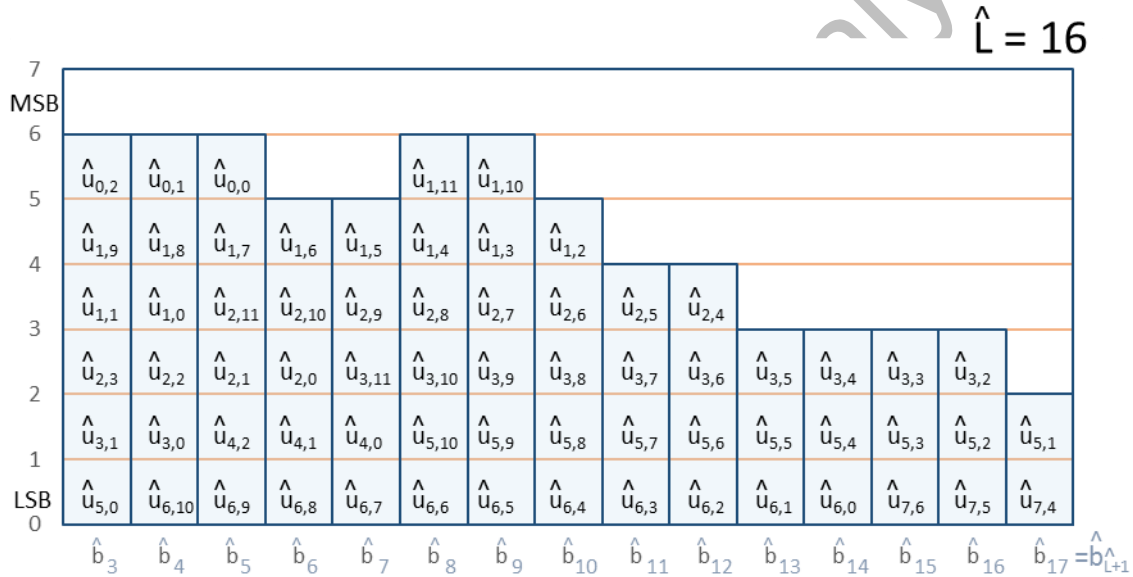
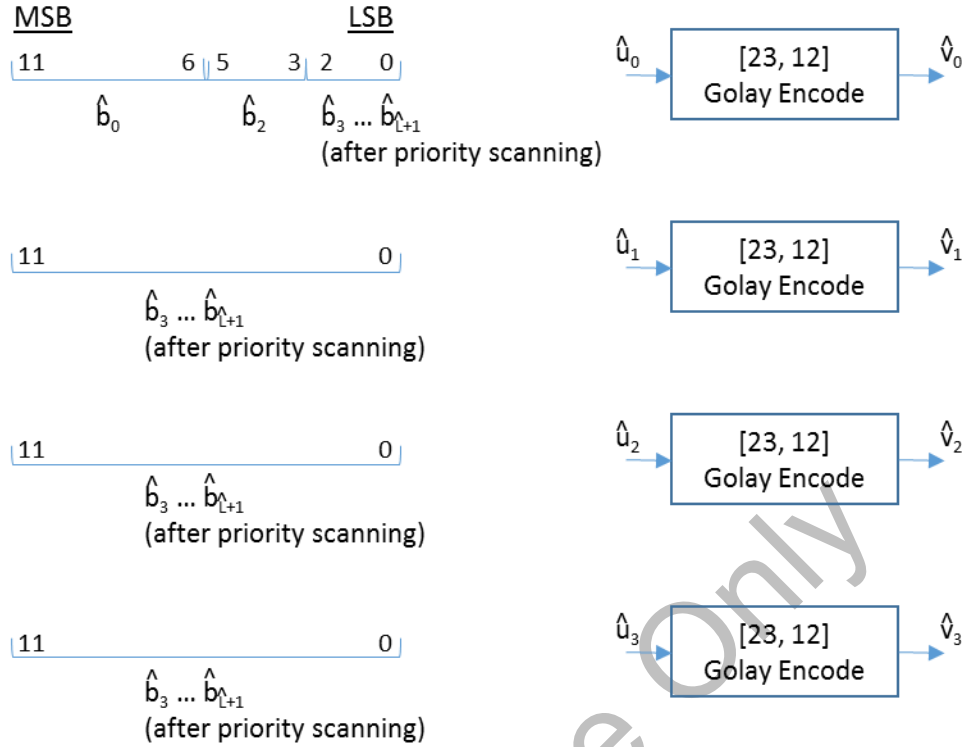


Figure 22 Priority Scanning of b_3 through b_{L+1}

7.2 Encryption

This document treats optional encryption and decryption as transparent elements and does not attempt to define the actual encryption process. However, Figures 20 and 21 depict an encryption and decryption element, respectively, in order to illustrate the proper placement of these elements in the IMBE vocoder. During encryption the bit vectors $\hat{u}_0, \dots, \hat{u}_7$ are combined bit-by-bit with an encryption sequence. This same sequence shall be used at the decoder to recover the bit vectors $\tilde{u}_0, \dots, \tilde{u}_7$. In order to be interoperable the encryption and decryption processes shall each use the same bit ordering. The standard ordering begins with the most significant bit (MSB) of u_0 and continues in order of significance until the least significant bit (LSB) of u_0 is reached. This is followed in order by the bit vectors $u_1, u_2, u_3, u_4, u_5, u_6$ and u_7 , respectively, where each bit vector proceeds from MSB to LSB. The reader is referred to the Common Air Interface and the Encryption Standard for more information on the encryption and decryption processes.

Figure 23 Formation of Code Vectors v_0 through v_3

7.3 Error Control Coding

At 7.2 kbps with a 20 ms frame size, 144 bits per frame are available for voice coding. The IMBE speech coder uses 88 of these bits to quantize the model parameters and provide synchronization, and the remaining 56 bits are used for forward error correction. The 56 error control bits are divided between four [23, 12] Golay codes and three [15, 11] Hamming codes. The reader is referred to references [8, 9] for more information on the encoding and decoding of Golay and Hamming codes.

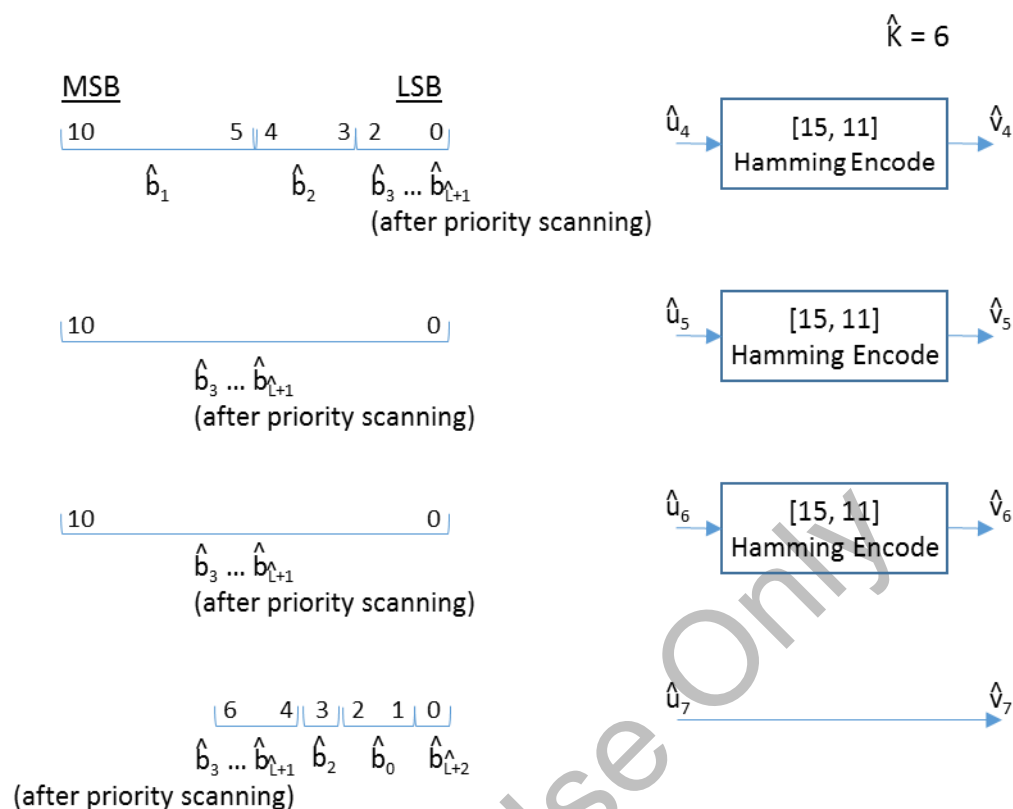
The generation of the eight code vectors $\hat{v}_i$ for $0 \leq i \leq 7$ is performed according to the following set of equations,

$$\hat{v}_i = \hat{u}_i \cdot g_G \quad \text{for } 0 \leq i \leq 3 \quad (81)$$

$$\hat{v}_i = \hat{u}_i \cdot g_H \quad \text{for } 4 \leq i \leq 6 \quad (82)$$

$$\hat{v}_7 = \hat{u}_7 \quad (83)$$

where the g_G and g_H are the generator matrices for the [23, 12] Golay code and the [15, 11] Hamming code, respectively. These are shown below where absent entries are assumed to equal zero. Note that all operations are modulo 2, and the vectors $\hat{v}_i$ and $\hat{u}_i$ are assumed to be row vectors, where the “left” most bit is the MSB. This convention is used throughout this section.

Figure 24 Formation of Code Vectors v_4 through v_7
$$g_G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$g_H = \begin{bmatrix} 1000000000001111 \\ 0100000000001110 \\ 0010000000001101 \\ 0001000000001100 \\ 0000100000001011 \\ 0000010000001010 \\ 0000001000001001 \\ 0000000100001001 \\ 0000000010000111 \\ 0000000010001110 \\ 0000000010010101 \\ 0000000001001111 \end{bmatrix}$$

7.4 Bit Modulation

The IMBE speech coder uses bit modulation keyed off the code vector $\hat{v}_0$ to provide a mechanism for detecting errors in $\hat{v}_0$ beyond the three errors that the [23, 12] Golay code can correct. Note that the term bit modulation in the context of this document refers to the presented method for multiplying (or modulating) each frame of code vectors by a data dependent pseudo-random sequence. The first step in this procedure is to generate a set of binary modulation vectors which are added (modulo 2) to the code vectors $\hat{v}_0$ through $\hat{v}_7$. The modulation vectors are generated from a pseudo-random sequence whose seed is derived from $\hat{u}_0$. Specifically, the sequence defined in the following equations is used,

$$p_r(0) = 16 \hat{u}_0 \quad (84)$$

$$p_r(n) = 173 p_r(n-1) + 13849 - 65536 \left\lfloor \frac{173 p_r(n-1) + 13849}{65536} \right\rfloor \quad (85)$$

where the bit vector $\hat{u}_0$ is interpreted as an unsigned 12 bit number in the range [0, 4095]. Equation (85) is used to recursively compute the pseudo-random sequence, $p_r(n)$, over the range $1 \leq n \leq 114$. Each element of this sequence can be interpreted as a 16 bit random number which is uniformly distributed over the interval [0, 65535]. Using this interpretation, a set of binary modulation vectors, denoted by $\hat{m}_0$ through $\hat{m}_7$, are generated from this sequence as shown below.

$$\hat{m}_0 = [0, 0, \dots, 0] \quad (86)$$

$$\hat{m}_1 = \left[\left\lfloor \frac{p_r(1)}{32768} \right\rfloor, \left\lfloor \frac{p_r(2)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(23)}{32768} \right\rfloor \right] \quad (87)$$

$$\hat{m}_2 = \left[\left\lfloor \frac{p_r(24)}{32768} \right\rfloor, \left\lfloor \frac{p_r(25)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(46)}{32768} \right\rfloor \right] \quad (88)$$

$$\hat{m}_3 = \left[\left\lfloor \frac{p_r(47)}{32768} \right\rfloor, \left\lfloor \frac{p_r(48)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(69)}{32768} \right\rfloor \right] \quad (89)$$

$$\hat{m}_4 = \left[\left\lfloor \frac{p_r(70)}{32768} \right\rfloor, \left\lfloor \frac{p_r(71)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(84)}{32768} \right\rfloor \right] \quad (90)$$

$$\hat{m}_5 = \left[\left\lfloor \frac{p_r(85)}{32768} \right\rfloor, \left\lfloor \frac{p_r(86)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(99)}{32768} \right\rfloor \right] \quad (91)$$

$$\hat{m}_6 = \left[\left\lfloor \frac{p_r(100)}{32768} \right\rfloor, \left\lfloor \frac{p_r(101)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(114)}{32768} \right\rfloor \right] \quad (92)$$

$$\hat{m}_7 = [0, 0, \dots, 0] \quad (93)$$

Once these modulation vectors have been computed in this manner, the modulated code vectors, $\hat{c}_i$ for $0 \leq i \leq 7$, are computed by adding (modulo 2) the code vectors to the modulation vectors.

$$\hat{c}_i = \hat{v}_i + \hat{m}_i \quad \text{for } 0 \leq i \leq 7 \quad (94)$$

One should note that the bit modulation performed by the IMBE encoder can be inverted by the decoder if $\tilde{c}_0$ does not contain any uncorrectable bit errors. In this case Golay decoding $\tilde{c}_0$, which always equals $\tilde{v}_0$ since $\tilde{m}_0 = 0$, will yield the correct value $\tilde{u}_0$. The decoder can then use $\tilde{u}_0$ to reconstruct the pseudo-random sequence and the modulation vectors $\tilde{m}_1$ through $\tilde{m}_7$. Subtracting these vectors from $\tilde{c}_1$ through $\tilde{c}_7$ will then yield the code vectors $\tilde{v}_1$ through $\tilde{v}_7$. At this point the remaining error control decoding can be performed. In the other case, where $\tilde{c}_0$ contains uncorrectable bit errors, the modulation cannot generally be inverted by the decoder. In this case the likely result of Golay decoding $\tilde{c}_0$ will be some $\tilde{u}_0$ which does not equal $\tilde{u}_0$. Consequently the decoder will initialize the pseudo-random sequence incorrectly, and the modulation vectors computed by the decoder will be uncorrelated with the modulation vectors used by the encoder. Using these incorrect modulation vectors to reconstruct the code vectors is essentially the same as passing $\tilde{v}_1, \dots, \tilde{v}_6$ through a 50 percent bit error rate (BER) channel. The IMBE decoder exploits the fact that, statistically, a 50 percent BER causes the Golay and Hamming codes employed on $\tilde{v}_1$ through $\tilde{v}_6$ to correct a number of errors which is near the maximum capability of the code. By counting the total number of errors which are corrected in all of these code vectors, the decoder is able to reliably detect frames in which $\tilde{c}_0$ is likely to contain uncorrectable bit errors. The decoder performs frame repeats during these frames in order to reduce the perceived degradation in the presence of bit errors. This is explained more fully in Sections 7.6 and 7.7.

7.5 Bit Interleaving

Intra-frame bit interleaving is used to spread short bursts of errors among several code words. The division of each frame of 144 bits into 72 dibit symbols is tabulated in Annex H. The minimum separation between any two bits of the same error correction code is 3 symbols. This document uses the notation $c_j(n)$ to designate the n -th bit of the modulated code vector $\hat{c}_j$ (or the demodulated code vector $\tilde{c}_j$). Note that bit $N-1$ (where N is the vector length) is the MSB of each vector and bit 0 is the LSB. The speech coder bits should be inserted into the frame format beginning with symbol 0 and ending with symbol 71. This is described more completely in the Common Air Interface specification.

7.6 Error Estimation

The IMBE speech decoder estimates the number of errors in each received data frame by computing the number of errors corrected by each of the [23, 12] Golay codes and [15, 11] Hamming codes. The number of errors for each code vector is denoted ε_i for $0 \leq i \leq 6$, where ε_i

refers to the number of bit errors which were corrected during the error decoding of $\tilde{u}_i$. From these error values two other error parameters are computed as shown below.

$$\varepsilon_T = \sum_{i=0}^6 \varepsilon_i \quad (95)$$

$$\varepsilon_R(0) = 0.95 * \varepsilon_R(-1) + 0.000365 \varepsilon_T \quad (96)$$

The parameter $\varepsilon_R(0)$ is the estimate of the error rate for the current frame, while $\varepsilon_R(-1)$ is the estimate of the error rate for the previous frame. These error parameters are used to control the frame repeat process described below, and to control the parametric smoothing functions described in Section 9.

7.7 Frame Repeats

The IMBE decoder examines each received data frame in order to detect and discard frames which are highly corrupted. A number of different fault conditions are checked and if any of these conditions indicate the current frame is invalid, then a frame repeat is performed. The IMBE speech encoder uses values of $\hat{b}_0$ in the range $0 \leq \hat{b}_0 \leq 207$ to represent valid pitch estimates. The remaining values of $\hat{b}_0$ are reserved for future expansion and are currently considered invalid. A frame repeat is performed by the decoder if it receives an invalid value of $\hat{b}_0$, or if both of the following two equations are true.

$$\varepsilon_0 \geq 2 \quad (97)$$

$$\varepsilon_T \geq 10 + 40 \varepsilon_R \quad (98)$$

These two equations are used to detect the incorrect bit demodulation which results if there are uncorrectable bit errors in $\tilde{c}_0$. The decoder performs a frame repeat by taking the following steps:

- 1) The current 144 bit received data frame is marked as invalid and subsequently ignored during future processing steps.
- 2) The IMBE model parameters for the current frame are set equal to the IMBE model parameters for the previous frame. Specifically, the following update expressions are computed.

$$\tilde{\omega}_0(0) = \tilde{\omega}_0(-1) \quad (99)$$

$$\tilde{L}(0) = \tilde{L}(-1) \quad (100)$$

$$\tilde{K}(0) = \tilde{K}(-1) \quad (101)$$

$$\tilde{v}_k(0) = \tilde{v}_k(-1) \quad \text{for } 1 \leq k \leq \tilde{K} \quad (102)$$

$$\tilde{M}_l(0) = \tilde{M}_l(-1) \quad \text{for } 1 \leq l \leq \tilde{L} \quad (103)$$

$$\bar{M}_l(0) = \bar{M}_l(-1) \quad \text{for } 1 \leq l \leq \tilde{L} \quad (104)$$

3) The repeated model parameters are used in all future processing wherever the current model parameters are required. This includes the synthesis of the current segment of speech as is described in Section 11.

7.8 Frame Muting

The IMBE decoder is required to mute in severe bit error environments for which $\varepsilon_R > 0.0875$. This capability causes the decoder to squelch its output if reliable communication cannot be supported.

The recommended muting method is to first compute the update equations as listed in step (2) of the frame repeat process (see Section 7.7). The decoder should then bypass the speech synthesis algorithm described in Section 11 and, instead, set the synthetic speech signal, $\tilde{s}(n)$ to random noise which is uniformly distributed over the interval $[-5, 5]$. This technique provides for a small amount of “comfort noise” as is typically done in telecommunication systems.

Limited Use Only

8 Full-Rate Spectral Amplitude Enhancement

The IMBE speech decoder attempts to improve the perceived quality of the synthesized speech by enhancing the spectral amplitudes. The unenhanced spectral amplitudes are required by future frames in the computation of Equation (77). However, the enhanced spectral amplitudes are used in speech synthesis. The spectral amplitude enhancement is accomplished by generating a set of spectral weights from the model parameters of the current frame. First R_{M0} and R_{M1} are calculated as shown below

$$R_{M0} = \sum_{l=1}^{\tilde{L}} \tilde{M}_l^2 \quad (105)$$

$$R_{M1} = \sum_{l=1}^{\tilde{L}} \tilde{M}_l^2 \cos(\tilde{\omega}_0 l) \quad (106)$$

Next, the parameters R_{M0} , and R_{M1} are used to calculate a set of weights, W_l , given by

$$W_l = \sqrt{\tilde{M}_l} \cdot \left[\frac{0.96 (R_{M0}^2 + R_{M1}^2 - 2 R_{M0} R_{M1} \cos(\tilde{\omega}_0 l))}{\tilde{\omega}_0 R_{M0} (R_{M0}^2 - R_{M1}^2)} \right]^{1/4} \quad \text{for } 1 \leq l \leq \tilde{L} \quad (107)$$

These weights are then used to enhance the spectral amplitudes for the current frame according to the relationship:

$$\bar{M}_l = \begin{cases} \tilde{M}_l & \text{if } 8l \leq \tilde{L} \\ 1.2 \cdot \tilde{M}_l & \text{else if } W_l > 1.2 \\ 0.5 \cdot \tilde{M}_l & \text{else if } W_l < 0.5 \\ W_l \tilde{M}_l & \text{otherwise} \end{cases} \quad \text{for } 1 \leq l \leq \tilde{L} \quad (108)$$

A final step is to scale the enhanced spectral amplitudes in order to remove any energy difference between the enhanced and unenhanced amplitudes. The correct scale factor, denoted by γ , is given below.

$$\gamma = \left[\frac{R_{M0}}{\sum_{l=1}^{\tilde{L}} |\bar{M}_l|^2} \right]^{1/2} \quad (109)$$

This scale factor is applied to each of the enhanced spectral amplitudes as shown in Equation (110).

$$\bar{M}_l = \gamma \cdot \bar{M}_l \quad \text{for } 1 \leq l \leq \tilde{L} \quad (110)$$

For notational simplicity this equation refers to both the scaled and unscaled spectral amplitudes as $\bar{M}_l$. This convention has been adopted since the unscaled amplitudes are discarded and only the scaled amplitudes are subsequently used by the decoder during parameter smoothing and speech synthesis.

The value of R_{M0} expressed in Equation (105) is a measure of the energy in the current frame. This value is used to update a local energy parameter in accordance with the following rule.

$$S_E(0) = \begin{cases} 0.95 S_E(-1) + 0.05 R_{M0} & \text{if } 0.95 S_E(-1) + 0.05 R_{M0} \geq 10000.0 \\ 10000.0 & \text{otherwise} \end{cases} \quad (111)$$

This equation generates the local energy parameter for the current frame, $S_E(0)$, from R_{M0} and the value of the local energy parameter from the previous frame $S_E(-1)$. The parameter $S_E(0)$ is used in the following section.

Limited Use Only

9 Adaptive Smoothing

As part of the error control process described in Section 7.6, the decoder estimates two error rate parameters, ε_T and ε_R , which measure the total number of errors and the local error rate for the current frame, respectively. These parameters are used by the decoder to adaptively smooth the decoded model parameters. The result is improved performance in high bit error environments.

The first parameters to be smoothed by the decoder are the V/UV decisions. First an adaptive threshold V_M is calculated using equation (112),

$$V_M = \begin{cases} \infty & \text{if } \varepsilon_R(0) \leq 0.005 \text{ and } \varepsilon_T \leq 4 \\ \frac{45.255 (S_E(0))^{0.375}}{\exp(277.26 \varepsilon_R(0))} & \text{else if } \varepsilon_R(0) \leq 0.0125 \text{ and } \varepsilon_4 = 0 \\ 1.414 (S_E(0))^{0.375} & \text{otherwise} \end{cases} \quad (112)$$

where the energy parameter $S_E(0)$ is defined in Equation (111) in Section 8. After the adaptive threshold is computed each enhanced spectral amplitude $\bar{M}_l$ for $1 \leq l \leq \tilde{L}$ is compared against V_M , and if $\bar{M}_l > V_M$ then the V/UV decision for that spectral amplitude is declared voiced, regardless of the decoded V/UV decision. Otherwise the decoded V/UV decision for that spectral amplitude is left unchanged. This process can be expressed mathematically as shown below.

$$\bar{v}_l = \begin{cases} 1 & \text{if } \bar{M}_l > V_M \\ \bar{v}_l & \text{otherwise} \end{cases} \quad \text{for } 1 \leq l \leq \tilde{L} \quad (113)$$

Once the V/UV decisions have been smoothed, the decoder adaptively smooths the spectral amplitudes $\bar{M}_l$ for $1 \leq l \leq \tilde{L}$. The spectral amplitude smoothing algorithm computes the following amplitude measure for the current segment.

$$A_M = \sum_{l=1}^{\tilde{L}} \bar{M}_l \quad (114)$$

Next an amplitude threshold is updated according to the following equation,

$$\tau_M(0) = \begin{cases} 20480 & \text{if } \varepsilon_R(0) \leq 0.005 \text{ and } \varepsilon_T(0) \leq 6 \\ 6000 - 300 \varepsilon_T + \tau_M(-1) & \text{otherwise} \end{cases} \quad (115)$$

where $\tau_M(0)$ and $\tau_M(-1)$ represent the value of the amplitude threshold for the current and previous frames respectively. The two parameters A_M and $\tau_M(0)$ are then used to compute a scale factor γ_M given below.

$$\gamma_M = \begin{cases} 1.0 & \text{if } \tau_M(0) > A_M \\ \tau_M(0)/A_M & \text{otherwise} \end{cases} \quad (116)$$

This scale factor is multiplied by each of the spectral amplitudes $\bar{M}_l$ for $1 \leq l \leq \tilde{L}$. Note that this step shall be completed after spectral amplitude enhancement has been performed using the methods of Section 8 and after V_M has been computed according to Equation (112). The correct sequence is shown in Figure 25.

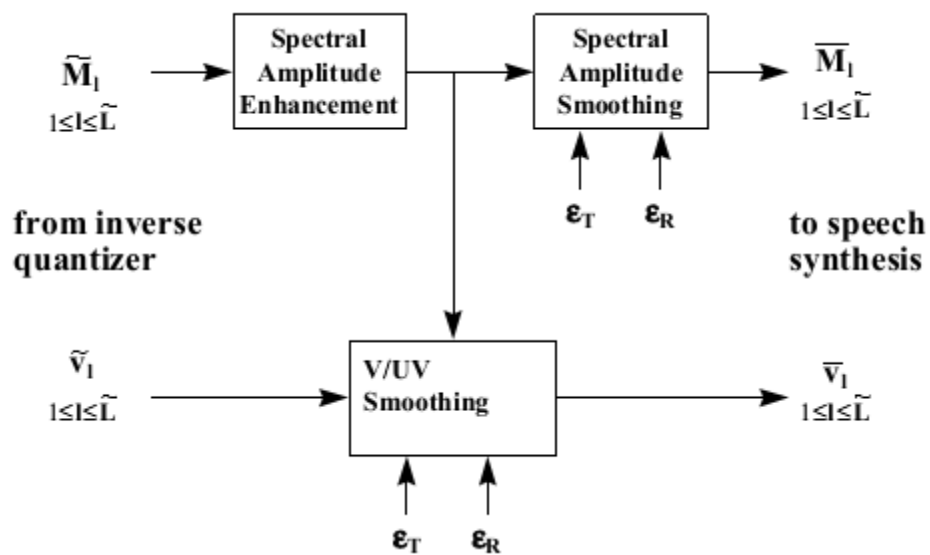


Figure 25 Parameter Enhancement and Smoothing

i	$\hat{f}_i$	$\hat{c}_{i,1} \dots \hat{c}_{i,\hat{f}_i}$
1	2	$\hat{T}_1, \hat{T}_2$
2	2	$\hat{T}_3, \hat{T}_4$
3	3	$\hat{T}_5, \hat{T}_6, \hat{T}_7$
4	3	$\hat{T}_8, \hat{T}_9, \hat{T}_{10}$
5	3	$\hat{T}_{11}, \hat{T}_{12}, \hat{T}_{13}$
6	3	$\hat{T}_{14}, \hat{T}_{15}, \hat{T}_{16}$

Table 5 Division of Prediction Residuals into Blocks in Encoding Example

10 Full-Rate Parameter Encoding Example

This section provides an example of the quantization and bit manipulation for a typical parameter frame. In this example the fundamental frequency is assumed to be equal to $\hat{\omega}_0 = 2\pi/35.125$. Since the values of $\hat{L}$ and $\hat{K}$ are related to $\hat{\omega}_0$ through equations (37) and (38), they are equal to $\hat{L}=16$ and $\hat{K}=6$. The remaining model parameters are left unspecified since they do not affect the numbers presented in this example.

The encoding of this example parameter frame proceeds as follows. First the fundamental frequency is encoded into the 8 bit value $\hat{b}_0$ using equation (45), and the 6 voiced / unvoiced decisions are encoded into the 6 bit value $\hat{b}_1$ using equation (49). The 16 spectral amplitude prediction residuals, $\hat{T}_l$ for $1 \leq l \leq 16$, are then formed using equations (52) through (55). Next, these prediction residuals are divided into six blocks where the lengths of each block, $\hat{f}_i$ for $1 \leq i \leq 6$, are shown in Table 5. The spectral amplitude prediction residuals are then divided into the six vectors $\hat{c}_{i,j}$ for $1 \leq i \leq 6$ and $1 \leq j \leq \hat{f}_i$. The first $\hat{f}_1$ elements of $\hat{T}_l$ form $\hat{c}_{1,j}$. The next $\hat{f}_2$ elements of $\hat{T}_l$ form $\hat{c}_{2,j}$, and so on. This is shown in Table 5. Each block $\hat{c}_{i,j}$ for $1 \leq i \leq 6$, is transformed with a $\hat{f}_i$ point DCT using equation (60) to produce the set of DCT coefficients $\hat{C}_{i,k}$ for $1 \leq k \leq \hat{f}_i$. The first DCT coefficient from each of the six blocks is used to form the gain vector $\hat{R}_l$. The gain vector is then transformed into the vector $\hat{G}_m$ using the six point DCT shown in Equation (61). The first element of the transformed gain vector, denoted by $\hat{G}_1$, is then quantized using the non-uniform quantizer tabulated in Annex E. The 6 bit value $\hat{b}_2$ is set equal to the index of the quantizer element which is closest to $\hat{G}_1$. The remaining five elements of the transformed gain vector are quantized using the uniform quantizers generated from Annex F with $\hat{L}=16$. The correct step sizes and bit allocation for $\hat{G}_2$ through $\hat{G}_6$ is shown in Table 6. Once the bit allocation and step sizes have been computed, the bit encodings $\hat{b}_3$ through $\hat{b}_7$ are generated using Equation (62).

m	$\hat{G}_m$	$\hat{B}_m$	$\hat{\Delta}_m$
2	$\hat{G}_2$	6	0.04650
3	$\hat{G}_3$	6	0.03015
4	$\hat{G}_4$	6	0.02520
5	$\hat{G}_5$	5	0.04060
6	$\hat{G}_6$	5	0.03696

Table 6 Example Bit Allocation and Step Size for the Transformed Gain Vector

After the gain vector has been quantized and encoded, the remaining bits are distributed among the ten higher order DCT coefficients, $\hat{C}_{i,k}$ for $1 \leq i \leq 6$ and $2 \leq k \leq \hat{f}_i$. This is done using Annex G and the resulting bit allocation is shown in Table 7. Each DCT coefficient is then quantized using equation

(63). The step sizes for these quantizers are computed using Tables 3 and 4, and the results are shown in Table 7.

Finally, the one bit synchronization value, denoted by $\hat{b}_{18}$, is generated as a sequence which alternates between 0 and 1 on each successive 20 ms voice frame. The 19 bit encodings, $\hat{b}_0$ through $\hat{b}_{18}$, are then rearranged into the eight bit vectors $\hat{u}_0$ through $\hat{u}_7$. This is accomplished using the procedure described in Section 7, and the result is shown in Tables 8 through 10. The convention in these tables is that the appropriate bit from the vector listed in the first two columns is set equal to the appropriate bit from the bit encoding listed in the last two columns, where the least significant bit corresponds to bit 0.

The four bit vectors $\hat{u}_0$ through $\hat{u}_3$ are each encoded with a [23, 12] Golay code into the code vectors $\hat{v}_0$ through $\hat{v}_3$, respectively. Similarly, the three bit vectors $\hat{u}_4$ through $\hat{u}_6$ are each encoded with a [15, 11] Hamming code into the code vectors $\hat{v}_4$ through $\hat{v}_6$, respectively. The vector $\hat{v}_7$ is set equal to $\hat{u}_7$. These code vectors are then modulated using Equations (84) through (94) to produce the modulated code vectors $\hat{c}_0$ through $\hat{c}_7$. The eight modulated code vectors are then interleaved as specified in Annex H, and finally the frame bits are embedded into the frame format in ascending order (i.e. bit t_1 first and t_{144} last).

m	$\hat{C}_{i,k}$	$\hat{B}_m$	$\hat{\Delta}_m$
8	$\hat{C}_{1,2}$	6	0.04605
9	$\hat{C}_{2,2}$	6	0.04605
10	$\hat{C}_{3,2}$	5	0.08596
11	$\hat{C}_{3,3}$	4	0.09640
12	$\hat{C}_{4,2}$	4	0.12280
13	$\hat{C}_{4,3}$	3	0.15665
14	$\hat{C}_{5,2}$	3	0.19955
15	$\hat{C}_{5,3}$	3	0.15665
16	$\hat{C}_{6,2}$	3	0.19955
17	$\hat{C}_{6,3}$	2	0.20485

Table 7 Example Bit Allocation and Step Size for Higher Order DCT Coefficients

Vector	Bit Number	Vector	Bit Number
$\hat{u}_0$	11	$\hat{b}_0$	7
$\hat{u}_0$	10	$\hat{b}_0$	6
$\hat{u}_0$	9	$\hat{b}_0$	5
$\hat{u}_0$	8	$\hat{b}_0$	4
$\hat{u}_0$	7	$\hat{b}_0$	3
$\hat{u}_0$	6	$\hat{b}_0$	2
$\hat{u}_0$	5	$\hat{b}_2$	5
$\hat{u}_0$	4	$\hat{b}_2$	4
$\hat{u}_0$	3	$\hat{b}_2$	3
$\hat{u}_0$	2	$\hat{b}_3$	5
$\hat{u}_0$	1	$\hat{b}_4$	5
$\hat{u}_0$	0	$\hat{b}_5$	5
$\hat{u}_1$	11	$\hat{b}_8$	5
$\hat{u}_1$	10	$\hat{b}_9$	5
$\hat{u}_1$	9	$\hat{b}_3$	4
$\hat{u}_1$	8	$\hat{b}_4$	4
$\hat{u}_1$	7	$\hat{b}_5$	4
$\hat{u}_1$	6	$\hat{b}_6$	4
$\hat{u}_1$	5	$\hat{b}_7$	4
$\hat{u}_1$	4	$\hat{b}_8$	4
$\hat{u}_1$	3	$\hat{b}_9$	4
$\hat{u}_1$	2	$\hat{b}_{10}$	4
$\hat{u}_1$	1	$\hat{b}_3$	3
$\hat{u}_1$	0	$\hat{b}_4$	3
$\hat{u}_2$	11	$\hat{b}_5$	3
$\hat{u}_2$	10	$\hat{b}_6$	3
$\hat{u}_2$	9	$\hat{b}_7$	3
$\hat{u}_2$	8	$\hat{b}_8$	3
$\hat{u}_2$	7	$\hat{b}_9$	3
$\hat{u}_2$	6	$\hat{b}_{10}$	3
$\hat{u}_2$	5	$\hat{b}_{11}$	3
$\hat{u}_2$	4	$\hat{b}_{12}$	3
$\hat{u}_2$	3	$\hat{b}_3$	2
$\hat{u}_2$	2	$\hat{b}_4$	2
$\hat{u}_2$	1	$\hat{b}_5$	2
$\hat{u}_2$	0	$\hat{b}_6$	2

Table 8 Construction of u_i in Encoding Example (1 of 3)

Vector	Bit Number	Vector	Bit Number
$\hat{u}_3$	11	$\hat{b}_7$	2
$\hat{u}_3$	10	$\hat{b}_8$	2
$\hat{u}_3$	9	$\hat{b}_9$	2
$\hat{u}_3$	8	$\hat{b}_{10}$	2
$\hat{u}_3$	7	$\hat{b}_{11}$	2
$\hat{u}_3$	6	$\hat{b}_{12}$	2
$\hat{u}_3$	5	$\hat{b}_{13}$	2
$\hat{u}_3$	4	$\hat{b}_{14}$	2
$\hat{u}_3$	3	$\hat{b}_{15}$	2
$\hat{u}_3$	2	$\hat{b}_{16}$	2
$\hat{u}_3$	1	$\hat{b}_3$	1
$\hat{u}_3$	0	$\hat{b}_4$	1
$\hat{u}_4$	10	$\hat{b}_1$	5
$\hat{u}_4$	9	$\hat{b}_1$	4
$\hat{u}_4$	8	$\hat{b}_1$	3
$\hat{u}_4$	7	$\hat{b}_1$	2
$\hat{u}_4$	6	$\hat{b}_1$	1
$\hat{u}_4$	5	$\hat{b}_1$	0
$\hat{u}_4$	4	$\hat{b}_2$	2
$\hat{u}_4$	3	$\hat{b}_2$	1
$\hat{u}_4$	2	$\hat{b}_5$	1
$\hat{u}_4$	1	$\hat{b}_6$	1
$\hat{u}_4$	0	$\hat{b}_7$	1
$\hat{u}_5$	10	$\hat{b}_8$	1
$\hat{u}_5$	9	$\hat{b}_9$	1
$\hat{u}_5$	8	$\hat{b}_{10}$	1
$\hat{u}_5$	7	$\hat{b}_{11}$	1
$\hat{u}_5$	6	$\hat{b}_{12}$	1
$\hat{u}_5$	5	$\hat{b}_{13}$	1
$\hat{u}_5$	4	$\hat{b}_{14}$	1
$\hat{u}_5$	3	$\hat{b}_{15}$	1
$\hat{u}_5$	2	$\hat{b}_{16}$	1
$\hat{u}_5$	1	$\hat{b}_{17}$	1
$\hat{u}_5$	0	$\hat{b}_3$	0

Table 9 Construction of u_i in Encoding Example (2 of 3)

Vector	Bit Number	Vector	Bit Number
$\hat{u}_6$	10	$\hat{b}_4$	0
$\hat{u}_6$	9	$\hat{b}_5$	0
$\hat{u}_6$	8	$\hat{b}_6$	0
$\hat{u}_6$	7	$\hat{b}_7$	0
$\hat{u}_6$	6	$\hat{b}_8$	0
$\hat{u}_6$	5	$\hat{b}_9$	0
$\hat{u}_6$	4	$\hat{b}_{10}$	0
$\hat{u}_6$	3	$\hat{b}_{11}$	0
$\hat{u}_6$	2	$\hat{b}_{12}$	0
$\hat{u}_6$	1	$\hat{b}_{13}$	0
$\hat{u}_6$	0	$\hat{b}_{14}$	0
$\hat{u}_7$	6	$\hat{b}_{15}$	0
$\hat{u}_7$	5	$\hat{b}_{16}$	1
$\hat{u}_7$	4	$\hat{b}_{17}$	0
$\hat{u}_7$	3	$\hat{b}_2$	0
$\hat{u}_7$	2	$\hat{b}_0$	1
$\hat{u}_7$	1	$\hat{b}_0$	0
$\hat{u}_7$	0	$\hat{b}_{18}$	0

Table 10 Construction of u_i in Encoding Example (3 of 3)

11 Speech Synthesis

As was discussed in Section 5, the IMBE speech coder estimates a set of model parameters for each speech frame. These parameters consist of the fundamental frequency $\hat{\omega}_0$, the V/UV decisions for each frequency band $\hat{v}_k$, and the spectral amplitudes $\hat{M}_l$. After the transmitted bits are received and decoded, a reconstructed set of model parameters is available for synthesizing speech. These reconstructed model parameters (after parameter enhancement and smoothing) are denoted $\tilde{\omega}_0$, $\tilde{v}_l$ and $\tilde{M}_l$, and they correspond to the reconstructed fundamental frequency, V/UV decisions and spectral amplitudes, respectively. In addition the parameter $\tilde{L}$, defined as the number of spectral amplitudes in the current frame, is generated from $\tilde{\omega}_0$ according to Equation (47). Because of a number of factors (such as quantization and channel errors) the reconstructed model parameters are not the same as the estimated model parameters $\hat{\omega}_0$, $\hat{v}_k$ and $\hat{M}_l$.

11.1 Speech Synthesis Notation

The IMBE speech synthesis algorithm uses the reconstructed model parameters to generate a speech signal which is perceptually similar to the original speech signal. For each new set of model parameters, the synthesis algorithm generates a 20 ms frame of speech, $\tilde{s}(n)$, which is interpolated between the previous set of model parameters and the newest or current set of model parameters. The notation $\tilde{L}(0)$, $\tilde{\omega}_0(0)$, $\tilde{v}_l(0)$, and $\tilde{M}_l(0)$ is used to denote the current set of reconstructed model parameters, while the notation $\tilde{L}(-1)$, $\tilde{\omega}_0(-1)$, $\tilde{v}_l(-1)$, and $\tilde{M}_l(-1)$ is used to denote the previous set of reconstructed model parameters. For each new set of model parameters, $\tilde{s}(n)$ is generated in the range $0 \leq n < N$, where N equals 160 samples (20ms.). This synthetic speech signal is the output of the IMBE voice coder and is suitable for digital to analog conversion with a sixteen bit converter.

The synthetic speech signal is divided into a voiced component $\tilde{s}_v(n)$ and an unvoiced component $\tilde{s}_{uv}(n)$. These two components are synthesized separately, as shown in Figure 26, and then summed to form $\tilde{s}(n)$. The unvoiced speech synthesis algorithm is discussed in Section 11.2 and the voiced speech synthesis algorithm is discussed in Section 11.3.

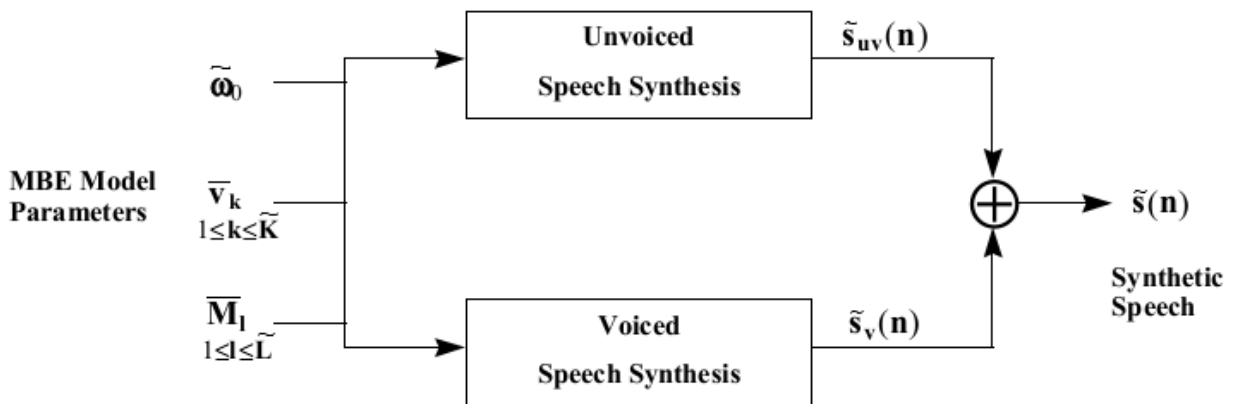


Figure 26 IMBE Speech Synthesis

11.2 Unvoiced Speech Synthesis

The energy from unvoiced spectral amplitudes is synthesized with an unvoiced speech synthesis algorithm. First a white noise sequence, $u(n)$, is generated. This noise sequence can have an arbitrary mean. A recommended noise sequence [10] can be generated as shown below.

$$u(n+1) = 171 u(n) + 11213 - 53125 \lfloor (171 u(n) + 11213)/53125 \rfloor \quad (117)$$

The noise sequence is initialized to $u(-105) = 3147$.

For each successive synthesis frame, $u(n)$ is shifted by 20 ms (160 samples) and windowed with $w_s(n)$, which is given in Annex I. Since $w_s(n)$ has a non-zero length of 209 samples, there is a 49 sample overlap between the noise signal used in successive synthesis frames. Once the noise sequence has been shifted and windowed, the 256 point Discrete Fourier Transform $U_w(m)$ is computed according to:

$$U_w(m) = \sum_{n=-104}^{104} u(n) w_s(n) e^{-j2\pi m n/256} \quad \text{for } -128 \leq m \leq 127 \quad (118)$$

The function $U_w(m)$ is generated in a manner which is analogous to $S_w(m)$ defined in Equation (29) except that $u(n)$ and $w_s(n)$ are used in place of $s(n)$ and $w_R(n)$.

The function $U_w(m)$ is then modified to create $\tilde{U}_w(m)$. For each l in the range $1 \leq l \leq \tilde{L}(0)$, $\tilde{U}_w(m)$ is computed according to Equation (119) if the l -th spectral amplitude is voiced, and according to Equation (120) if the l -th spectral amplitude is unvoiced.

$$\tilde{U}_w(m) = 0 \quad \text{for } [\tilde{a}_l] \leq |m| \leq [\tilde{b}_l] \quad (119)$$

$$\tilde{U}_w(m) = \frac{\gamma_w \bar{M}_l(0) U_w(m)}{\left[\frac{\sum_{\eta=[\tilde{a}_l]}^{[\tilde{b}_l]-1} |U_w(\eta)|^2}{[\tilde{b}_l] - [\tilde{a}_l]} \right]^{1/2}} \quad \text{for } [\tilde{a}_l] \leq |m| \leq [\tilde{b}_l] \quad (120)$$

The unvoiced scaling coefficient γ_w is a function of the synthesis window $w_s(n)$ and the pitch refinement window $w_R(n)$. It is computed according to the formula:

$$\gamma_w = \left[\sum_{n=-110}^{110} w_R(n) \right] \cdot \left[\frac{\sum_{n=-104}^{104} w_s^2(n)}{\sum_{n=-110}^{110} w_R^2(n)} \right]^{1/2} \quad (121)$$

The frequency bands edges $\tilde{a}_l$ and $\tilde{b}_l$ are computed from $\tilde{\omega}_0$ according to equations (122) and (123), respectively.

$$\tilde{a}_l = \frac{256}{2\pi} (l - 0.5) \cdot \tilde{\omega}_0 \quad (122)$$

$$\tilde{b}_l = \frac{256}{2\pi} (l + 0.5) \cdot \tilde{\omega}_0 \quad (123)$$

Finally, the very low frequency and very high frequency components of $\tilde{U}_w(m)$ are set equal to zero as shown in the following equation.

$$\tilde{U}_w(m) = \begin{cases} 0 & \text{for } |m| < [\tilde{a}_1] \\ 0 & \text{for } [\tilde{b}_L] \leq |m| \leq 128 \end{cases} \quad (124)$$

The sequence $\tilde{u}_w(n)$, defined as the 256 point Inverse Discrete Fourier Transform of $\tilde{U}_w(m)$, is the unvoiced speech for the current frame. The sequence $\tilde{u}_w(n)$ is computed as shown in the following equation.

$$\tilde{u}_w(n) = \frac{1}{256} \sum_{m=-128}^{127} \tilde{U}_w(m) e^{-j2\pi m n/256} \quad \text{for } -128 \leq n \leq 127 \quad (125)$$

In order to generate $\tilde{s}_{uv}(n)$, $\tilde{u}_w(n)$ should be combined with the unvoiced speech from the previous frame. This is accomplished using the Weighted Overlap Add algorithm described in [4]. If $\tilde{u}_w(n, 0)$ is used to denote the unvoiced speech for the current frame, and $\tilde{u}_w(n, -1)$ is used to denote the unvoiced speech for the previous frame, then $\tilde{s}_{uv}(n)$ is given by

$$\tilde{s}_{uv}(n) = \frac{w_s(n) \tilde{u}_w(n, -1) + w_s(n-N) \tilde{u}_w(n-N, 0)}{w_s^2(n) + w_s^2(n-N)} \quad \text{for } 0 \leq n < N \quad (126)$$

In this equation $w_s(n)$ is assumed to be zero outside the range $-105 \leq n \leq 105$, and $\tilde{u}_w(n, 0)$ and $\tilde{u}_w(n, -1)$ are assumed to be zero outside the range $-128 \leq n \leq 127$.

11.3 Voiced Speech Synthesis

The voiced speech is synthesized by summing a voiced signal for each spectral amplitude according to the following equation.

$$\tilde{s}_v(n) = \sum_{l=1}^{\max[\tilde{L}(-1), \tilde{L}(0)]} 2 \cdot \tilde{s}_{v,l}(n) \quad \text{for } 0 \leq n < N \quad (127)$$

The reader is referred to references [1, 3] for background information on the algorithm described in this section. The voiced synthesis algorithm attempts to match the l -th spectral amplitude of the current frame with the l -th spectral amplitude of the previous frame. The algorithm assumes that all spectral amplitudes outside the allowed range are equal to zero as shown in Equations (128) and (129).

$$\bar{M}_l(0) = 0 \quad \text{for } l > \tilde{L}(0) \quad (128)$$

$$\bar{M}_l(-1) = 0 \quad \text{for } l > \tilde{L}(-1) \quad (129)$$

In addition it assumes that these spectral amplitudes are unvoiced. These assumptions are needed for the case where the number of spectral amplitudes in the current frame is not equal to the number of spectral amplitudes in the previous frame (i.e. $\tilde{L}(0) \neq \tilde{L}(-1)$).

The signal $\tilde{s}_{v,l}(n)$ is computed differently for each spectral amplitude. If the l -th spectral amplitude is unvoiced for both the previous and current speech frame then $\tilde{s}_{v,l}(n)$ is set equal to zero as shown in the following equation. In this case the energy in this region of the spectrum is completely synthesized by the unvoiced synthesis algorithm described in the previous section.

$$\tilde{s}_{v,l}(n) = 0 \quad \text{for } 0 \leq n < N \quad (130)$$

Alternatively, if the l -th spectral amplitude is unvoiced for the current frame and voiced for the previous frame, then $\tilde{s}_{v,l}(n)$ is given by the following equation. In this case the energy in this region of the spectrum transitions from the voiced synthesis algorithm to the unvoiced synthesis algorithm.

$$\tilde{s}_{v,l}(n) = w_s(n) \bar{M}_l(-1) \cos[\tilde{\omega}_0(-1) n l + \varphi_l(-1)] \quad \text{for } 0 \leq n < N \quad (131)$$

Similarly, if the l -th spectral amplitude is voiced for the current frame and unvoiced for the previous frame then $\tilde{s}_{v,l}(n)$ is given by the following equation. In this case the energy in this region of the spectrum transitions from the unvoiced synthesis algorithm to the voiced synthesis algorithm.

$$\tilde{s}_{v,l}(n) = w_s(n - N) \bar{M}_l(0) \cos[\tilde{\omega}_0(0) (n - N) l + \varphi_l(0)] \quad \text{for } 0 \leq n < N \quad (132)$$

Otherwise, if the l -th spectral amplitude is voiced for both the current and the previous frame, and if either $l \geq 8$ or $|\tilde{\omega}_0(0) - \tilde{\omega}_0(-1)| \geq 0.1 \tilde{\omega}_0(0)$, then $\tilde{s}_{v,l}(n)$ is given by the following equation. In this case the energy in this region of the spectrum is completely synthesized by the voiced synthesis algorithm.

$$\begin{aligned} \tilde{s}_{v,l}(n) = & w_s(n) \bar{M}_l(-1) \cos[\tilde{\omega}_0(-1) n l + \varphi_l(-1)] \\ & + w_s(n - N) \bar{M}_l(0) \cos[\tilde{\omega}_0(0) (n - N) l + \varphi_l(0)] \end{aligned} \quad (133)$$

The variable n is restricted to the range $0 \leq n < N$. The synthesis window $w_s(n)$ used in Equations (131), (132) and (133) is assumed to be equal to zero outside the range $-105 \leq n \leq 105$.

A final rule is used if the l -th spectral amplitude is voiced for both the current and the previous frame, and if both $l < 8$ and $|\tilde{\omega}_0(0) - \tilde{\omega}_0(-1)| < 0.1 \tilde{\omega}_0(0)$. In this case $\tilde{s}_{v,l}(n)$ is given by the following equation, and the energy in this region of the spectrum is completely synthesized by the voiced synthesis algorithm.

$$\tilde{s}_{v,l}(n) = a_l(n) \cos[\theta_l(n)] \quad \text{for } 0 \leq n < N \quad (134)$$

The amplitude function $a_l(n)$ is given by,

$$a_l(n) = \bar{M}_l(-1) + \frac{n}{N} [\bar{M}_l(0) - \bar{M}_l(-1)] \quad (135)$$

and the phase function $\theta_l(n)$ is given by Equations (136) through (138).

$$\theta_l(n) = \varphi_l(-1) + [\tilde{\omega}_0(-1) \cdot l + \Delta\omega_l(0)] n + [\tilde{\omega}_0(0) \cdot l - \tilde{\omega}_0(-1)] \cdot \frac{ln^2}{2N} \quad (136)$$

$$\Delta\varphi_l(n) = \varphi_l(0) - \varphi_l(-1) - [\tilde{\omega}_0(-1) + \tilde{\omega}_0(0)] \cdot \frac{lN}{2} \quad (137)$$

$$\Delta\omega_l(0) = \frac{1}{N} \left[\Delta\varphi_l(0) - 2\pi \left\lfloor \frac{\Delta\varphi_l(0) + \pi}{2\pi} \right\rfloor \right] \quad (138)$$

The phase parameter φ_l which is used in the above equations shall be updated for each frame using Equations (139) through (141). The notation $\varphi_l(0)$ and $\psi_l(0)$ refers to the parameter values in the current frame, while $\varphi_l(-1)$ and $\psi_l(-1)$ denotes their counterparts in the previous frame.

$$\psi_l(0) = \psi_l(-1) + [\tilde{\omega}_0(-1) + \tilde{\omega}_0(0)] \cdot \frac{lN}{2} \quad \text{for } 0 \leq l \leq 56 \quad (139)$$

$$\varphi_l(0) = \begin{cases} \psi_l(0) & \text{for } 1 \leq l \leq \lfloor \tilde{L}/4 \rfloor \\ \psi_l(0) + \frac{\tilde{L}_{uv}(0) \cdot \rho_l(0)}{\tilde{L}(0)} & \text{for } \lfloor \tilde{L}/4 \rfloor < l \leq \max(\tilde{L}(-1), \tilde{L}(0)) \end{cases} \quad (140)$$

The parameter $\tilde{L}_{uv}(0)$ is equal to the number of unvoiced spectral amplitudes in the current frame, and the parameter $\rho_l(0)$ used in equation (140) is defined to be a random number which is uniformly distributed in the interval $[-\pi, \pi)$. This random number can be generated using the following equation,

$$\rho_l(0) = \frac{2\pi}{53125} u(l) - \pi \quad (141)$$

where $u(l)$ refers to shifted noise sequence for the current frame, which is described in Section 11.2.

Note that $\psi_l(0)$ shall be updated every frame using Equation (139) for $1 \leq l \leq 56$, regardless of the value of $\tilde{L}$ or the value of the V/UV decisions.

Once $\tilde{s}_{v,l}(n)$ is generated for each spectral amplitude the complete voiced component is generated according to equation (127). The synthetic speech signal is then computed by summing the voiced and unvoiced components as shown in equation (142).

$$\tilde{s}(n) = \tilde{s}_{uv}(n) + \tilde{s}_v(n) \quad \text{for } 0 \leq n < N \quad (142)$$

This completes the IMBE speech synthesis algorithm.

<i>Algorithm</i>	<i>Delay (ms)</i>
Analysis	73.75
Quantization	0.0
FEC / Interleaving	0.0
Reconstruction	0.0
Synthesis	6.25

Table 11 Breakdown of Algorithmic Delay

12 Additional Notes

The total algorithmic delay is 80 ms. This does not include any processing delay or transmission delay. The breakdown of the delay is shown in Table 11. The analysis delay is due to the filtering, windowing and two frame look-ahead used in the initial pitch estimation algorithm. The synthesis delay is introduced by the manner in which the synthesis algorithm smoothly transitions between the parameters estimated for consecutive speech frames.

In a few of the figures and flow charts, the variable $\bar{x}$ is equivalent to the variable $\tilde{x}$. For example the variable $\bar{v}$ in Figure 15 refers to the variable $\tilde{v}$ in the text. This notational discrepancy is a consequence of the graphical software used to produce this document.

Quantizer Value	Model Parameter	Bits Per Frame
b_0	Fundamental Frequency	7
b_1	V/UV Decisions	5
b_2	Gain	5
b_3	PRBA24 Vector	9
b_4	PRBA58 Vector	7
b_5	HOC1 Vector	5
b_6	HOC2 Vector	4
b_7	HOC3 Vector	4
b_8	HOC4 Vector	3
Total		49

Table 12 Half-Rate Voice/Silence Frame Bit Allocation

13 Half-Rate Vocoder Quantization

This section describes the final parameter estimation and quantization used by the Half-Rate Vocoder for each voice frame.

The analysis of each speech frame described in Chapter 5 generates a set of MBE model parameters consisting of the fundamental frequency, $\hat{\omega}_0$, the V/UV decisions, $\hat{v}_k$ for $1 \leq k \leq \hat{K}$, and the spectral amplitudes, $\hat{M}_l$ for $1 \leq l \leq \hat{L}$. The number of V/UV decisions, $\hat{K}$, and the number of spectral magnitudes, $\hat{L}$, varies from frame-to-frame as a function of the fundamental frequency. Since the Half-Rate vocoder is designed to operate at 3.6 kbps with a 20 ms frame length, 72 bits per frame are available for encoding the model parameters. Of these 72 bits, 23 are reserved for error control as is discussed in Clause 14.2 of this document, and the remaining 49 bits are divided among the model parameters as shown in Table 12.

This section describes the manner in which these bits are used to quantize, encode, decode and reconstruct the model parameters. In Clause 13.1 the encoding and decoding of the fundamental frequency is discussed, while Clause 13.2 discusses the encoding and decoding of the V/UV decisions. Clause 13.3 discusses the quantization and encoding of the spectral amplitudes, and Clause 13.4 discusses the decoding and reconstruction of the spectral amplitudes. Reference [13] provides general information on many of the techniques used in this section.

At the encoder, the output of the Half-Rate quantizer is a set of nine quantizer values, $\hat{b}_0, \hat{b}_1, \dots, \hat{b}_8$ which contain the 49 bits used to quantize the frame parameters. At the decoder, the corresponding received quantizer values, $\tilde{b}_0, \tilde{b}_1, \dots, \tilde{b}_8$ contain the 49 input bits used to reconstruct the MBE parameters for the frame. The numbers of bits assigned to each quantizer value are shown in Table 12, where b_i refers to both $\hat{b}_i$ in the encoder as well as $\tilde{b}_i$ in the decoder. This table applies to both voice frames and silence frames. Note that for tone frames, this table does not apply, and instead tone frames are formatted as described in Section 16.

13.1 Fundamental Frequency Encoding and Decoding

When voice is present, the fundamental frequency is estimated in the interval $\frac{2\pi}{123.125} \leq \hat{\omega}_0 \leq \frac{2\pi}{19.875}$ and is encoded by setting $\hat{b}_0$ equal to the 7-bit index corresponding to the entry in Annex L that is closest to $\hat{\omega}_0$. Note that values of $\hat{b}_0 \geq 120$ are reserved for non-voice frames as shown in Table 14. If the frame is a silence frame, then $\hat{b}_0 = 124$.

value	bits
0	000 0000
1	000 0001
2	000 0010
.	.
.	.
.	.
127	111 1111

Table 13 Seven Bit Binary Representation

Frame Type	b_0
voice	0 – 119
erasure	120 – 123
silence	124 – 125
tone	126 – 127

Table 14 Vocoder Frame Types

The selected quantizer value $\hat{b}_0$ is represented with 7 bits using the unsigned binary representation shown in Table 13. This representation is used throughout the vocoder to convert quantized values into a specific bit pattern.

At the receiver, the frame type is decoded first by checking $\tilde{b}_0$ against Table 14 to determine whether the current frame is a voice frame, erasure frame, silence frame or tone frame. If the frame is a voice frame, then the fundamental frequency is decoded by setting $\tilde{\omega}_0$ equal to the value shown in Annex L corresponding to the 7-bit index $n = \tilde{b}_0$. In addition $\tilde{b}_0$ is used to calculate $\tilde{L}$, the number of spectral amplitudes, as shown in Annex L.

If the frame is determined to be a silence frame, then the values of $\tilde{\omega}_0$, $\tilde{L}$, and $\tilde{v}_l$ are set as shown in the following equations:

$$\tilde{\omega}_0 = \frac{2\pi}{32} \quad (143)$$

$$\tilde{L} = 14 \quad (144)$$

$$\tilde{v}_l = 0 \quad (145)$$

The spectral amplitudes are then decoded using the same procedure as for voice frames (see Clause 13.4).

If the frame is determined to be a tone frame, then a set of MBE speech model parameters are not decoded for the frame. Instead a set of tone parameters are decoded from the tone frame as described in Clause 16.2.

If the frame is determined to be an erasure frame, then the frame is considered invalid (i.e. erased) and a set of MBE speech model parameters is not decoded for the frame. Instead a frame repeat is performed as described in Clause 14.6. Note that for erasure frames, only the frame type needs to be decoded, and the remaining bits in the frame are ignored.

13.2 Voiced / Unvoiced Decision Encoding and Decoding

The V/UV decisions $\hat{v}_k$, for $1 \leq k \leq \hat{K}$, are binary values that classify each of the $\hat{K}$ frequency bands as either voiced or unvoiced. These values are encoded into the 5-bit quantizer value $\hat{b}_1$ using the vector table shown in Annex M. Each of the first 17 vectors, $v(n)$ for $0 \leq n < 17$ are evaluated using distance Equation (146) and for voice frames $\hat{b}_1$ is set equal to the index n of the vector that yields the minimum distance:

$$E_n = \sum_{l=1}^{\hat{L}} |\hat{M}_l|^2 |\hat{v}_{k_l} - v_{j_l}(n)|^2 \quad (146)$$

where $\hat{v}_k$ are the estimated voicing decisions, and where the indices k_l and j_l are computed as follows:

$$j_l = \left\lfloor \frac{16 \hat{\omega}_0}{2\pi} \right\rfloor \quad (147)$$

$$k_l = \begin{cases} \left\lfloor \frac{l+2}{3} \right\rfloor & \text{if } l \leq 36 \\ 12 & \text{otherwise} \end{cases} \quad (148)$$

For silence frames $\hat{b}_1 = 0$.

For voice frames, the decoder reconstructs the V/UV decisions using the received 5-bit quantizer value $\tilde{b}_1$. This value is used as an index $n = \tilde{b}_1$ to select the corresponding 8 element vector $v(n)$ from Annex M. The voicing decisions, $\tilde{v}_l$ for $1 \leq l \leq \tilde{L}$, are then reconstructed from the selected vector using the following relationship:

$$\tilde{v}_l = v_{j_l}(n) \quad (149)$$

where j_l is the index computed according to Equation (147). For silence frames, the voicing decisions are reconstructed according to Equation (145).

Note that the decoder reconstructs the V/UV decisions for each spectral amplitude $1 \leq l \leq \tilde{L}$, rather than for each voicing band $1 \leq k \leq \tilde{K}$. This is a departure from the V/UV convention used by the encoder, which used a single V/UV decision to represent an entire frequency band. Instead the decoder assigns a separate V/UV decision for each spectral amplitude. This same procedure (i.e. computing the voicing decisions corresponding to each spectral amplitude) is also done by the encoder for use in computing the log spectral magnitudes according to Equation (150) in Clause 13.3.

13.3 Spectral Amplitudes Encoding

The process described in this Clause and the corresponding decoding Clause 13.4 is used to encode and decode, respectively, the spectral amplitudes for both voice frames and silence frames.

For the purpose of the following discussion $\hat{L}(0)$ or $\hat{L}$ refer to the number of harmonics in the current frame, while $\hat{L}(-1)$ refers to the number of harmonics in the previous frame. Similarly, $\hat{\Lambda}_l(0)$ (or $\hat{\Lambda}_l$) for $1 \leq l \leq \hat{L}(0)$ refers to the unquantized spectral amplitudes of the current frame, while $\hat{\Lambda}_l(-1)$ for $1 \leq l \leq \hat{L}(-1)$ refers to the quantized spectral amplitudes of the last voice frame.

The spectral amplitudes $\hat{M}_l$, for $1 \leq l \leq \hat{L}$, are real values which shall be quantized prior to encoding. This is accomplished by first computing the log spectral amplitudes, $\hat{\Lambda}_l$, for $1 \leq l \leq \hat{L}$, as follows:

$$\hat{\Lambda}_l = \begin{cases} \log_2 \hat{M}_l + 0.5 \log_2 \hat{L} & \text{if } \tilde{v}_l = 1 \\ \log_2 \hat{M}_l + 0.5 \log_2(\tilde{\omega}_0 \cdot \hat{L}) + 2.289 & \text{otherwise} \end{cases} \quad (150)$$

where $\tilde{v}_l$ is the voicing decision corresponding to the l -th spectral amplitude that is computed by the encoder. The gain is then computed from the log spectral parameters as follows:

$$\hat{\gamma}(0) = \frac{1}{\hat{L}} \sum_{l=1}^{\hat{L}} \hat{\Lambda}_l \quad (151)$$

and the differential gain is then computed as:

$$\hat{\Delta}_\gamma = \hat{\gamma}(0) - 0.5 \tilde{\gamma}(-1) \quad (152)$$

Where $\hat{\gamma}(0)$ is the computed gain for the current frame while $\tilde{\gamma}(-1)$ is the reconstructed gain from the previous frame. The differential gain $\hat{\Delta}_\gamma$ is then quantized using the 5 bit quantizer shown in Annex O. The quantizer value $\hat{b}_2$ is determined as the 5 bit index n corresponding to the quantizer level from the table in Annex O that is closest to $\hat{\Delta}_\gamma$. The value of $\tilde{\gamma}(-1)$ should be initialized to 0.

Once the gain has been computed and quantized, the log spectral amplitudes, $\hat{\Lambda}_l$ for $1 \leq l \leq \hat{L}$, are then used to compute the prediction residuals $\hat{T}_l$ for $1 \leq l \leq \hat{L}$, according to Equations (153) through (157).

$$\hat{k}_l = \frac{\hat{L}(-1)}{\hat{L}(0)} l \quad (153)$$

$$\hat{\delta}_l = \hat{k}_l - \lfloor \hat{k}_l \rfloor \quad (154)$$

$$\begin{aligned} \hat{T}_l &= \hat{\Lambda}_l(0) - 0.65 (1 - \hat{\delta}_l) \tilde{\Lambda}_{\lfloor \hat{k}_l \rfloor}(-1) - 0.65 \hat{\delta}_l \tilde{\Lambda}_{\lfloor \hat{k}_l \rfloor + 1}(-1) \\ &\quad + \frac{0.65}{\hat{L}(0)} \sum_{\lambda=1}^{\hat{L}(0)} (1 - \hat{\delta}_\lambda) \tilde{\Lambda}_{\lfloor \hat{k}_\lambda \rfloor}(-1) + \hat{\delta}_\lambda \tilde{\Lambda}_{\lfloor \hat{k}_\lambda \rfloor + 1}(-1) \end{aligned} \quad (155)$$

In order to form $\hat{T}_l$ using Equations (153) through (155), the following assumptions are made:

$$\tilde{\Lambda}_0(-1) = \tilde{\Lambda}_1(-1) \quad (156)$$

$$\tilde{\Lambda}_l(-1) = \tilde{\Lambda}_{\hat{L}(-1)}(-1) \quad \text{for } l > \hat{L}(-1) \quad (157)$$

Upon initialization $\tilde{\Lambda}_l(-1)$ should be set equal to 1.0 for all l , and $\hat{L}(-1) = 15$.

Once the $\hat{L}$ prediction residuals have been computed, they are divided into 4 blocks. The length of each block, denoted $\hat{J}_i$ for $1 \leq i \leq 4$, is determined as a function of $\hat{L}$ in accordance with Annex N.

The first or lowest frequency block is denoted by $\hat{c}_{1,j}$ for $1 \leq j \leq \hat{f}_1$, and it consists of the first $\hat{f}_1$ consecutive elements of $\hat{T}_l$ (i.e. $1 \leq l \leq \hat{f}_1$). The second block is denoted by $\hat{c}_{2,j}$ for $1 \leq j \leq \hat{f}_2$, and it consists of the next $\hat{f}_2$ consecutive elements of $\hat{T}_l$ (i.e. $\hat{f}_1 + 1 \leq l \leq \hat{f}_1 + \hat{f}_2$). This continues through the fourth or highest frequency block, which is denoted by $\hat{c}_{4,j}$ for $1 \leq j \leq \hat{f}_4$. It consists of the last $\hat{f}_4$ consecutive elements of $\hat{T}_l$ (i.e. $\hat{L} + 1 - \hat{f}_4 \leq l \leq \hat{L}$).

Each of the four blocks is transformed using a Discrete Cosine Transform (DCT), which is discussed in [7]. The length of the DCT for the i -th block is equal to $\hat{f}_i$. The DCT coefficients are denoted by $\hat{C}_{i,k}$, where $1 \leq i \leq 4$ refers to the block number, and $1 \leq k \leq \hat{f}_i$ refers to the particular coefficient within each block. The formula for the computation of these DCT coefficients is as follows:

$$\hat{C}_{i,k} = \frac{1}{\hat{f}_i} \sum_{j=1}^{\hat{f}_i} \hat{c}_{i,j} \cos \left[\frac{\pi(k-1)(j-1/2)}{\hat{f}_i} \right] \quad (158)$$

The DCT coefficients from each of the four blocks are then divided into two groups. The first group consists of the first two DCT coefficients from each of the four blocks. These first coefficients are used to form an eight element Predictive Residual Block Average (PRBA) vector, $\hat{R}_i$ for $1 \leq i \leq 8$, as follows:

$$\hat{R}_1 = \hat{C}_{1,1} + \sqrt{2} \hat{C}_{1,2} \quad (159)$$

$$\hat{R}_2 = \hat{C}_{1,1} - \sqrt{2} \hat{C}_{1,2} \quad (160)$$

$$\hat{R}_3 = \hat{C}_{2,1} + \sqrt{2} \hat{C}_{2,2} \quad (161)$$

$$\hat{R}_4 = \hat{C}_{2,1} - \sqrt{2} \hat{C}_{2,2} \quad (162)$$

$$\hat{R}_5 = \hat{C}_{3,1} + \sqrt{2} \hat{C}_{3,2} \quad (163)$$

$$\hat{R}_6 = \hat{C}_{3,1} - \sqrt{2} \hat{C}_{3,2} \quad (164)$$

$$\hat{R}_7 = \hat{C}_{4,1} + \sqrt{2} \hat{C}_{4,2} \quad (165)$$

$$\hat{R}_8 = \hat{C}_{4,1} - \sqrt{2} \hat{C}_{4,2} \quad (166)$$

The quantization of the PRBA vector is discussed in Clause 13.3.1. The second group consists of the remaining higher order DCT coefficients from each block. These coefficients correspond to $\hat{C}_{i,j}$, for $1 \leq i \leq 4$ and $3 \leq j \leq \hat{f}_i$. Note that if $\hat{f}_i = 2$, then there are no higher order DCT coefficients in the i -th block. The quantization of the higher order DCT coefficients is discussed in Clause 13.3.2.

One important feature of the spectral amplitude encoding algorithm is that the spectral amplitude information is transmitted differentially. Specifically, a prediction residual is transmitted which measures the change in the spectral envelope between the current frame and the previous frame. In order for a differential scheme of this type to work properly, the encoder needs to simulate the operation of the decoder and normally use the reconstructed spectral amplitudes from the previous frame to predict the spectral amplitudes of the current frame. The Half-Rate spectral amplitude encoder simulates the spectral amplitude decoder by setting $\tilde{L} = \hat{L}$ and then reconstructing the spectral amplitudes as discussed in Clause 13.4 and the corresponding voicing decisions for each spectral amplitude as discussed in Clause 13.2. These reconstructed spectral amplitudes are then

saved by the encoder for use in quantizing the spectral amplitudes of the next frame. An exception to this reconstruct and update process occurs for silence frames, tone frames and erasure frames in which case the saved reconstructed spectral amplitudes are not updated with the reconstructed spectral amplitudes from the current frame. This ensures that only voice frames (i.e. not silence frames, tone frames or erasure frames) are used in predicting future spectral amplitudes.

13.3.1 Encoding the PRBA Vector

The quantization of the PRBA vector begins with an eight point DCT of $\hat{R}_i$ for $1 \leq i \leq 8$ as shown in the following equation:

$$\hat{G}_m = \frac{1}{8} \sum_{i=1}^8 \hat{R}_i \cos \left[\frac{\pi(m-1)(j-1/2)}{8} \right] \quad \text{for } 1 \leq m \leq 8 \quad (167)$$

The resulting vector, denoted by $\hat{G}_m$ for $1 \leq m \leq 8$, is vector quantized in two parts. The first element, $\hat{G}_1$, is discarded. The next three elements of the PRBA vector (i.e. $[\hat{G}_2, \hat{G}_3, \hat{G}_4]$) are jointly quantized using the 9-bit vector quantizer shown in Annex P. The quantizer value $\hat{b}_3$ is determined as the 9-bit index n corresponding to the quantizer vector from the table in Annex P which is closest (i.e. has the minimum mean-squared error) to the three element vector $[\hat{G}_2, \hat{G}_3, \hat{G}_4]$. The final four elements of the PRBA vector (i.e. $[\hat{G}_5, \hat{G}_6, \hat{G}_7, \hat{G}_8]$) are jointly quantized using the 7-bit vector quantizer shown in Annex Q. The quantizer value $\hat{b}_4$ is determined as the 7-bit index n corresponding to the quantizer vector from the table in Annex Q which is closest (i.e. has the minimum mean-squared error) to the four element vector $[\hat{G}_5, \hat{G}_6, \hat{G}_7, \hat{G}_8]$.

13.3.2 Encoding the Higher Order DCT Coefficients

Once the gain and the PRBA vector has been quantized, the higher order DCT coefficients from each of the four blocks are quantized to form the four quantizer values $\hat{b}_5, \hat{b}_6, \hat{b}_7$, and $\hat{b}_8$. The number of bits for each of these quantizer values is specified in Table 12.

The quantizer value $\hat{b}_{i+4}$, for $1 \leq i \leq 4$, is the quantizer output for the i -th block. These quantizer values are determined by first forming a set of Higher Order Coefficient (HOC) vectors $\hat{H}_{i,j} = \hat{C}_{i,j+2}$, for $1 \leq i \leq 4$ and $1 \leq j \leq \hat{J}_i - 2$. However, if $\hat{J}_i \leq 2$ for any i , then an HOC vector is not formed for that block and the corresponding quantizer value $\hat{b}_{i+4} = 0$.

Once the 4 HOC vectors are formed in this manner, then each HOC vector is vector quantized using the specified vector quantization table shown in Annex R. A 5-bit (32 vector) quantization table is used to determine $\hat{b}_5$ from $\hat{H}_{1,j}$; a 4-bit (16 vector) quantization table is used to determine $\hat{b}_6$ from $\hat{H}_{2,j}$; another 4-bit (16 vector) quantization table is used to determine $\hat{b}_7$ from $\hat{H}_{3,j}$; and a 3-bit (8 vector) quantization table is used to determine $\hat{b}_8$ from $\hat{H}_{4,j}$. Each of these vector quantization tables is listed separately in Annex R. The quantizer value for each HOC vector, is determined by setting $\hat{b}_{i+4}$ to the index of the quantizer vector from the specified table in Annex R which is closest (i.e. has the minimum mean-squared error) to the HOC vector $\hat{H}_{i,j}$ for $1 \leq i \leq 4$. Note that each quantization vector shown in Annex R is 4 elements long, while the length of the HOC vectors can vary from 1 to nearly 20 (ignoring the zero length HOC vectors). If an HOC vector has a non-zero length less than 4, then only the first $\hat{J}_i - 2$ elements of each quantization vector are used in computing the minimum mean squared error and the remaining elements from each quantization

vector are ignored. Similarly, if an HOC vector has a length greater than 4, then only the first 4 elements of that HOC vector are used in computing the minimum mean squared error, and the remaining elements in the HOC vector are ignored.

13.4 Spectral Amplitudes Decoding

In order for the decoder to reconstruct the spectral amplitudes, the parameter $\tilde{L}$ shall first be computed from $\tilde{b}_0$ using Annex L, and the V/UV decisions $\tilde{v}_l$ for $1 \leq l \leq \tilde{L}$ shall be computed from $\tilde{b}_1$ using Equation (149). Next the spectral amplitudes are decoded and reconstructed by inverting the quantization and encoding procedure described above.

Reconstruction of the spectral amplitudes is accomplished by dividing them into 4 DCT blocks, where the length of each block, $\tilde{J}_i$ for $1 \leq i \leq 4$ is set depending on $\tilde{L}$ according to Annex N.

13.4.1 Decoding the Gain

The first spectral parameter to be decoded is the differential gain, $\tilde{\Delta}_\gamma$ using the quantizer value $\tilde{b}_2$. This is accomplished by setting $\tilde{\Delta}_\gamma$ equal to the quantization level from Annex O corresponding to the index $n = \tilde{b}_2$. The decoded gain for the current frame, $\tilde{\gamma}(0)$, is then reconstructed as follows:

$$\tilde{\gamma}(0) = \tilde{\Delta}_\gamma + 0.5 \tilde{\gamma}(-1) \quad (168)$$

where $\tilde{\gamma}(-1)$ is the reconstructed gain from the last valid voice frame, not including any past tone, silence or erasure frames, and where $\tilde{\gamma}(-1)$ is initialized to 0.

13.4.2 Decoding the PRBA Vector

Once the gain has been decoded, the transformed PRBA vector, denoted by $\tilde{G}_m$ for $1 \leq m \leq 8$ is decoded using the quantizer values $\tilde{b}_3$ and $\tilde{b}_4$. The first element of the transformed PRBA vector is set equal to 0 (i.e. $\tilde{G}_1 = 0$). The next three elements of the transformed PRBA vector, $[\tilde{G}_2, \tilde{G}_3, \tilde{G}_4]$ are set equal to the quantization vector from the table in Annex P corresponding to index $n = \tilde{b}_3$. The final four elements of the transformed PRBA vector, $[\tilde{G}_5, \tilde{G}_6, \tilde{G}_7, \tilde{G}_8]$ are set equal to the quantization vector from the table in Annex Q corresponding to index $n = \tilde{b}_4$.

Once the transformed PRBA vector has been reconstructed in this manner, then the PRBA vector $\tilde{R}_m$ for $1 \leq m \leq 8$ is computed through an inverse DCT of $\tilde{G}_m$ as shown in the following equation:

$$\tilde{R}_i = \sum_{m=1}^8 \alpha(m) \tilde{G}_m \cos \left[\frac{\pi(m-1)(i-1/2)}{8} \right] \quad \text{for } 1 \leq i \leq 8 \quad (169)$$

$$\alpha(m) = \begin{cases} 1 & \text{if } m = 1 \\ 2 & \text{otherwise} \end{cases} \quad (170)$$

The first two elements of each of the four DCT blocks, denoted by $\tilde{C}_{i,k}$ for $1 \leq i \leq 4$ and $1 \leq k \leq 2$ are then computed from the PRBA vector as follows:

$$\tilde{C}_{1,1} = \frac{1}{2} (\tilde{R}_1 + \tilde{R}_2) \quad (171)$$

$$\tilde{C}_{1,2} = \frac{\sqrt{2}}{4} (\tilde{R}_1 - \tilde{R}_2) \quad (172)$$

$$\tilde{C}_{2,1} = \frac{1}{2} (\tilde{R}_3 + \tilde{R}_4) \quad (173)$$

$$\tilde{C}_{2,2} = \frac{\sqrt{2}}{4} (\tilde{R}_3 - \tilde{R}_4) \quad (174)$$

$$\tilde{C}_{3,1} = \frac{1}{2} (\tilde{R}_5 + \tilde{R}_6) \quad (175)$$

$$\tilde{C}_{3,2} = \frac{\sqrt{2}}{4} (\tilde{R}_5 - \tilde{R}_6) \quad (176)$$

$$\tilde{C}_{4,1} = \frac{1}{2} (\tilde{R}_7 + \tilde{R}_8) \quad (177)$$

$$\tilde{C}_{4,2} = \frac{\sqrt{2}}{4} (\tilde{R}_7 - \tilde{R}_8) \quad (178)$$

Reconstruction of the remaining elements of $\tilde{C}_{i,k}$ is discussed in the Clause 13.4.3.

13.4.3 Decoding the Higher Order DCT Coefficients

The higher order DCT coefficients, which are denoted by $\tilde{C}_{i,k}$ for $1 \leq i \leq 4$ and $2 \leq k \leq \tilde{J}_i$ are reconstructed from the quantizer values $\tilde{b}_5, \tilde{b}_6, \tilde{b}_7$, and $\tilde{b}_8$. For the i -th block, an HOC vector, denoted by $\tilde{H}_{i,j}$ is first reconstructed as the quantization vector from the specified table in Annex R corresponding to the index $n = \tilde{b}_{i+4}$.

The higher order DCT coefficients, $\tilde{C}_{i,k}$ for $1 \leq i \leq 4$ and $2 \leq k \leq \tilde{J}_i$, are then reconstructed from the HOC vectors according to the following equation:

$$\tilde{C}_{i,k} = \begin{cases} \tilde{H}_{i,k-2} & \text{for } 2 \leq k \leq \tilde{J}_i \text{ and } k \leq 4 \\ 0 & \text{otherwise} \end{cases} \quad (179)$$

Note that if $\tilde{J}_i \leq 2$, then there are no higher order DCT coefficients to reconstruct, and this step is skipped for the i -th block.

Once all the DCT coefficients $\tilde{C}_{i,k}$ have been reconstructed, an inverse DCT is computed on each of the four blocks to form the vectors $\tilde{c}_{i,j}$. This is done using the following equations for $1 \leq i \leq 4$.

$$\tilde{c}_{i,j} = \sum_{k=1}^{\tilde{J}_i} \alpha(k) \tilde{C}_{i,k} \cos \left[\frac{\pi(k-1)(j-1/2)}{\tilde{J}_i} \right] \quad \text{for } 1 \leq j \leq \tilde{J}_i \quad (180)$$

$$\alpha(k) = \begin{cases} 1 & \text{if } k = 1 \\ 2 & \text{otherwise} \end{cases} \quad (181)$$

The four inverse transformed blocks $\tilde{c}_{i,j}$ are then joined to form a single vector of length $\tilde{L}$, which is denoted $\tilde{T}_l$ for $1 \leq l \leq \tilde{L}$. The vector $\tilde{T}_l$ corresponds to the reconstructed spectral amplitude prediction residuals. The adopted convention is that the first $\tilde{J}_1$ elements of $\tilde{T}_l$ are equal to $\tilde{c}_{1,j}$ for

$1 \leq j \leq \tilde{J}_1$. The next $\tilde{J}_2$ elements of $\tilde{T}_l$ are equal to $\tilde{c}_{2,j}$ for $1 \leq j \leq \tilde{J}_2$. This continues until the last $\tilde{J}_4$ elements of $\tilde{T}_l$ are equal to $\tilde{c}_{4,j}$ for $1 \leq j \leq \tilde{J}_4$.

Once the vector $\tilde{T}_l$ has been reconstructed in this manner, then the reconstructed log spectral amplitudes for the current frame, denoted by $\tilde{\Lambda}_l(0)$ for $1 \leq l \leq \tilde{L}$, are computed using the following equations:

$$\tilde{k}_l = \frac{\tilde{L}(-1)}{\tilde{L}(0)} l \quad (182)$$

$$\tilde{\delta}_l = \tilde{k}_l - \lfloor \tilde{k}_l \rfloor \quad (183)$$

$$\tilde{\Gamma} = \tilde{\gamma}(0) - 0.5 \log_2 \tilde{L} - \frac{1}{\tilde{L}} \sum_{\lambda=1}^{\tilde{L}} \tilde{T}_\lambda \quad (184)$$

$$\begin{aligned} \tilde{\Lambda}_l(0) = & \tilde{T}_l + 0.65 (1 - \tilde{\delta}_l) \tilde{\Lambda}_{\lfloor \tilde{k}_l \rfloor}(-1) + 0.65 \tilde{\delta}_l \tilde{\Lambda}_{\lfloor \tilde{k}_l \rfloor + 1}(-1) \\ & - \frac{0.65}{\tilde{L}(0)} \sum_{\lambda=1}^{\tilde{L}(0)} (1 - \tilde{\delta}_\lambda) \tilde{\Lambda}_{\lfloor \tilde{k}_\lambda \rfloor}(-1) + \tilde{\delta}_\lambda \tilde{\Lambda}_{\lfloor \tilde{k}_\lambda \rfloor + 1}(-1) + \tilde{\Gamma} \end{aligned} \quad (185)$$

where $\tilde{\Lambda}_l(-1)$ denotes the reconstructed log spectral magnitudes from the last voice frame, not including any tone, silence, erasure or invalid frames. In applying Equations (182) through (185), the following assumptions are made:

$$\tilde{\Lambda}_0(-1) = \tilde{\Lambda}_1(-1) \quad (186)$$

$$\tilde{\Lambda}_l(-1) = \tilde{\Lambda}_{\tilde{L}(-1)}(-1) \quad \text{for } l > \tilde{L}(-1) \quad (187)$$

In addition, it is assumed that upon initialization $\tilde{\Lambda}_l(-1) = 1$ for all l , and $\tilde{L}(-1) = 15$.

Finally, the spectral amplitudes for the current frame, denoted by $\tilde{M}_l$, $1 \leq l \leq \tilde{L}$ are reconstructed from $\tilde{\Lambda}_l$ using the following equation:

$$\tilde{M}_l = \begin{cases} \exp(0.693 \tilde{\Lambda}_l) & \text{if } \tilde{v}_l = 1 \\ \frac{0.2046}{\sqrt{\tilde{\omega}_0}} \exp(0.693 \tilde{\Lambda}_l) & \text{otherwise} \end{cases} \quad (188)$$

where $\tilde{v}_l$ is the voicing decision corresponding to the l -th spectral amplitude that is computed by the decoder. The reconstructed spectral amplitudes $\tilde{M}_l$ are then used by the synthesis algorithm, as described in Chapter 11 to produce the synthetic digital speech output from the Half-Rate decoder.

One final note is that the Half-Rate encoder uses the value $\hat{L}$ to determine the four DCT block sizes, $\hat{J}_i$, while at the decoder the corresponding block sizes, denoted by $\tilde{J}_i$ are determined by the value $\tilde{L}$. In order to ensure proper operation it is necessary that these two values be equal (i.e. $\hat{L} = \tilde{L}$). The encoder and decoder are designed to ensure this property except in the presence of a very large number of bit errors. In addition, the decoder is designed to detect frames where a large number of bit errors may prevent the generation of the correct bit allocation and quantizer step sizes. In this case, the decoder discards the bits for the current frame and repeats the parameters from the previous frame. This is discussed in more detail in later clauses of this document.

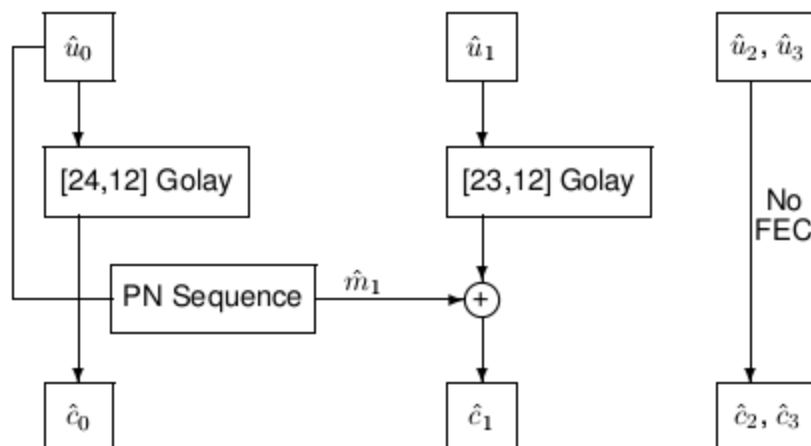


Figure 27 Half-Rate Code Vector Construction

14 Half-Rate Bit Manipulations

The Half-Rate vocoder uses a number of different bit manipulations in order to increase its robustness to channel degradations. The quantizer values, $\hat{b}_0, \hat{b}_1, \dots, \hat{b}_8$, are first prioritized into a set of four bit vectors, denoted by values, $\hat{u}_0, \hat{u}_1, \dots, \hat{u}_3$. These vectors are protected with error control codes, consisting of one [24, 12] Golay code and one [23, 12] Golay code, and modulated to produce a set of code vectors denoted by $\hat{c}_0, \dots, \hat{c}_3$. The error control codes add redundancy by increasing the number of bits per voice frame from 49 (2450 bps) to 72 (3600 bps). The construction of the code vectors from the quantizer values is depicted in Figure 27. Intra-frame bit interleaving is applied to these code vectors to form the transmitted bit stream for the Half-Rate Vocoder.

The Half-Rate decoder reverses the bit manipulations performed by the encoder. First the decoder de-interleaves each frame of 72 bits to obtain the four code vectors $\hat{c}_0, \dots, \hat{c}_3$. The decoder then demodulates and error control decodes these code vectors to produce the decoded bit vectors $\tilde{u}_0, \dots, \tilde{u}_3$. In order to ensure sufficient performance it is necessary that the decoder decode all error control codes up to their maximum error correction/detection capability, and soft-decision decoding is recommended to further improve the robustness to bit errors (soft-decision decoding was used in the MOS evaluation of the Half-Rate Vocoder performed by TIA in 2003). Note that there are a number of well-established methods for decoding of the Golay codes, and specific methods are not presented in this document.

Once the decoder has performed error correction and detection, it shall rearrange the bit vectors to reconstruct the quantizer values, denoted by $\tilde{b}_0, \tilde{b}_1, \dots, \tilde{b}_8$. These values are then used to reconstruct the MBE model parameters as described in Section 13, and the resulting parameters are finally used to synthesize the current frame of speech.

One should note that the Half-Rate decoder employs a number of different mechanisms to improve performance in the presence of bit errors. These mechanisms consist first of error control codes, which are able to remove a significant number of errors. In addition, the Half-Rate decoder uses bit modulation combined with frame repeats and frame mutes to detect and discard highly corrupted

frames. Finally, the Half-Rate decoder uses adaptive smoothing to reduce the perceived effect of any remaining errors. These mechanisms are all discussed in the following clauses of this description.

14.1 Bit Prioritization

The first bit manipulation performed by the Half-Rate encoder is a rearrangement of the quantizer values $\hat{b}_0, \hat{b}_1, \dots, \hat{b}_8$ into a set of 4 prioritized bit vectors denoted by $\hat{u}_0, \hat{u}_1, \dots, \hat{u}_3$. The bit vectors $\hat{u}_0$ and $\hat{u}_1$ are both 12 bits long, while the bit vector $\hat{u}_2$ is 11 bits long, and the bit vector $\hat{u}_3$ is 14 bits long. Throughout this clause the convention has been adopted that $\hat{b}(N-1)$, where N is the vector length, represents bit $N-1$ which is the most significant bit (MSB), and $\hat{b}(0)$ represents bit 0 which is the least significant bit (LSB).

The prioritization of the quantizer values into the set of bit vectors begins with $\hat{u}_0$. The four most significant bits of $\hat{u}_0$ (i.e. bits 11 through 8) are set equal to the four most significant bits of $\hat{b}_0$ (i.e. bits 6 through 3). The next four most significant bits of $\hat{u}_0$ (i.e. bits 7 through 4) are set equal to the four most significant bits of $\hat{b}_1$ (i.e. bits 4 through 1). The remaining four bits of $\hat{u}_0$ (i.e. bits 3 through 0) are generated from the four most significant bits of $\hat{b}_2$ (i.e. bits 4 through 1). The construction of bit vector $\hat{u}_0$ is further depicted in Table 15, where the top row indicates the bit position in $\hat{u}_0$ (bit 11 is the MSB), and the bottom row represents the corresponding quantizer bits.

11	8	7	4	3	0
$\hat{b}_0(6, 5, 4, 3)$		$\hat{b}_1(4, 3, 2, 1)$		$\hat{b}_2(4, 3, 2, 1)$	

Table 15 Construction of Bit Vector u_0

The prioritization of the quantizer values continues with bit vector $\hat{u}_1$. The eight most significant bits of $\hat{u}_1$ (i.e. bits 11 through 4) are set equal to the eight most significant bits of $\hat{b}_3$ (i.e. bits 8 through 1). The four least significant bits of $\hat{u}_1$ (i.e. bits 3 through 0) are set equal to the four most significant bits of $\hat{b}_4$ (i.e. bits 6 through 3). The construction of bit vector $\hat{u}_1$ is further depicted in Table 16.

11	4	3	0
$\hat{b}_3(8, 7, 6, 5, 4, 3, 2, 1)$		$\hat{b}_4(6, 5, 4, 3)$	

Table 16 Construction of Bit Vector u_1

Next bit vector $\hat{u}_2$ is constructed starting with the four most significant bits of $\hat{b}_5$ (i.e. bits 4 through 1), followed by the three most significant bits of $\hat{b}_6$ (i.e. bits 3 through 1), then followed by the three most significant bits of $\hat{b}_7$ (i.e. bits 3 through 1), and finishing with the most significant bit of $\hat{b}_8$ (i.e. bit 2). The construction of bit vector $\hat{u}_2$ is further depicted in Table 17.

10	7	6	4	3	1	0
$\hat{b}_5(4, 3, 2, 1)$		$\hat{b}_6(3, 2, 1)$		$\hat{b}_7(3, 2, 1)$		$\hat{b}_8(2)$

Table 17 Construction of Bit Vector u_2

Finally, bit vector $\hat{u}_3$ is constructed from the remaining bits starting with the least significant bit of $\hat{b}_1$ (i.e. bit 0), and continuing as shown in Table 18.

13	12	11	9	8	7	5	4	3	2	1	0
$\hat{b}_1(0)$	$\hat{b}_2(0)$	$\hat{b}_0(2, 1, 0)$	$\hat{b}_3(0)$	$\hat{b}_4(2, 1, 0)$	$\hat{b}_5(0)$	$\hat{b}_6(0)$	$\hat{b}_7(0)$	$\hat{b}_8(1, 0)$			

Table 18 Construction of Bit Vector u_3

14.2 Error Control Coding

At 3.6 kbps with a 20 ms frame size, 72 bits per frame are available for transmitting voice over the air interface. The Half-Rate vocoder uses 49 of these bits to quantize the model parameters as previously described. The remaining 23 bits are used for forward error correction. These 23 error control bits are divided between one [24, 12] Golay code and one [23, 12] Golay code. These two codes are identical except the [24, 12] Golay code includes one additional parity bit in the LSB position. The reader is referred to references [8] and [9] for more information on the encoding and decoding of Golay codes.

The generation of the four code vectors $\hat{c}_i$ for $0 \leq i \leq 3$ is performed using the following set of equations to map each bit vector, $\hat{u}_i$, to a corresponding code vector, $\hat{c}_i$:

$$\hat{c}_0 = \hat{u}_0 G_{24,12} \quad (189)$$

$$\hat{c}_1 = [\hat{u}_1 G_{23,12}] + \hat{m}_1 \quad (190)$$

$$\hat{c}_2 = \hat{u}_2 \quad (191)$$

$$\hat{c}_3 = \hat{u}_3 \quad (192)$$

Where the $G_{24,12}$ and $G_{23,12}$ are the generator matrices for the extended and regular Golay codes shown below, and the modulation vector $\hat{m}_1$ applied in Equation (190) is described in the following clause. Note all operations are modulo 2. The construction of the code vectors is further depicted in Figure 27.

$$G_{24,12} = \begin{bmatrix} 100000000000110001110101 \\ 0100000000000011000111011 \\ 0010000000000111101101000 \\ 000100000000011110110100 \\ 00001000000001111011010 \\ 00000100000001111011010 \\ 0000001000000110110011001 \\ 000000010000011011001101 \\ 0000000010000001101100111 \\ 0000000001000110111000110 \\ 0000000000100101010010111 \\ 0000000000010100100111110 \\ 0000000000001100011101011 \end{bmatrix}$$

$G_{23,12} =$

1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	0	1	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	1	0
0	0	1	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	0	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	0	0
0	0	0	0	1	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	0	1
0	0	0	0	0	1	0	0	0	0	0	1	1	0	1	1	0	0	1	1	0	0
0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	0

Note that in these generator matrices absent entries are assumed to equal 0, and the input and output vectors are assumed to be row vectors, where the “left” most bit is the MSB. This convention is used throughout this clause.

14.3 Bit Modulation

The Half-Rate vocoder uses a pseudo-random bit pattern, keyed off the bit vector $\hat{u}_0$ to compute a modulation vector $\hat{m}_1$ that is applied in the computation of $\hat{c}_1$ according to Equation (190). This modulation step provides a mechanism for detecting errors in $\hat{u}_0$ beyond the error correction / detection capability of the [24, 12] Golay code. Note that the term bit modulation in the context of this document refers to the presented method for multiplying (or modulating) a code vector by a data dependent pseudo-random sequence. The first step in this procedure is to generate the modulation vector $\hat{m}_1$ from a pseudo-random sequence whose seed is derived from $\hat{u}_0$. Specifically, the sequence defined in the following equations is used,

$$p_r(0) = 16 \hat{u}_0 \quad (193)$$

$$p_r(n) = 173 p_r(n-1) + 13849 - 65536 \left\lfloor \frac{173 p_r(n-1) + 13849}{65536} \right\rfloor \quad (194)$$

where the bit vector $\hat{u}_0$ is interpreted as an unsigned 12 bit number in the range [0, 4095]. Equation (194) is then used to recursively compute the pseudo-random sequence, $p_r(n)$, over the range $1 \leq n \leq 23$. Each element of this sequence can be interpreted as a 16 bit random number which is uniformly distributed over the interval [0, 65535]. Using this interpretation, a binary modulation vector $\hat{m}_1$ is generated from this sequence as shown below.

$$\hat{m}_1 = \left[\left\lfloor \frac{p_r(1)}{32768} \right\rfloor, \left\lfloor \frac{p_r(2)}{32768} \right\rfloor, \dots, \left\lfloor \frac{p_r(23)}{32768} \right\rfloor \right] \quad (195)$$

The modulation vector $\hat{m}_1$ is then added (modulo 2) to the result of the [23, 12] Golay code to produce $\hat{c}_1$ as described in Equation (190).

One should note that the bit modulation performed by the Half-Rate encoder can be inverted by the decoder if $\tilde{c}_0$ does not contain any uncorrectable bit errors. In this case, Golay decoding $\tilde{c}_0$ will yield the correct value of $\tilde{u}_0$. The decoder can then use $\tilde{u}_0$ to reconstruct the pseudo-random sequence

and the modulation vector $\tilde{m}_1$. Subtracting this modulation vector (modulo 2) from $\tilde{c}_1$ and Golay decoding the result will then yield the decoded bit vector $\tilde{u}_1$. The remaining bit vectors are recovered using the relationships $\tilde{u}_2 = \tilde{c}_2$ and $\tilde{u}_3 = \tilde{c}_3$ without any FEC decoding.

One should also note that in the other case, where $\tilde{c}_0$ contains uncorrectable bit errors, the modulation cannot generally be inverted by the decoder. In this case the likely result of Golay decoding $\tilde{c}_0$ will be some $\tilde{u}_0$ which does not equal $\hat{u}_0$. Consequently, the decoder will initialize the pseudo-random sequence incorrectly, and the modulation vector computed by the decoder will be uncorrelated with the modulation vectors used by the encoder. Using these incorrect modulation vectors to reconstruct the code vectors is essentially the same as passing $\tilde{c}_1$ through a 50 percent bit error rate (BER) channel. The Half-Rate decoder exploits the fact that, statistically, a 50 percent BER causes the [23, 12] Golay decoder to correct a number of errors which is near the maximum capability of the code (i.e. near 3). By counting the total number of errors which are corrected in all of these code vectors, the decoder is able to detect many frames in which $\tilde{c}_0$ is likely to contain uncorrectable bit errors. The decoder performs frame repeats during these frames in order to reduce the perceived degradation in the presence of bit errors. This is explained more fully in Clauses 14.5 and 14.6.

14.4 Bit Interleaving

Intra-frame bit interleaving is used to spread short bursts of errors among several code vectors (i.e. between $\tilde{u}_0$ and $\tilde{u}_1$). The division of each frame of 72 bits into 36 dibit symbols is shown in Annex S, where Position 0 is the first bit of each voice frame (in time) to be transmitted and received. Note this annex uses the notation $c_j(n)$ to designate the n -th bit of the modulated code vector $\hat{c}_j$ (or the demodulated code vector $\tilde{c}_j$), where bit $N - 1$ (assuming a vector length of N) is the MSB of each vector and bit 0 is the LSB.

14.5 Error Estimation

The Half-Rate speech decoder estimates the number of errors in each received data frame by computing the number of errors corrected by the two Golay codes. The number of bit errors that were corrected or detected during FEC decoding of the i -th code is denoted ε_i and index i ranges over $0 \leq i \leq 1$. From these two error values two other error parameters are computed as shown below.

$$\varepsilon_T = \varepsilon_0 + \varepsilon_1 \quad (196)$$

$$\varepsilon_R(0) = 0.95 \varepsilon_R(-1) + 0.001064 \varepsilon_T \quad (197)$$

The parameter $\varepsilon_R(0)$ is the estimate of the error rate for the current frame, while $\varepsilon_R(-1)$ is the estimate of the error rate for the previous frame. Note that $\varepsilon_R(-1) = 0$ upon initialization of the decoder. These error parameters are used to control the frame repeat process described below.

14.6 Frame Repeats

The Half-Rate decoder examines each received data frame in order to detect and discard frames which are highly corrupted. A number of different fault conditions are checked and if any of these conditions indicate the current frame is invalid, then a frame repeat is performed.

The Half-Rate speech encoder uses values of $\hat{b}_0$ to denote different frame types as shown in Table 14. If a value of $\hat{b}_0$ is received in the range $120 \leq \hat{b}_0 \leq 123$ then the decoder interprets the frame as an erasure frame and performs a frame repeat. Similarly, a frame repeat is performed by the decoder if $\varepsilon_0 \geq 4$, or if both of the following two equations are true.

$$\varepsilon_0 \geq 2 \quad (198)$$

$$\varepsilon_T \geq 6 \quad (199)$$

These two equations serve to detect the incorrect bit demodulation which results if there are uncorrectable bit errors in $\tilde{c}_0$.

The decoder performs a frame repeat by taking the following steps:

- 1) The current received voice frame is marked as invalid and subsequently ignored during future processing steps.
- 2) The MBE speech model parameters for the current frame are set equal to the decoded model parameters for the previous frame. Specifically, for voice frames the following update expressions are computed.

$$\tilde{\omega}_0(0) = \tilde{\omega}_0(-1) \quad (200)$$

$$\tilde{L}(0) = \tilde{L}(-1) \quad (201)$$

$$\tilde{K}(0) = \tilde{K}(-1) \quad (202)$$

$$\tilde{v}_l(0) = \tilde{v}_l(-1) \quad \text{for } 1 \leq l \leq \tilde{L}(-1) \quad (203)$$

$$\tilde{M}_l(0) = \tilde{M}_l(-1) \quad \text{for } 1 \leq l \leq \tilde{L}(-1) \quad (204)$$

$$\bar{M}_l(0) = \bar{M}_l(-1) \quad \text{for } 1 \leq l \leq \tilde{L}(-1) \quad (205)$$

- 3) The repeated model parameters are used in all future processing wherever the current model parameters are required. This includes the synthesis of the current segment of speech as is described in Chapter 11.

14.7 Frame Muting

The Half-Rate decoder is required to mute in severe bit error environments for which $\varepsilon_R > 0.096$ or if 4 consecutive voice frames are determined to be invalid (muting would occur instead of performing the 4-th consecutive frame repeat). This procedure causes the Half-Rate decoder to squelch its output if reliable communications cannot be supported.

The recommended muting method is to first compute the update equations as listed in step (2) of the frame repeat process (see Clause 14.6). The decoder should then bypass the speech synthesis procedure described in Chapter 11, and alternately set the synthetic speech signal, $\tilde{s}(n)$ to random noise which is uniformly distributed over the interval $[-5,5]$. This technique provides for a small amount of “comfort noise”.

15 Half-Rate Spectral Amplitude Enhancement

The Half-Rate speech decoder attempts to improve the perceived quality of the synthesized speech by enhancing the spectral amplitudes. These enhanced spectral amplitudes are used throughout speech synthesis. The spectral amplitude enhancement is accomplished using the procedure described in Chapter 8.

Limited Use Only

16 Tone Frames

The Half-Rate vocoder supports the transmission of tone frames to allow transmission and regeneration of a tone signal through the vocoder. Supported tones include Dual Tone Multi-Frequency (DTMF) and KNOX tones, as well as certain single-frequency and call progress tones. The supported tone signals are shown in Table 19.

In order to pass tones (single or DTMF), the voice encoder should repeatedly detect whether the incoming signal $s(n)$ is either voice or a tone signal. If voice is detected, then the voice signal is encoded and transmitted as specified in the Section 13 of this document. However, if a tone is detected, then a special “Tone Frame” bit sequence is transmitted from Table 19 which specifies the Tone Index I_D and the amplitude of the tone which was detected.

Similarly, the voice decoder should check whether the received data for the frame contains a special Tone Frame. This is done by decoding the first six bits or $\tilde{u}_0$. If these 6 bits equal 63 as shown in the first line of Table 20, then the frame is a Tone Frame. Note this is equivalent to decoding the frame type from the quantizer value $\tilde{b}_0$ and checking that the frame type is set to “tone” (i.e. [126, 127]) as described in Clause 13.1. If the frame type is determined to be a Tone Frame, then the decoder decodes the tone index and amplitude and synthesizes a tone signal with the correct amplitude and frequency as its output.

Note that the format of the bit vectors is different for a Tone Frame compared to a Voice or Silence Frame. Consequently, for Tone Frames, the frame format is determined by Table 20. Table 12 and Tables 15 through Table 18 should be ignored.

The remaining clauses of this document discuss further details of the Half-Rate Vocoder operation in the presence of tones. Clause 16.1 addresses tone detection, Clause 16.2 discusses the bit format used to transmit Tone Frames and Clause 16.3 discusses tone regeneration.

16.1 Tone Detection

The Half-Rate Vocoder divides an input signal $s(n)$ into 20 ms segments and then analyzes each segment to extract the MBE model parameters. In order to support the required tone features, a separate tone detection stage shall be added to the voice encoder. Note: tone detection is an optional feature and not required for interoperability.

The tone detection algorithm shall analyze the input signal every 20 ms and determine whether the input signal for the current frame corresponds to voice or a tone. This is typically performed by analyzing the input signal $s(n)$ every 20 ms. Many standard techniques for tone detection can be found in the literature and the reader is referred elsewhere for more information on this subject. Regardless of the method, the algorithm used to perform tone detection shall determine whether the input signal is voice or a supported tone signal during each 20 ms voice frame. If a tone is detected it shall identify which tone is present according to the Tone Index shown in Table 19 and estimate the amplitude of the tone. Furthermore, if a single tone is detected it shall estimate the center frequency of the tone.

16.2 Tone Transmission

The Half-Rate Vocoder supports transmission of tone information through the use of a dedicated Tone Frame, where the bit format of a tone frame is shown in Table 20. Note that the first 6 bits of the bit vector $\hat{u}_0$ are always set equal to [1, 1,..., 1] (=decimal 63) for a Tone Frame. This equates to a value of $\hat{b}_0$ in the range $120 \leq \hat{b}_0 \leq 127$ and forces the two MSB's of $\hat{b}_1$ to 1. The result is that this allows the frame type to be positively identified as a Tone Frame at the decoder.

Tone Index I_D	Tone Type	Frequency 1 f_1	Frequency 2 f_2
0-4	Invalid	N/A	N/A
5	Single Frequency	156.25	N/A
6	Single Frequency	187.5	N/A
7 - 122	Single Frequency	31.25 I_D	N/A
123-127	Invalid	N/A	N/A
128	DTMF "0"	1336	941
129	DTMF "1"	1209	697
130	DTMF "2"	1336	697
131	DTMF "3"	1477	697
132	DTMF "4"	1209	770
133	DTMF "5"	1336	770
134	DTMF "6"	1477	770
135	DTMF "7"	1209	852
136	DTMF "8"	1336	852
137	DTMF "9"	1477	852
138	DTMF "A"	1633	697
139	DTMF "B"	1633	770
140	DTMF "C"	1633	852
141	DTMF "D"	1633	941
142	DTMF "*"	1209	941
143	DTMF "#"	1477	941
144	KNOX "0"	1162	820
145	KNOX "1"	1052	606
146	KNOX "2"	1162	606
147	KNOX "3"	1279	606
148	KNOX "4"	1052	672
149	KNOX "5"	1162	672
150	KNOX "6"	1279	672
151	KNOX "7"	1052	743
152	KNOX "8"	1162	743
153	KNOX "9"	1279	743
154	KNOX "A"	1430	606
155	KNOX "B"	1430	672
156	KNOX "C"	1430	743
157	KNOX "D"	1430	820
158	KNOX "*"	1052	820
159	KNOX "#"	1279	820
160	Call Progress	440	350
161	Call Progress	480	440
162	Call Progress	620	480
163	Call Progress	490	350
164-254	Invalid	N/A	N/A
255	Zero Amplitude	N/A	N/A

Table 19 Half-Rate Vocoder: Supported Tone Signals

Bit Vector	Value
$\hat{u}_0(11, 10, 9, 8, 7, 6)$	63
$\hat{u}_0(5, 4, 3, 2, 1, 0)$	$A_D(6, 5, 4, 3, 2, 1)$
$\hat{u}_1(11, 10, 9, 8, 7, 6, 5, 4)$	$I_D(7, 6, 5, 4, 3, 2, 1, 0)$
$\hat{u}_1(3, 2, 1, 0)$	$I_D(7, 6, 5, 4)$
$\hat{u}_2(10, 9, 8, 7)$	$I_D(3, 2, 1, 0)$
$\hat{u}_2(6, 5, 4, 3, 2, 1, 0)$	$I_D(7, 6, 5, 4, 3, 2, 1)$
$\hat{u}_3(13)$	$I_D(0)$
$\hat{u}_3(12, 11, 10, 9, 8, 7, 6, 5)$	$I_D(7, 6, 5, 4, 3, 2, 1, 0)$
$\hat{u}_3(4)$	$A_D(0)$
$\hat{u}_3(3, 2, 1, 0)$	0

Table 20 Tone Frame Format

The amplitude of the tone is represented through the log amplitude parameter A_D , which is a 7-bit parameter that spans the range $0 \leq A_D \leq 127$. This parameter is scaled such that $A_D = 127$ for a tone at the maximum sinusoidal input level of the A-to-D (+3.17 dBm0), and scaled such that $A_D = 0$ for a tone at -87.13 dBm0. This equates to an amplitude step size of 0.711 dB. The 6 MSBs of A_D are transmitted in the last 6 bits of $\hat{u}_0$, and the LSB of A_D is transmitted in bit 4 of $\hat{u}_3$.

The tone index I_D is repeated 4 times within the bit vectors $\hat{u}_1, \hat{u}_2, \hat{u}_3$ to provide an extra measure of redundancy. The last 4 bits of $\hat{u}_3$ are set to 0 for all Tone Frames.

Tone Frames are transmitted using the same FEC, bit modulation and interleaving as voice frames. Consequently, the encoder and decoder apply the same methods described in Clauses 14.2, 14.3, and 14.4 for all Tone Frames.

16.3 Tone Regeneration

In the event the decoder receives a Tone Frame with a valid tone index, then it shall synthesize a tone signal for the current frame using the decoded tone amplitude and frequency parameters. This synthesized tone signal is output by the decoder for the current frame instead of the voice signal described in Chapter 11. If the decoder receives a Tone Frame with $I_D = 255$, then the amplitude of the synthesized tone signal is set to zero, regardless of the received value of A_D .

If the decoder receives a Tone Frame with an invalid tone index, then the frame is considered an erasure frame and a frame repeat is performed as indicated in Clause 14.6.

Half-Rate Frame Type	Full-Rate Frame Type
voice	voice
erasure	invalid
silence	voice
tone	voice

Table 21 Frame Type Conversion

17 Parametric Rate Conversion

The Half-Rate vocoder and the Full-Rate vocoder described in this document share the same MBE model parameters for every 20 ms frame. As a result parametric rate conversion can be used to convert between Full-Rate and Half-Rate bit streams.

Parametric rate conversion is accomplished by decoding and reconstructing the MBE model parameters at the input rate and then quantizing and re-encoding these MBE parameters at the output rate. A major advantage of this method of performing rate conversion is that the signal is not converted back to speech as part of the conversion process (i.e. it does not use tandeming). Instead the MBE model parameters are used as the intermediate stage between bit rates, and the synthesis of speech from the MBE model parameters, and the analysis (i.e. extraction) of the model parameters from speech is not performed. This approach avoids the added delay and much of the distortion normally associated with tandeming.

The details of the method are straightforward. The input bit stream is de-interleaved, FEC decoding and the quantizer values $\tilde{b}_n$ are formed for each frame. The quantizer values are then used to reconstruct the MBE model parameters: $\tilde{L}$, $\tilde{\omega}_0$, $\tilde{v}_l$, and $\tilde{M}_l$. These model parameters are then re-quantized into new quantizer values $\hat{b}_m$ at the output rate. These quantizer values are used to form the new bit vectors, which are FEC encoded and interleaved to produce the output bit stream.

As shown in Table 14, the Half-Rate Vocoder identifies four separate frame types which may be received. In contrast the Full-Rate vocoder only supports voice frames and invalid frames ($\tilde{b}_0 \geq 208$). Conversion of Half-Rate frames to Full-Rate frames works as shown in Table 21. Conversion from Full-Rate frames to Half-Rate frames works in the same manner except only the first two lines of Table 21 are used.

To convert a Half-Rate erasure frame to a Full-Rate invalid frame, the value of $\tilde{b}_0$ in the Full-Rate frame is set to 240 and all other bit vectors can be set to zero since they are ignored at the receiver. Similarly to convert a Full-Rate invalid frame to a Half-Rate erasure frame, the value of $\tilde{b}_0$ in the Half-Rate frame is set to 120 and all other bit vectors can be set to zero.

To convert a Half-Rate silence frame to a Full-Rate voice frame, the reconstructed MBE model parameters: $\tilde{L}$, $\tilde{\omega}_0$, $\tilde{v}_l$, and $\tilde{M}_l$ are just quantized using the Full-Rate quantizer in the same manner as if the frame was a voice frame.

To convert a Half-Rate tone frame to a Full-Rate voice frame, an approximate set of MBE model parameters is first generated for the Tone Frame to provide a reasonable fit between the frequency components of the tone and the generated MBE spectral parameters. Annex T lists the tone frame

parameters, f_0 , l_1 , and l_2 corresponding to each possible value of the received tone index I_D . Using the value of f_0 , the MBE parameters $\tilde{\omega}_0$ and $\tilde{L}$ are generated as follows:

$$\tilde{\omega}_0 = \frac{2\pi}{8000} f_0 \quad (206)$$

$$\tilde{L} = \left\lfloor \frac{3812.5}{f_0} \right\rfloor \quad (207)$$

The tone frame parameters l_1 and l_2 from Annex T are then used to determine the voicing decisions $\tilde{v}_l$ and spectral amplitudes $\tilde{M}_l$ according to Equations (208) and (209), respectively.

$$\tilde{v}_l = \begin{cases} 1 & \text{if } l = l_1 \text{ or } l = l_2 \\ 0 & \text{otherwise} \end{cases} \quad (208)$$

$$\tilde{M}_l = \begin{cases} 16384 \cdot 10^{0.03555(A_D - 127)} & \text{if } l = l_1 \text{ or } l = l_2 \\ 0 & \text{otherwise} \end{cases} \quad (209)$$

Once the approximate set of MBE model parameters have been generated in this manner, then the frame is quantized using the Full-Rate quantizer in the same manner as if the frame was a voice frame.

Annex A Variable Initialization

<i>Variable</i>	<i>Initial Value</i>
P_{-1}	100
P_{-2}	100
$E_{-1}(P)$	0 or all P
$E_{-2}(P)$	0 for all P
$\xi_{\max}$	100000
$\tilde{\omega}_0(-1)$	0.02985π
$\tilde{M}_l(-1)$	1 for all l
$\bar{M}_l(-1)$	0 for all l
$\tilde{L}(-1)$	30
$\tilde{K}(-1)$	10
$\tilde{v}_k(-1)$	0 for all k
$\bar{v}_k(-1)$	0 for all k
ε_R	0.0
S_E	75000
$u(n)$	$u(-105) = 3147$
$\tilde{u}_w(n,-1)$	0 for all n
$\varphi_l(-1)$	0 for all l
$\psi_l(-1)$	0 for all l

Annex B Initial Pitch Estimation Window

n	$w_I(n)$	n	$w_I(n)$	n	$w_I(n)$	n	$w_I(n)$
-150	0.00270174	-110	0.02113326	-70	0.05359428	-30	0.08371301
-149	0.00295485	-109	0.02181198	-69	0.05446896	-29	0.08424166
-148	0.00321783	-108	0.02250030	-68	0.05534208	-28	0.08475496
-147	0.00349080	-107	0.02319806	-67	0.05621328	-27	0.08525265
-146	0.00377385	-106	0.02390508	-66	0.05708218	-26	0.08573449
-145	0.00406710	-105	0.02462121	-65	0.05794841	-25	0.08620022
-144	0.00437064	-104	0.02534627	-64	0.05881159	-24	0.08664962
-143	0.00468457	-103	0.02608007	-63	0.05967135	-23	0.08708246
-142	0.00500898	-102	0.02682242	-62	0.06052732	-22	0.08749853
-141	0.00534396	-101	0.02757312	-61	0.06137912	-21	0.08789761
-140	0.00568957	-100	0.02833197	-60	0.06222637	-20	0.08827951
-139	0.00604590	-99	0.02909874	-59	0.06306869	-19	0.08864402
-138	0.00641300	-98	0.02987323	-58	0.06390572	-18	0.08899098
-137	0.00679095	-97	0.03065520	-57	0.06473706	-17	0.08932019
-136	0.00717979	-96	0.03144442	-56	0.06556235	-16	0.08963150
-135	0.00757957	-95	0.03224064	-55	0.06638121	-15	0.08992474
-134	0.00799034	-94	0.03304362	-54	0.06719326	-14	0.09019977
-133	0.00841213	-93	0.03385311	-53	0.06799814	-13	0.09045644
-132	0.00884496	-92	0.03466884	-52	0.06879548	-12	0.09069464
-131	0.00928887	-91	0.03549055	-51	0.06958490	-11	0.09091422
-130	0.00974387	-90	0.03631796	-50	0.07036603	-10	0.09111509
-129	0.01020996	-89	0.03715079	-49	0.07113852	-9	0.09129714
-128	0.01068715	-88	0.03798876	-48	0.07190201	-8	0.09146028
-127	0.01117544	-87	0.03883158	-47	0.07265612	-7	0.09160442
-126	0.01167480	-86	0.03967895	-46	0.07340051	-6	0.09172950
-125	0.01218523	-85	0.04053057	-45	0.07413482	-5	0.09183544
-124	0.01270669	-84	0.04138613	-44	0.07485870	-4	0.09192219
-123	0.01323915	-83	0.04224531	-43	0.07557182	-3	0.09198971
-122	0.01378257	-82	0.04310781	-42	0.07627382	-2	0.09203796
-121	0.01433691	-81	0.04397329	-41	0.07696436	-1	0.09206692
-120	0.01490209	-80	0.04484143	-40	0.07764313	0	0.09207658
-119	0.01547807	-79	0.04571191	-39	0.07830978	1	0.09206692
-118	0.01606477	-78	0.04658437	-38	0.07896400	2	0.09203796
-117	0.01666211	-77	0.04745849	-37	0.07960546	3	0.09198971
-116	0.01727001	-76	0.04833392	-36	0.08023385	4	0.09192219
-115	0.01788837	-75	0.04921031	-35	0.08084887	5	0.09183544
-114	0.01851708	-74	0.05008731	-34	0.08145022	6	0.09172950
-113	0.01915605	-73	0.05096456	-33	0.08203759	7	0.09160442
-112	0.01980516	-72	0.05184172	-32	0.08261070	8	0.09146028
-111	0.02046427	-71	0.05271841	-31	0.08316926	9	0.09129714

n	$w_l(n)$	n	$w_l(n)$	n	$w_l(n)$	n	$w_l(n)$
10	0.09111509	50	0.07036603	90	0.03631796	130	0.00974387
11	0.09091422	51	0.06958490	91	0.03549055	131	0.00928887
12	0.09069464	52	0.06879548	92	0.03466884	132	0.00884496
13	0.09045644	53	0.06799814	93	0.03385311	133	0.00841213
14	0.09019977	54	0.06719326	94	0.03304362	134	0.00799034
15	0.08992474	55	0.06638121	95	0.03224064	135	0.00757957
16	0.08963150	56	0.06556235	96	0.03144442	136	0.00717979
17	0.08932019	57	0.06473706	97	0.03065520	137	0.00679095
18	0.08899098	58	0.06390572	98	0.02987323	138	0.00641300
19	0.08864402	59	0.06306869	99	0.02909874	139	0.00604590
20	0.08827951	60	0.06222637	100	0.02833197	140	0.00568957
21	0.08789761	61	0.06137912	101	0.02757312	141	0.00534396
22	0.08749853	62	0.06052732	102	0.02682242	142	0.00500898
23	0.08708246	63	0.05967135	103	0.02608007	143	0.00468457
24	0.08664962	64	0.05881159	104	0.02534627	144	0.00437064
25	0.08620022	65	0.05794841	105	0.02462121	145	0.00406710
26	0.08573449	66	0.05708218	106	0.02390508	146	0.00377385
27	0.08525265	67	0.05621328	107	0.02319806	147	0.00349080
28	0.08475496	68	0.05534208	108	0.02250030	148	0.00321783
29	0.08424166	69	0.05446896	109	0.02181198	149	0.00295485
30	0.08371301	70	0.05359428	110	0.02113326	150	0.00270174
31	0.08316926	71	0.05271841	111	0.02046427		
32	0.08261070	72	0.05184172	112	0.01980516		
33	0.08203759	73	0.05096456	113	0.01915605		
34	0.08145022	74	0.05008731	114	0.01851708		
35	0.08084887	75	0.04921031	115	0.01788837		
36	0.08023385	76	0.04833392	116	0.01727001		
37	0.07960546	77	0.04745849	117	0.01666211		
38	0.07896400	78	0.04658437	118	0.01606477		
39	0.07830978	79	0.04571191	119	0.01547807		
40	0.07764313	80	0.04484143	120	0.01490209		
41	0.07696436	81	0.04397329	121	0.01433691		
42	0.07627382	82	0.04310781	122	0.01378257		
43	0.07557182	83	0.04224531	123	0.01323915		
44	0.07485870	84	0.04138613	124	0.01270669		
45	0.07413482	85	0.04053057	125	0.01218523		
46	0.07340051	86	0.03967895	126	0.01167480		
47	0.07265612	87	0.03883158	127	0.01117544		
48	0.07190201	88	0.03798876	128	0.01068715		
49	0.07113852	89	0.03715079	129	0.01020996		

Annex C Pitch Refinement Window

n	$w_R(n)$	n	$w_R(n)$	n	$w_R(n)$	n	$w_R(n)$
-110	0.014873	-78	0.205355	-46	0.607067	-14	0.956477
-109	0.017397	-77	0.215294	-45	0.620807	-13	0.962377
-108	0.020102	-76	0.225466	-44	0.634490	-12	0.967867
-107	0.022995	-75	0.235869	-43	0.648105	-11	0.972940
-106	0.026081	-74	0.246497	-42	0.661638	-10	0.977592
-105	0.029365	-73	0.257347	-41	0.675076	-9	0.981817
-104	0.032852	-72	0.268413	-40	0.688407	-8	0.985610
-103	0.036546	-71	0.279689	-39	0.701616	-7	0.988967
-102	0.040451	-70	0.291171	-38	0.714692	-6	0.991884
-101	0.044573	-69	0.302851	-37	0.727621	-5	0.994358
-100	0.048915	-68	0.314724	-36	0.740390	-4	0.996386
-99	0.053482	-67	0.326782	-35	0.752986	-3	0.997966
-98	0.058277	-66	0.339018	-34	0.765397	-2	0.999096
-97	0.063303	-65	0.351425	-33	0.777610	-1	0.999774
-96	0.068563	-64	0.363994	-32	0.789612	0	1.000000
-95	0.074062	-63	0.376718	-31	0.801391	1	0.999774
-94	0.079801	-62	0.389588	-30	0.812935	2	0.999096
-93	0.085782	-61	0.402594	-29	0.824231	3	0.997966
-92	0.092009	-60	0.415727	-28	0.835267	4	0.996386
-91	0.098483	-59	0.428978	-27	0.846033	5	0.994358
-90	0.105205	-58	0.442337	-26	0.856516	6	0.991884
-89	0.112176	-57	0.455793	-25	0.866705	7	0.988967
-88	0.119398	-56	0.469336	-24	0.876589	8	0.985610
-87	0.126872	-55	0.482956	-23	0.886157	9	0.981817
-86	0.134596	-54	0.496640	-22	0.895400	10	0.977592
-85	0.142572	-53	0.510379	-21	0.904307	11	0.972940
-84	0.150799	-52	0.524160	-20	0.912868	12	0.967867
-83	0.159276	-51	0.537971	-19	0.921074	13	0.962377
-82	0.168001	-50	0.551802	-18	0.928917	14	0.956477
-81	0.176974	-49	0.565639	-17	0.936386	15	0.950174
-80	0.186192	-48	0.579470	-16	0.943474	16	0.943474
-79	0.195653	-47	0.593284	-15	0.950174	17	0.936386

n	$w_R(n)$	n	$w_R(n)$	n	$w_R(n)$
18	0.928917	50	0.551802	82	0.168001
19	0.921074	51	0.537971	83	0.159276
20	0.912868	52	0.524160	84	0.150799
21	0.904307	53	0.510379	85	0.142572
22	0.895400	54	0.496640	86	0.134596
23	0.886157	55	0.482956	87	0.126872
24	0.876589	56	0.469336	88	0.119398
25	0.866705	57	0.455793	89	0.112176
26	0.856516	58	0.442337	90	0.105205
27	0.846033	59	0.428978	91	0.098483
28	0.835267	60	0.415727	92	0.092009
29	0.824231	61	0.402594	93	0.085782
30	0.812935	62	0.389588	94	0.079801
31	0.801391	63	0.376718	95	0.074062
32	0.789612	64	0.363994	96	0.068563
33	0.777610	65	0.351425	97	0.063303
34	0.765397	66	0.339018	98	0.058277
35	0.752986	67	0.326782	99	0.053482
36	0.740390	68	0.314724	100	0.048915
37	0.727621	69	0.302851	101	0.044573
38	0.714692	70	0.291171	102	0.040451
39	0.701616	71	0.279689	103	0.036546
40	0.688407	72	0.268413	104	0.032852
41	0.675076	73	0.257347	105	0.029365
42	0.661638	74	0.246497	106	0.026081
43	0.648105	75	0.235869	107	0.022995
44	0.634490	76	0.225466	108	0.020102
45	0.620807	77	0.215294	109	0.017397
46	0.607067	78	0.205355	110	0.014873
47	0.593284	79	0.195653		
48	0.579470	80	0.186192		
49	0.565639	81	0.176974		

Annex D FIR Low Pass Filter

n	$h_{LPF}(n)$
-10	-0.002898
-9	-0.002831
-8	0.005666
-7	0.016601
-6	0.008800
-5	-0.026955
-4	-0.055990
-3	-0.015116
-2	0.118754
-1	0.278990
0	0.351338
1	0.278990
2	0.118754
3	-0.015116
4	-0.055990
5	-0.026955
6	0.008800
7	0.016601
8	0.005666
9	-0.002831
10	-0.002898

Annex E Gain Quantizer Levels

$\hat{b}_2$	Quantizer Level	$\hat{b}_2$	Quantizer Level
0	-2.842205	32	2.653909
1	-2.694235	33	2.780654
2	-2.558260	34	2.925355
3	-2.382850	35	3.076390
4	-2.221042	36	3.220825
5	-2.095574	37	3.402869
6	-1.980845	38	3.585096
7	-1.836058	39	3.784606
8	-1.645556	40	3.955521
9	-1.417658	41	4.155636
10	-1.261301	42	4.314009
11	-1.125631	43	4.444150
12	-0.958207	44	4.577542
13	-0.781591	45	4.735552
14	-0.555837	46	4.909493
15	-0.346976	47	5.085264
16	-0.147249	48	5.254767
17	0.027755	49	5.411894
18	0.211495	50	5.568094
19	0.388380	51	5.738523
20	0.552873	52	5.919215
21	0.737223	53	6.087701
22	0.932197	54	6.280685
23	1.139032	55	6.464201
24	1.320955	56	6.647736
25	1.483433	57	6.834672
26	1.648297	58	7.022583
27	1.801447	59	7.211777
28	1.942731	60	7.471016
29	2.118613	61	7.738948
30	2.321486	62	8.124863
31	2.504443	63	8.695827

Annex F Bit Allocation and Step Size for Transformed Gain Vector

$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$	$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$
9	$\hat{G}_2$	$\hat{b}_3$	10	0.003100	15	$\hat{G}_2$	$\hat{b}_3$	7	0.024800
9	$\hat{G}_3$	$\hat{b}_4$	9	0.004020	15	$\hat{G}_3$	$\hat{b}_4$	6	0.030150
9	$\hat{G}_4$	$\hat{b}_5$	9	0.003360	15	$\hat{G}_4$	$\hat{b}_5$	6	0.025200
9	$\hat{G}_5$	$\hat{b}_6$	9	0.002900	15	$\hat{G}_5$	$\hat{b}_6$	6	0.021750
9	$\hat{G}_6$	$\hat{b}_7$	9	0.002640	15	$\hat{G}_6$	$\hat{b}_7$	5	0.036960
10	$\hat{G}_2$	$\hat{b}_3$	9	0.006200	16	$\hat{G}_2$	$\hat{b}_3$	6	0.046500
10	$\hat{G}_3$	$\hat{b}_4$	9	0.004020	16	$\hat{G}_3$	$\hat{b}_4$	6	0.030150
10	$\hat{G}_4$	$\hat{b}_5$	8	0.006720	16	$\hat{G}_4$	$\hat{b}_5$	6	0.025200
10	$\hat{G}_5$	$\hat{b}_6$	8	0.005800	16	$\hat{G}_5$	$\hat{b}_6$	5	0.040600
10	$\hat{G}_6$	$\hat{b}_7$	8	0.005280	16	$\hat{G}_6$	$\hat{b}_7$	5	0.036960
11	$\hat{G}_2$	$\hat{b}_3$	8	0.012400	17	$\hat{G}_2$	$\hat{b}_3$	6	0.046500
11	$\hat{G}_3$	$\hat{b}_4$	8	0.008040	17	$\hat{G}_3$	$\hat{b}_4$	6	0.030150
11	$\hat{G}_4$	$\hat{b}_5$	8	0.006720	17	$\hat{G}_4$	$\hat{b}_5$	5	0.047040
11	$\hat{G}_5$	$\hat{b}_6$	7	0.011600	17	$\hat{G}_5$	$\hat{b}_6$	5	0.040600
11	$\hat{G}_6$	$\hat{b}_7$	7	0.010560	17	$\hat{G}_6$	$\hat{b}_7$	5	0.036960
12	$\hat{G}_2$	$\hat{b}_3$	8	0.012400	18	$\hat{G}_2$	$\hat{b}_3$	6	0.046500
12	$\hat{G}_3$	$\hat{b}_4$	7	0.016080	18	$\hat{G}_3$	$\hat{b}_4$	5	0.056280
12	$\hat{G}_4$	$\hat{b}_5$	7	0.013440	18	$\hat{G}_4$	$\hat{b}_5$	5	0.047040
12	$\hat{G}_5$	$\hat{b}_6$	7	0.011600	18	$\hat{G}_5$	$\hat{b}_6$	5	0.040600
12	$\hat{G}_6$	$\hat{b}_7$	7	0.010560	18	$\hat{G}_6$	$\hat{b}_7$	5	0.036960
13	$\hat{G}_2$	$\hat{b}_3$	7	0.024800	19	$\hat{G}_2$	$\hat{b}_3$	6	0.046500
13	$\hat{G}_3$	$\hat{b}_4$	7	0.016080	19	$\hat{G}_3$	$\hat{b}_4$	5	0.056280
13	$\hat{G}_4$	$\hat{b}_5$	7	0.013440	19	$\hat{G}_4$	$\hat{b}_5$	5	0.047040
13	$\hat{G}_5$	$\hat{b}_6$	6	0.021750	19	$\hat{G}_5$	$\hat{b}_6$	4	0.058000
13	$\hat{G}_6$	$\hat{b}_7$	6	0.019800	19	$\hat{G}_6$	$\hat{b}_7$	4	0.052800
14	$\hat{G}_2$	$\hat{b}_3$	7	0.024800	20	$\hat{G}_2$	$\hat{b}_3$	6	0.046500
14	$\hat{G}_3$	$\hat{b}_4$	6	0.030150	20	$\hat{G}_3$	$\hat{b}_4$	5	0.056280
14	$\hat{G}_4$	$\hat{b}_5$	6	0.025200	20	$\hat{G}_4$	$\hat{b}_5$	5	0.047040
14	$\hat{G}_5$	$\hat{b}_6$	6	0.021750	20	$\hat{G}_5$	$\hat{b}_6$	4	0.058000
14	$\hat{G}_6$	$\hat{b}_7$	6	0.019800	20	$\hat{G}_6$	$\hat{b}_7$	4	0.052800

$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$	$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$
21	$\hat{G}_2$	$\hat{b}_3$	5	0.086800	27	$\hat{G}_2$	$\hat{b}_3$	5	0.086800
21	$\hat{G}_3$	$\hat{b}_4$	5	0.056280	27	$\hat{G}_3$	$\hat{b}_4$	4	0.080400
21	$\hat{G}_4$	$\hat{b}_5$	5	0.047040	27	$\hat{G}_4$	$\hat{b}_5$	4	0.067200
21	$\hat{G}_5$	$\hat{b}_6$	4	0.058000	27	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
21	$\hat{G}_6$	$\hat{b}_7$	4	0.052800	27	$\hat{G}_6$	$\hat{b}_7$	3	0.085800
22	$\hat{G}_2$	$\hat{b}_3$	5	0.086800	28	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
22	$\hat{G}_3$	$\hat{b}_4$	5	0.056280	28	$\hat{G}_3$	$\hat{b}_4$	4	0.080400
22	$\hat{G}_4$	$\hat{b}_5$	4	0.067200	28	$\hat{G}_4$	$\hat{b}_5$	4	0.067200
22	$\hat{G}_5$	$\hat{b}_6$	4	0.058000	28	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
22	$\hat{G}_6$	$\hat{b}_7$	4	0.052800	28	$\hat{G}_6$	$\hat{b}_7$	3	0.085800
23	$\hat{G}_2$	$\hat{b}_3$	5	0.086800	29	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
23	$\hat{G}_3$	$\hat{b}_4$	4	0.080400	29	$\hat{G}_3$	$\hat{b}_4$	4	0.080400
23	$\hat{G}_4$	$\hat{b}_5$	4	0.067200	29	$\hat{G}_4$	$\hat{b}_5$	4	0.067200
23	$\hat{G}_5$	$\hat{b}_6$	4	0.058000	29	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
23	$\hat{G}_6$	$\hat{b}_7$	4	0.052800	29	$\hat{G}_6$	$\hat{b}_7$	3	0.085800
24	$\hat{G}_2$	$\hat{b}_3$	5	0.086800	30	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
24	$\hat{G}_3$	$\hat{b}_4$	4	0.080400	30	$\hat{G}_3$	$\hat{b}_4$	4	0.080400
24	$\hat{G}_4$	$\hat{b}_5$	4	0.067200	30	$\hat{G}_4$	$\hat{b}_5$	4	0.067200
24	$\hat{G}_5$	$\hat{b}_6$	4	0.058000	30	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
24	$\hat{G}_6$	$\hat{b}_7$	4	0.052800	30	$\hat{G}_6$	$\hat{b}_7$	3	0.085800
25	$\hat{G}_2$	$\hat{b}_3$	5	0.086800	31	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
25	$\hat{G}_3$	$\hat{b}_4$	4	0.080400	31	$\hat{G}_3$	$\hat{b}_4$	4	0.080400
25	$\hat{G}_4$	$\hat{b}_5$	4	0.067200	31	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
25	$\hat{G}_5$	$\hat{b}_6$	4	0.058000	31	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
25	$\hat{G}_6$	$\hat{b}_7$	3	0.085800	31	$\hat{G}_6$	$\hat{b}_7$	3	0.085800
26	$\hat{G}_2$	$\hat{b}_3$	5	0.086800	32	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
26	$\hat{G}_3$	$\hat{b}_4$	4	0.080400	32	$\hat{G}_3$	$\hat{b}_4$	4	0.080400
26	$\hat{G}_4$	$\hat{b}_5$	4	0.067200	32	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
26	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	32	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
26	$\hat{G}_6$	$\hat{b}_7$	3	0.085800	32	$\hat{G}_6$	$\hat{b}_7$	3	0.085800

$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$	$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$
33	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	39	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
33	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	39	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
33	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	39	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
33	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	39	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
33	$\hat{G}_6$	$\hat{b}_7$	3	0.085800	39	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
34	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	40	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
34	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	40	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
34	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	40	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
34	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	40	$\hat{G}_5$	$\hat{b}_6$	3	0.094250
34	$\hat{G}_6$	$\hat{b}_7$	3	0.085800	40	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
35	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	41	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
35	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	41	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
35	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	41	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
35	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	41	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
35	$\hat{G}_6$	$\hat{b}_7$	3	0.085800	41	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
36	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	42	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
36	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	42	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
36	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	42	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
36	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	42	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
36	$\hat{G}_6$	$\hat{b}_7$	3	0.085800	42	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
37	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	43	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
37	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	43	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
37	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	43	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
37	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	43	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
37	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	43	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
38	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	44	$\hat{G}_2$	$\hat{b}_3$	4	0.124000
38	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	44	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
38	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	44	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
38	$\hat{G}_5$	$\hat{b}_6$	3	0.094250	44	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
38	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	44	$\hat{G}_6$	$\hat{b}_7$	2	0.112200

$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$	$\hat{L}$	$\hat{G}_{m-1}$	$\hat{b}_m$	$\hat{B}_m$	$\hat{\Delta}_m$
45	$\hat{G}_2$	$\hat{b}_3$	4	0.124000	51	$\hat{G}_2$	$\hat{b}_3$	3	0.201500
45	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	51	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
45	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	51	$\hat{G}_4$	$\hat{b}_5$	3	0.109200
45	$\hat{G}_5$	$\hat{b}_6$	2	0.123250	51	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
45	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	51	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
46	$\hat{G}_2$	$\hat{b}_3$	3	0.201500	52	$\hat{G}_2$	$\hat{b}_3$	3	0.201500
46	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	52	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
46	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	52	$\hat{G}_4$	$\hat{b}_5$	2	0.142800
46	$\hat{G}_5$	$\hat{b}_6$	2	0.123250	52	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
46	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	52	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
47	$\hat{G}_2$	$\hat{b}_3$	3	0.201500	53	$\hat{G}_2$	$\hat{b}_3$	3	0.201500
47	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	53	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
47	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	53	$\hat{G}_4$	$\hat{b}_5$	2	0.142800
47	$\hat{G}_5$	$\hat{b}_6$	2	0.123250	53	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
47	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	53	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
48	$\hat{G}_2$	$\hat{b}_3$	3	0.201500	54	$\hat{G}_2$	$\hat{b}_3$	3	0.201500
48	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	54	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
48	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	54	$\hat{G}_4$	$\hat{b}_5$	2	0.142800
48	$\hat{G}_5$	$\hat{b}_6$	2	0.123250	54	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
48	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	54	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
49	$\hat{G}_2$	$\hat{b}_3$	3	0.201500	55	$\hat{G}_2$	$\hat{b}_3$	3	0.201500
49	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	55	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
49	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	55	$\hat{G}_4$	$\hat{b}_5$	2	0.142800
49	$\hat{G}_5$	$\hat{b}_6$	2	0.123250	55	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
49	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	55	$\hat{G}_6$	$\hat{b}_7$	2	0.112200
50	$\hat{G}_2$	$\hat{b}_3$	3	0.201500	56	$\hat{G}_2$	$\hat{b}_3$	3	0.201500
50	$\hat{G}_3$	$\hat{b}_4$	3	0.130650	56	$\hat{G}_3$	$\hat{b}_4$	3	0.130650
50	$\hat{G}_4$	$\hat{b}_5$	3	0.109200	56	$\hat{G}_4$	$\hat{b}_5$	2	0.142800
50	$\hat{G}_5$	$\hat{b}_6$	2	0.123250	56	$\hat{G}_5$	$\hat{b}_6$	2	0.123250
50	$\hat{G}_6$	$\hat{b}_7$	2	0.112200	56	$\hat{G}_6$	$\hat{b}_7$	2	0.112200

Annex G Bit Allocation for Higher Order DCT Coefficients

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
9	$C_{4,2}$	b_8	9	15	$C_{1,2}$	b_8	6	18	$C_{1,2}$	b_8	5
9	$C_{5,2}$	b_9	8	15	$C_{2,2}$	b_9	7	18	$C_{1,3}$	b_9	4
9	$C_{6,2}$	b_{10}	7	15	$C_{3,2}$	b_{10}	5	18	$C_{2,2}$	b_{10}	5
				15	$C_{4,2}$	b_{11}	4	18	$C_{2,3}$	b_{11}	5
10	$C_{3,2}$	b_8	9	15	$C_{4,3}$	b_{12}	4	18	$C_{3,2}$	b_{12}	4
10	$C_{4,2}$	b_9	7	15	$C_{5,2}$	b_{13}	3	18	$C_{3,3}$	b_{13}	3
10	$C_{5,2}$	b_{10}	6	15	$C_{5,3}$	b_{14}	3	18	$C_{4,2}$	b_{14}	3
10	$C_{6,2}$	b_{11}	5	15	$C_{6,2}$	b_{15}	3	18	$C_{4,3}$	b_{15}	3
				15	$C_{6,3}$	b_{16}	3	18	$C_{5,2}$	b_{16}	3
11	$C_{2,2}$	b_8	9					18	$C_{5,3}$	b_{17}	2
11	$C_{3,2}$	b_9	7	16	$C_{1,2}$	b_8	6	18	$C_{6,2}$	b_{18}	2
11	$C_{4,2}$	b_{10}	6	16	$C_{2,2}$	b_9	6	18	$C_{6,3}$	b_{19}	2
11	$C_{5,2}$	b_{11}	5	16	$C_{3,2}$	b_{10}	5				
11	$C_{6,2}$	b_{12}	4	16	$C_{3,3}$	b_{11}	4	19	$C_{1,2}$	b_8	5
				16	$C_{4,2}$	b_{12}	4	19	$C_{1,3}$	b_9	4
12	$C_{1,2}$	b_8	8	16	$C_{4,3}$	b_{13}	3	19	$C_{2,2}$	b_{10}	5
12	$C_{2,2}$	b_9	7	16	$C_{5,2}$	b_{14}	3	19	$C_{2,3}$	b_{11}	4
12	$C_{3,2}$	b_{10}	6	16	$C_{5,3}$	b_{15}	3	19	$C_{3,2}$	b_{12}	4
12	$C_{4,2}$	b_{11}	5	16	$C_{6,2}$	b_{16}	3	19	$C_{3,3}$	b_{13}	3
12	$C_{5,2}$	b_{12}	4	16	$C_{6,3}$	b_{17}	2	19	$C_{4,2}$	b_{14}	3
12	$C_{6,2}$	b_{13}	3					19	$C_{4,3}$	b_{15}	3
				17	$C_{1,2}$	b_8	5	19	$C_{5,2}$	b_{16}	3
13	$C_{1,2}$	b_8	7	17	$C_{2,2}$	b_9	5	19	$C_{5,3}$	b_{17}	2
13	$C_{2,2}$	b_9	7	17	$C_{2,3}$	b_{10}	5	19	$C_{6,2}$	b_{18}	3
13	$C_{3,2}$	b_{10}	6	17	$C_{3,2}$	b_{11}	4	19	$C_{6,3}$	b_{19}	2
13	$C_{4,2}$	b_{11}	5	17	$C_{3,3}$	b_{12}	4	19	$C_{6,4}$	b_{20}	1
13	$C_{5,2}$	b_{12}	4	17	$C_{4,2}$	b_{13}	4				
13	$C_{6,2}$	b_{13}	3	17	$C_{4,3}$	b_{14}	3				
13	$C_{6,3}$	b_{14}	3	17	$C_{5,2}$	b_{15}	3				
				17	$C_{5,3}$	b_{16}	2				
14	$C_{1,2}$	b_8	7	17	$C_{6,2}$	b_{17}	3				
14	$C_{2,2}$	b_9	7	17	$C_{6,3}$	b_{18}	2				
14	$C_{3,2}$	b_{10}	5								
14	$C_{4,2}$	b_{11}	4								
14	$C_{5,2}$	b_{12}	4								
14	$C_{5,3}$	b_{13}	3								
14	$C_{6,2}$	b_{14}	4								
14	$C_{6,3}$	b_{15}	3								

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
20	$C_{1,2}$	b_8	5	22	$C_{1,2}$	b_8	4	24	$C_{1,2}$	b_8	4
20	$C_{1,3}$	b_9	4	22	$C_{1,3}$	b_9	4	24	$C_{1,3}$	b_9	3
20	$C_{2,2}$	b_{10}	5	22	$C_{2,2}$	b_{10}	4	24	$C_{1,4}$	b_{10}	3
20	$C_{2,3}$	b_{11}	4	22	$C_{2,3}$	b_{11}	4	24	$C_{2,2}$	b_{11}	4
20	$C_{3,2}$	b_{12}	4	22	$C_{3,2}$	b_{12}	4	24	$C_{2,3}$	b_{12}	3
20	$C_{3,3}$	b_{13}	3	22	$C_{3,3}$	b_{13}	3	24	$C_{2,4}$	b_{13}	3
20	$C_{4,2}$	b_{14}	3	22	$C_{3,4}$	b_{14}	2	24	$C_{3,2}$	b_{14}	3
20	$C_{4,3}$	b_{15}	2	22	$C_{4,2}$	b_{15}	3	24	$C_{3,3}$	b_{15}	3
20	$C_{5,2}$	b_{16}	3	22	$C_{4,3}$	b_{16}	2	24	$C_{3,4}$	b_{16}	2
20	$C_{5,3}$	b_{17}	2	22	$C_{4,4}$	b_{17}	2	24	$C_{4,2}$	b_{17}	3
20	$C_{5,4}$	b_{18}	1	22	$C_{5,2}$	b_{18}	3	24	$C_{4,3}$	b_{18}	2
20	$C_{6,2}$	b_{19}	3	22	$C_{5,3}$	b_{19}	2	24	$C_{4,4}$	b_{19}	1
20	$C_{6,3}$	b_{20}	2	22	$C_{5,4}$	b_{20}	1	24	$C_{5,2}$	b_{20}	2
20	$C_{6,4}$	b_{21}	1	22	$C_{6,2}$	b_{21}	2	24	$C_{5,3}$	b_{21}	2
				22	$C_{6,3}$	b_{22}	2	24	$C_{5,4}$	b_{22}	1
				22	$C_{6,4}$	b_{23}	1	24	$C_{6,2}$	b_{23}	2
21	$C_{1,2}$	b_8	4					24	$C_{6,3}$	b_{24}	2
21	$C_{1,3}$	b_9	4	23	$C_{1,2}$	b_8	4	24	$C_{6,4}$	b_{25}	1
21	$C_{2,2}$	b_{10}	5	23	$C_{1,3}$	b_9	3				
21	$C_{2,3}$	b_{11}	4	23	$C_{2,2}$	b_{10}	4	25	$C_{1,2}$	b_8	4
21	$C_{3,2}$	b_{12}	4	23	$C_{2,3}$	b_{11}	4	25	$C_{1,3}$	b_9	3
21	$C_{3,3}$	b_{13}	3	23	$C_{2,4}$	b_{12}	3	25	$C_{1,4}$	b_{10}	3
21	$C_{4,2}$	b_{14}	3	23	$C_{3,2}$	b_{13}	4	25	$C_{2,2}$	b_{11}	4
21	$C_{4,3}$	b_{15}	2	23	$C_{3,3}$	b_{14}	3	25	$C_{2,3}$	b_{12}	3
21	$C_{4,4}$	b_{16}	2	23	$C_{3,4}$	b_{15}	2	25	$C_{2,4}$	b_{13}	3
21	$C_{5,2}$	b_{17}	3	23	$C_{4,2}$	b_{16}	3	25	$C_{3,2}$	b_{14}	3
21	$C_{5,3}$	b_{18}	2	23	$C_{4,3}$	b_{17}	2	25	$C_{3,3}$	b_{15}	3
21	$C_{5,4}$	b_{19}	1	23	$C_{4,4}$	b_{18}	2	25	$C_{3,4}$	b_{16}	2
21	$C_{6,2}$	b_{20}	3	23	$C_{5,2}$	b_{19}	2	25	$C_{4,2}$	b_{17}	3
21	$C_{6,3}$	b_{21}	2	23	$C_{5,3}$	b_{20}	2	25	$C_{4,3}$	b_{18}	2
21	$C_{6,4}$	b_{22}	1	23	$C_{5,4}$	b_{21}	1	25	$C_{4,4}$	b_{19}	1
				23	$C_{6,2}$	b_{22}	2	25	$C_{5,2}$	b_{20}	2
				23	$C_{6,3}$	b_{23}	2	25	$C_{5,3}$	b_{21}	2
				23	$C_{6,4}$	b_{24}	1	25	$C_{5,4}$	b_{22}	1
								25	$C_{6,2}$	b_{23}	2
								25	$C_{6,3}$	b_{24}	1
								25	$C_{6,4}$	b_{25}	1
								25	$C_{6,5}$	b_{26}	1

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
26	$C_{1,2}$	b_8	4	27	$C_{1,2}$	b_8	4	28	$C_{1,2}$	b_8	4
26	$C_{1,3}$	b_9	3	27	$C_{1,3}$	b_9	3	28	$C_{1,3}$	b_9	3
26	$C_{1,4}$	b_{10}	3	27	$C_{1,4}$	b_{10}	2	28	$C_{1,4}$	b_{10}	2
26	$C_{2,2}$	b_{11}	4	27	$C_{2,2}$	b_{11}	4	28	$C_{2,2}$	b_{11}	4
26	$C_{2,3}$	b_{12}	3	27	$C_{2,3}$	b_{12}	3	28	$C_{2,3}$	b_{12}	3
26	$C_{2,4}$	b_{13}	3	27	$C_{2,4}$	b_{13}	2	28	$C_{2,4}$	b_{13}	2
26	$C_{3,2}$	b_{14}	3	27	$C_{3,2}$	b_{14}	3	28	$C_{3,2}$	b_{14}	3
26	$C_{3,3}$	b_{15}	2	27	$C_{3,3}$	b_{15}	2	28	$C_{3,3}$	b_{15}	2
26	$C_{3,4}$	b_{16}	2	27	$C_{3,4}$	b_{16}	2	28	$C_{3,4}$	b_{16}	2
26	$C_{4,2}$	b_{17}	3	27	$C_{4,2}$	b_{17}	3	28	$C_{3,5}$	b_{17}	2
26	$C_{4,3}$	b_{18}	2	27	$C_{4,3}$	b_{18}	2	28	$C_{4,2}$	b_{18}	3
26	$C_{4,4}$	b_{19}	1	27	$C_{4,4}$	b_{19}	2	28	$C_{4,3}$	b_{19}	2
26	$C_{5,2}$	b_{20}	2	27	$C_{4,5}$	b_{20}	1	28	$C_{4,4}$	b_{20}	1
26	$C_{5,3}$	b_{21}	2	27	$C_{5,2}$	b_{21}	2	28	$C_{4,5}$	b_{21}	1
26	$C_{5,4}$	b_{22}	1	27	$C_{5,3}$	b_{22}	2	28	$C_{5,2}$	b_{22}	2
26	$C_{5,5}$	b_{23}	1	27	$C_{5,4}$	b_{23}	1	28	$C_{5,3}$	b_{23}	2
26	$C_{6,2}$	b_{24}	2	27	$C_{5,5}$	b_{24}	1	28	$C_{5,4}$	b_{24}	1
26	$C_{6,3}$	b_{25}	2	27	$C_{6,2}$	b_{25}	2	28	$C_{5,5}$	b_{25}	1
26	$C_{6,4}$	b_{26}	1	27	$C_{6,3}$	b_{26}	2	28	$C_{6,2}$	b_{26}	2
26	$C_{6,5}$	b_{27}	1	27	$C_{6,4}$	b_{27}	1	28	$C_{6,3}$	b_{27}	1
				27	$C_{6,5}$	b_{28}	1	28	$C_{6,4}$	b_{28}	1
								28	$C_{6,5}$	b_{29}	1

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
29	$C_{1,2}$	b_8	3	30	$C_{1,2}$	b_8	3	31	$C_{1,2}$	b_8	3
29	$C_{1,3}$	b_9	3	30	$C_{1,3}$	b_9	3	31	$C_{1,3}$	b_9	3
29	$C_{1,4}$	b_{10}	2	30	$C_{1,4}$	b_{10}	2	31	$C_{1,4}$	b_{10}	2
29	$C_{2,2}$	b_{11}	4	30	$C_{1,5}$	b_{11}	2	31	$C_{1,5}$	b_{11}	2
29	$C_{2,3}$	b_{12}	3	30	$C_{2,2}$	b_{12}	3	31	$C_{2,2}$	b_{12}	3
29	$C_{2,4}$	b_{13}	2	30	$C_{2,3}$	b_{13}	3	31	$C_{2,3}$	b_{13}	3
29	$C_{2,5}$	b_{14}	2	30	$C_{2,4}$	b_{14}	2	31	$C_{2,4}$	b_{14}	2
29	$C_{3,2}$	b_{15}	3	30	$C_{2,5}$	b_{15}	2	31	$C_{2,5}$	b_{15}	2
29	$C_{3,3}$	b_{16}	2	30	$C_{3,2}$	b_{16}	3	31	$C_{3,2}$	b_{16}	3
29	$C_{3,4}$	b_{17}	2	30	$C_{3,3}$	b_{17}	2	31	$C_{3,3}$	b_{17}	2
29	$C_{3,5}$	b_{18}	2	30	$C_{3,4}$	b_{18}	2	31	$C_{3,4}$	b_{18}	2
29	$C_{4,2}$	b_{19}	3	30	$C_{3,5}$	b_{19}	1	31	$C_{3,5}$	b_{19}	1
29	$C_{4,3}$	b_{20}	2	30	$C_{4,2}$	b_{20}	3	31	$C_{4,2}$	b_{20}	2
29	$C_{4,4}$	b_{21}	1	30	$C_{4,3}$	b_{21}	2	31	$C_{4,3}$	b_{21}	2
29	$C_{4,5}$	b_{22}	1	30	$C_{4,4}$	b_{22}	1	31	$C_{4,4}$	b_{22}	1
29	$C_{5,2}$	b_{23}	2	30	$C_{4,5}$	b_{23}	1	31	$C_{4,5}$	b_{23}	1
29	$C_{5,3}$	b_{24}	1	30	$C_{5,2}$	b_{24}	2	31	$C_{5,2}$	b_{24}	2
29	$C_{5,4}$	b_{25}	1	30	$C_{5,3}$	b_{25}	1	31	$C_{5,3}$	b_{25}	1
29	$C_{5,5}$	b_{26}	1	30	$C_{5,4}$	b_{26}	1	31	$C_{5,4}$	b_{26}	1
29	$C_{6,2}$	b_{27}	2	30	$C_{5,5}$	b_{27}	1	31	$C_{5,5}$	b_{27}	1
29	$C_{6,3}$	b_{28}	1	30	$C_{6,2}$	b_{28}	2	31	$C_{6,2}$	b_{28}	2
29	$C_{6,4}$	b_{29}	1	30	$C_{6,3}$	b_{29}	1	31	$C_{6,3}$	b_{29}	1
29	$C_{6,5}$	b_{30}	1	30	$C_{6,4}$	b_{30}	1	31	$C_{6,4}$	b_{30}	1
				30	$C_{6,5}$	b_{31}	1	31	$C_{6,5}$	b_{31}	1
								31	$C_{6,6}$	b_{32}	1

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
32	$C_{1,2}$	b_8	3	33	$C_{1,2}$	b_8	3	34	$C_{1,2}$	b_8	3
32	$C_{1,3}$	b_9	3	33	$C_{1,3}$	b_9	3	34	$C_{1,3}$	b_9	2
32	$C_{1,4}$	b_{10}	2	33	$C_{1,4}$	b_{10}	2	34	$C_{1,4}$	b_{10}	2
32	$C_{1,5}$	b_{11}	2	33	$C_{1,5}$	b_{11}	2	34	$C_{1,5}$	b_{11}	2
32	$C_{2,2}$	b_{12}	3	33	$C_{2,2}$	b_{12}	3	34	$C_{2,2}$	b_{12}	3
32	$C_{2,3}$	b_{13}	3	33	$C_{2,3}$	b_{13}	3	34	$C_{2,3}$	b_{13}	2
32	$C_{2,4}$	b_{14}	2	33	$C_{2,4}$	b_{14}	2	34	$C_{2,4}$	b_{14}	2
32	$C_{2,5}$	b_{15}	2	33	$C_{2,5}$	b_{15}	2	34	$C_{2,5}$	b_{15}	2
32	$C_{3,2}$	b_{16}	3	33	$C_{3,2}$	b_{16}	3	34	$C_{3,2}$	b_{16}	3
32	$C_{3,3}$	b_{17}	2	33	$C_{3,3}$	b_{17}	2	34	$C_{3,3}$	b_{17}	2
32	$C_{3,4}$	b_{18}	2	33	$C_{3,4}$	b_{18}	1	34	$C_{3,4}$	b_{18}	2
32	$C_{3,5}$	b_{19}	1	33	$C_{3,5}$	b_{19}	1	34	$C_{3,5}$	b_{19}	1
32	$C_{4,2}$	b_{20}	2	33	$C_{4,2}$	b_{20}	2	34	$C_{3,6}$	b_{20}	1
32	$C_{4,3}$	b_{21}	2	33	$C_{4,3}$	b_{21}	2	34	$C_{4,2}$	b_{21}	2
32	$C_{4,4}$	b_{22}	1	33	$C_{4,4}$	b_{22}	1	34	$C_{4,3}$	b_{22}	2
32	$C_{4,5}$	b_{23}	1	33	$C_{4,5}$	b_{23}	1	34	$C_{4,4}$	b_{23}	1
32	$C_{5,2}$	b_{24}	2	33	$C_{4,6}$	b_{24}	1	34	$C_{4,5}$	b_{24}	1
32	$C_{5,3}$	b_{25}	1	33	$C_{5,2}$	b_{25}	2	34	$C_{4,6}$	b_{25}	1
32	$C_{5,4}$	b_{26}	1	33	$C_{5,3}$	b_{26}	1	34	$C_{5,2}$	b_{26}	2
32	$C_{5,5}$	b_{27}	1	33	$C_{5,4}$	b_{27}	1	34	$C_{5,3}$	b_{27}	1
32	$C_{5,6}$	b_{28}	1	33	$C_{5,5}$	b_{28}	1	34	$C_{5,4}$	b_{28}	1
32	$C_{6,2}$	b_{29}	2	33	$C_{5,6}$	b_{29}	1	34	$C_{5,5}$	b_{29}	1
32	$C_{6,3}$	b_{30}	1	33	$C_{6,2}$	b_{30}	2	34	$C_{5,6}$	b_{30}	1
32	$C_{6,4}$	b_{31}	1	33	$C_{6,3}$	b_{31}	1	34	$C_{6,2}$	b_{31}	2
32	$C_{6,5}$	b_{32}	1	33	$C_{6,4}$	b_{32}	1	34	$C_{6,3}$	b_{32}	1
32	$C_{6,6}$	b_{33}	0	33	$C_{6,5}$	b_{33}	1	34	$C_{6,4}$	b_{33}	1
				33	$C_{6,6}$	b_{34}	1	34	$C_{6,5}$	b_{34}	1
								34	$C_{6,6}$	b_{35}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
35	$C_{1,2}$	b_8	3	36	$C_{1,2}$	b_8	3	37	$C_{1,2}$	b_8	3
35	$C_{1,3}$	b_9	2	36	$C_{1,3}$	b_9	2	37	$C_{1,3}$	b_9	2
35	$C_{1,4}$	b_{10}	2	36	$C_{1,4}$	b_{10}	2	37	$C_{1,4}$	b_{10}	2
35	$C_{1,5}$	b_{11}	2	36	$C_{1,5}$	b_{11}	2	37	$C_{1,5}$	b_{11}	2
35	$C_{2,2}$	b_{12}	3	36	$C_{1,6}$	b_{12}	1	37	$C_{1,6}$	b_{12}	1
35	$C_{2,3}$	b_{13}	2	36	$C_{2,2}$	b_{13}	3	37	$C_{2,2}$	b_{13}	3
35	$C_{2,4}$	b_{14}	2	36	$C_{2,3}$	b_{14}	2	37	$C_{2,3}$	b_{14}	2
35	$C_{2,5}$	b_{15}	2	36	$C_{2,4}$	b_{15}	2	37	$C_{2,4}$	b_{15}	2
35	$C_{2,6}$	b_{16}	2	36	$C_{2,5}$	b_{16}	2	37	$C_{2,5}$	b_{16}	2
35	$C_{3,2}$	b_{17}	3	36	$C_{2,6}$	b_{17}	1	37	$C_{2,6}$	b_{17}	2
35	$C_{3,3}$	b_{18}	2	36	$C_{3,2}$	b_{18}	3	37	$C_{3,2}$	b_{18}	3
35	$C_{3,4}$	b_{19}	1	36	$C_{3,3}$	b_{19}	2	37	$C_{3,3}$	b_{19}	2
35	$C_{3,5}$	b_{20}	1	36	$C_{3,4}$	b_{20}	1	37	$C_{3,4}$	b_{20}	1
35	$C_{3,6}$	b_{21}	1	36	$C_{3,5}$	b_{21}	1	37	$C_{3,5}$	b_{21}	1
35	$C_{4,2}$	b_{22}	2	36	$C_{3,6}$	b_{22}	1	37	$C_{3,6}$	b_{22}	1
35	$C_{4,3}$	b_{23}	2	36	$C_{4,2}$	b_{23}	2	37	$C_{4,2}$	b_{23}	2
35	$C_{4,4}$	b_{24}	1	36	$C_{4,3}$	b_{24}	2	37	$C_{4,3}$	b_{24}	1
35	$C_{4,5}$	b_{25}	1	36	$C_{4,4}$	b_{25}	1	37	$C_{4,4}$	b_{25}	1
35	$C_{4,6}$	b_{26}	1	36	$C_{4,5}$	b_{26}	1	37	$C_{4,5}$	b_{26}	1
35	$C_{5,2}$	b_{27}	2	36	$C_{4,6}$	b_{27}	1	37	$C_{4,6}$	b_{27}	1
35	$C_{5,3}$	b_{28}	1	36	$C_{5,2}$	b_{28}	2	37	$C_{5,2}$	b_{28}	2
35	$C_{5,4}$	b_{29}	1	36	$C_{5,3}$	b_{29}	1	37	$C_{5,3}$	b_{29}	1
35	$C_{5,5}$	b_{30}	1	36	$C_{5,4}$	b_{30}	1	37	$C_{5,4}$	b_{30}	1
35	$C_{5,6}$	b_{31}	0	36	$C_{5,5}$	b_{31}	1	37	$C_{5,5}$	b_{31}	1
35	$C_{6,2}$	b_{32}	2	36	$C_{5,6}$	b_{32}	0	37	$C_{5,6}$	b_{32}	0
35	$C_{6,3}$	b_{33}	1	36	$C_{6,2}$	b_{33}	2	37	$C_{6,2}$	b_{33}	2
35	$C_{6,4}$	b_{34}	1	36	$C_{6,3}$	b_{34}	1	37	$C_{6,3}$	b_{34}	1
35	$C_{6,5}$	b_{35}	1	36	$C_{6,4}$	b_{35}	1	37	$C_{6,4}$	b_{35}	1
35	$C_{6,6}$	b_{36}	0	36	$C_{6,5}$	b_{36}	1	37	$C_{6,5}$	b_{36}	1
				36	$C_{6,6}$	b_{37}	0	37	$C_{6,6}$	b_{37}	1
								37	$C_{6,7}$	b_{38}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
38	$C_{1,2}$	b_8	3	39	$C_{1,2}$	b_8	3	40	$C_{1,2}$	b_8	3
38	$C_{1,3}$	b_9	2	39	$C_{1,3}$	b_9	2	40	$C_{1,3}$	b_9	2
38	$C_{1,4}$	b_{10}	2	39	$C_{1,4}$	b_{10}	2	40	$C_{1,4}$	b_{10}	2
38	$C_{1,5}$	b_{11}	2	39	$C_{1,5}$	b_{11}	2	40	$C_{1,5}$	b_{11}	2
38	$C_{1,6}$	b_{12}	1	39	$C_{1,6}$	b_{12}	1	40	$C_{1,6}$	b_{12}	1
38	$C_{2,2}$	b_{13}	3	39	$C_{2,2}$	b_{13}	3	40	$C_{2,2}$	b_{13}	3
38	$C_{2,3}$	b_{14}	2	39	$C_{2,3}$	b_{14}	2	40	$C_{2,3}$	b_{14}	2
38	$C_{2,4}$	b_{15}	2	39	$C_{2,4}$	b_{15}	2	40	$C_{2,4}$	b_{15}	2
38	$C_{2,5}$	b_{16}	2	39	$C_{2,5}$	b_{16}	2	40	$C_{2,5}$	b_{16}	1
38	$C_{2,6}$	b_{17}	1	39	$C_{2,6}$	b_{17}	1	40	$C_{2,6}$	b_{17}	1
38	$C_{3,2}$	b_{18}	3	39	$C_{3,2}$	b_{18}	3	40	$C_{3,2}$	b_{18}	3
38	$C_{3,3}$	b_{19}	2	39	$C_{3,3}$	b_{19}	2	40	$C_{3,3}$	b_{19}	2
38	$C_{3,4}$	b_{20}	1	39	$C_{3,4}$	b_{20}	1	40	$C_{3,4}$	b_{20}	1
38	$C_{3,5}$	b_{21}	1	39	$C_{3,5}$	b_{21}	1	40	$C_{3,5}$	b_{21}	1
38	$C_{3,6}$	b_{22}	1	39	$C_{3,6}$	b_{22}	1	40	$C_{3,6}$	b_{22}	1
38	$C_{4,2}$	b_{23}	2	39	$C_{4,2}$	b_{23}	2	40	$C_{3,7}$	b_{23}	1
38	$C_{4,3}$	b_{24}	1	39	$C_{4,3}$	b_{24}	2	40	$C_{4,2}$	b_{24}	2
38	$C_{4,4}$	b_{25}	1	39	$C_{4,4}$	b_{25}	1	40	$C_{4,3}$	b_{25}	2
38	$C_{4,5}$	b_{26}	1	39	$C_{4,5}$	b_{26}	1	40	$C_{4,4}$	b_{26}	1
38	$C_{4,6}$	b_{27}	1	39	$C_{4,6}$	b_{27}	1	40	$C_{4,5}$	b_{27}	1
38	$C_{5,2}$	b_{28}	2	39	$C_{4,7}$	b_{28}	0	40	$C_{4,6}$	b_{28}	1
38	$C_{5,3}$	b_{29}	1	39	$C_{5,2}$	b_{29}	2	40	$C_{4,7}$	b_{29}	0
38	$C_{5,4}$	b_{30}	1	39	$C_{5,3}$	b_{30}	1	40	$C_{5,2}$	b_{30}	2
38	$C_{5,5}$	b_{31}	1	39	$C_{5,4}$	b_{31}	1	40	$C_{5,3}$	b_{31}	1
38	$C_{5,6}$	b_{32}	1	39	$C_{5,5}$	b_{32}	1	40	$C_{5,4}$	b_{32}	1
38	$C_{5,7}$	b_{33}	0	39	$C_{5,6}$	b_{33}	1	40	$C_{5,5}$	b_{33}	1
38	$C_{6,2}$	b_{34}	2	39	$C_{5,7}$	b_{34}	0	40	$C_{5,6}$	b_{34}	1
38	$C_{6,3}$	b_{35}	1	39	$C_{6,2}$	b_{35}	2	40	$C_{5,7}$	b_{35}	0
38	$C_{6,4}$	b_{36}	1	39	$C_{6,3}$	b_{36}	1	40	$C_{6,2}$	b_{36}	2
38	$C_{6,5}$	b_{37}	1	39	$C_{6,4}$	b_{37}	1	40	$C_{6,3}$	b_{37}	1
38	$C_{6,6}$	b_{38}	1	39	$C_{6,5}$	b_{38}	1	40	$C_{6,4}$	b_{38}	1
38	$C_{6,7}$	b_{39}	0	39	$C_{6,6}$	b_{39}	0	40	$C_{6,5}$	b_{39}	1
				39	$C_{6,7}$	b_{40}	0	40	$C_{6,6}$	b_{40}	0
								40	$C_{6,7}$	b_{41}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
41	$C_{1,2}$	b_8	3	42	$C_{1,2}$	b_8	3	43	$C_{1,2}$	b_8	3
41	$C_{1,3}$	b_9	2	42	$C_{1,3}$	b_9	2	43	$C_{1,3}$	b_9	2
41	$C_{1,4}$	b_{10}	2	42	$C_{1,4}$	b_{10}	2	43	$C_{1,4}$	b_{10}	2
41	$C_{1,5}$	b_{11}	1	42	$C_{1,5}$	b_{11}	2	43	$C_{1,5}$	b_{11}	2
41	$C_{1,6}$	b_{12}	1	42	$C_{1,6}$	b_{12}	1	43	$C_{1,6}$	b_{12}	1
41	$C_{2,2}$	b_{13}	3	42	$C_{1,7}$	b_{13}	1	43	$C_{1,7}$	b_{13}	1
41	$C_{2,3}$	b_{14}	2	42	$C_{2,2}$	b_{14}	3	43	$C_{2,2}$	b_{14}	3
41	$C_{2,4}$	b_{15}	2	42	$C_{2,3}$	b_{15}	2	43	$C_{2,3}$	b_{15}	2
41	$C_{2,5}$	b_{16}	2	42	$C_{2,4}$	b_{16}	2	43	$C_{2,4}$	b_{16}	2
41	$C_{2,6}$	b_{17}	1	42	$C_{2,5}$	b_{17}	2	43	$C_{2,5}$	b_{17}	2
41	$C_{2,7}$	b_{18}	1	42	$C_{2,6}$	b_{18}	1	43	$C_{2,6}$	b_{18}	1
41	$C_{3,2}$	b_{19}	3	42	$C_{2,7}$	b_{19}	1	43	$C_{2,7}$	b_{19}	1
41	$C_{3,3}$	b_{20}	2	42	$C_{3,2}$	b_{20}	2	43	$C_{3,2}$	b_{20}	2
41	$C_{3,4}$	b_{21}	1	42	$C_{3,3}$	b_{21}	2	43	$C_{3,3}$	b_{21}	2
41	$C_{3,5}$	b_{22}	1	42	$C_{3,4}$	b_{22}	1	43	$C_{3,4}$	b_{22}	1
41	$C_{3,6}$	b_{23}	1	42	$C_{3,5}$	b_{23}	1	43	$C_{3,5}$	b_{23}	1
41	$C_{3,7}$	b_{24}	1	42	$C_{3,6}$	b_{24}	1	43	$C_{3,6}$	b_{24}	1
41	$C_{4,2}$	b_{25}	2	42	$C_{3,7}$	b_{25}	1	43	$C_{3,7}$	b_{25}	1
41	$C_{4,3}$	b_{26}	2	42	$C_{4,2}$	b_{26}	2	43	$C_{4,2}$	b_{26}	2
41	$C_{4,4}$	b_{27}	1	42	$C_{4,3}$	b_{27}	2	43	$C_{4,3}$	b_{27}	1
41	$C_{4,5}$	b_{28}	1	42	$C_{4,4}$	b_{28}	1	43	$C_{4,4}$	b_{28}	1
41	$C_{4,6}$	b_{29}	1	42	$C_{4,5}$	b_{29}	1	43	$C_{4,5}$	b_{29}	1
41	$C_{4,7}$	b_{30}	0	42	$C_{4,6}$	b_{30}	1	43	$C_{4,6}$	b_{30}	1
41	$C_{5,2}$	b_{31}	2	42	$C_{4,7}$	b_{31}	0	43	$C_{4,7}$	b_{31}	0
41	$C_{5,3}$	b_{32}	1	42	$C_{5,2}$	b_{32}	2	43	$C_{5,2}$	b_{32}	2
41	$C_{5,4}$	b_{33}	1	42	$C_{5,3}$	b_{33}	1	43	$C_{5,3}$	b_{33}	1
41	$C_{5,5}$	b_{34}	1	42	$C_{5,4}$	b_{34}	1	43	$C_{5,4}$	b_{34}	1
41	$C_{5,6}$	b_{35}	1	42	$C_{5,5}$	b_{35}	1	43	$C_{5,5}$	b_{35}	1
41	$C_{5,7}$	b_{36}	0	42	$C_{5,6}$	b_{36}	0	43	$C_{5,6}$	b_{36}	0
41	$C_{6,2}$	b_{37}	2	42	$C_{5,7}$	b_{37}	0	43	$C_{5,7}$	b_{37}	0
41	$C_{6,3}$	b_{38}	1	42	$C_{6,2}$	b_{38}	2	43	$C_{6,2}$	b_{38}	2
41	$C_{6,4}$	b_{39}	1	42	$C_{6,3}$	b_{39}	1	43	$C_{6,3}$	b_{39}	1
41	$C_{6,5}$	b_{40}	1	42	$C_{6,4}$	b_{40}	1	43	$C_{6,4}$	b_{40}	1
41	$C_{6,6}$	b_{41}	0	42	$C_{6,5}$	b_{41}	1	43	$C_{6,5}$	b_{41}	1
41	$C_{6,7}$	b_{42}	0	42	$C_{6,6}$	b_{42}	0	43	$C_{6,6}$	b_{42}	1
				42	$C_{6,7}$	b_{43}	0	43	$C_{6,7}$	b_{43}	0
								43	$C_{6,8}$	b_{44}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
44	$C_{1,2}$	b_8	3	45	$C_{1,2}$	b_8	3	46	$C_{1,2}$	b_8	3
44	$C_{1,3}$	b_9	2	45	$C_{1,3}$	b_9	2	46	$C_{1,3}$	b_9	2
44	$C_{1,4}$	b_{10}	2	45	$C_{1,4}$	b_{10}	2	46	$C_{1,4}$	b_{10}	2
44	$C_{1,5}$	b_{11}	1	45	$C_{1,5}$	b_{11}	1	46	$C_{1,5}$	b_{11}	1
44	$C_{1,6}$	b_{12}	1	45	$C_{1,6}$	b_{12}	1	46	$C_{1,6}$	b_{12}	1
44	$C_{1,7}$	b_{13}	1	45	$C_{1,7}$	b_{13}	1	46	$C_{1,7}$	b_{13}	1
44	$C_{2,2}$	b_{14}	3	45	$C_{2,2}$	b_{14}	3	46	$C_{2,2}$	b_{14}	3
44	$C_{2,3}$	b_{15}	2	45	$C_{2,3}$	b_{15}	2	46	$C_{2,3}$	b_{15}	2
44	$C_{2,4}$	b_{16}	2	45	$C_{2,4}$	b_{16}	2	46	$C_{2,4}$	b_{16}	2
44	$C_{2,5}$	b_{17}	2	45	$C_{2,5}$	b_{17}	1	46	$C_{2,5}$	b_{17}	1
44	$C_{2,6}$	b_{18}	1	45	$C_{2,6}$	b_{18}	1	46	$C_{2,6}$	b_{18}	1
44	$C_{2,7}$	b_{19}	1	45	$C_{2,7}$	b_{19}	1	46	$C_{2,7}$	b_{19}	1
44	$C_{3,2}$	b_{20}	2	45	$C_{3,2}$	b_{20}	2	46	$C_{3,2}$	b_{20}	2
44	$C_{3,3}$	b_{21}	2	45	$C_{3,3}$	b_{21}	2	46	$C_{3,3}$	b_{21}	2
44	$C_{3,4}$	b_{22}	1	45	$C_{3,4}$	b_{22}	1	46	$C_{3,4}$	b_{22}	1
44	$C_{3,5}$	b_{23}	1	45	$C_{3,5}$	b_{23}	1	46	$C_{3,5}$	b_{23}	1
44	$C_{3,6}$	b_{24}	1	45	$C_{3,6}$	b_{24}	1	46	$C_{3,6}$	b_{24}	1
44	$C_{3,7}$	b_{25}	1	45	$C_{3,7}$	b_{25}	1	46	$C_{3,7}$	b_{25}	1
44	$C_{4,2}$	b_{26}	2	45	$C_{4,2}$	b_{26}	2	46	$C_{3,8}$	b_{26}	1
44	$C_{4,3}$	b_{27}	1	45	$C_{4,3}$	b_{27}	2	46	$C_{4,2}$	b_{27}	2
44	$C_{4,4}$	b_{28}	1	45	$C_{4,4}$	b_{28}	1	46	$C_{4,3}$	b_{28}	2
44	$C_{4,5}$	b_{29}	1	45	$C_{4,5}$	b_{29}	1	46	$C_{4,4}$	b_{29}	1
44	$C_{4,6}$	b_{30}	1	45	$C_{4,6}$	b_{30}	1	46	$C_{4,5}$	b_{30}	1
44	$C_{4,7}$	b_{31}	0	45	$C_{4,7}$	b_{31}	0	46	$C_{4,6}$	b_{31}	1
44	$C_{5,2}$	b_{32}	2	45	$C_{4,8}$	b_{32}	0	46	$C_{4,7}$	b_{32}	0
44	$C_{5,3}$	b_{33}	1	45	$C_{5,2}$	b_{33}	2	46	$C_{4,8}$	b_{33}	0
44	$C_{5,4}$	b_{34}	1	45	$C_{5,3}$	b_{34}	1	46	$C_{5,2}$	b_{34}	2
44	$C_{5,5}$	b_{35}	1	45	$C_{5,4}$	b_{35}	1	46	$C_{5,3}$	b_{35}	1
44	$C_{5,6}$	b_{36}	1	45	$C_{5,5}$	b_{36}	1	46	$C_{5,4}$	b_{36}	1
44	$C_{5,7}$	b_{37}	0	45	$C_{5,6}$	b_{37}	1	46	$C_{5,5}$	b_{37}	1
44	$C_{5,8}$	b_{38}	0	45	$C_{5,7}$	b_{38}	0	46	$C_{5,6}$	b_{38}	1
44	$C_{6,2}$	b_{39}	2	45	$C_{5,8}$	b_{39}	0	46	$C_{5,7}$	b_{39}	0
44	$C_{6,3}$	b_{40}	1	45	$C_{6,2}$	b_{40}	2	46	$C_{5,8}$	b_{40}	0
44	$C_{6,4}$	b_{41}	1	45	$C_{6,3}$	b_{41}	1	46	$C_{6,2}$	b_{41}	2
44	$C_{6,5}$	b_{42}	1	45	$C_{6,4}$	b_{42}	1	46	$C_{6,3}$	b_{42}	1
44	$C_{6,6}$	b_{43}	1	45	$C_{6,5}$	b_{43}	1	46	$C_{6,4}$	b_{43}	1
44	$C_{6,7}$	b_{44}	0	45	$C_{6,6}$	b_{44}	1	46	$C_{6,5}$	b_{44}	1
44	$C_{6,8}$	b_{45}	0	45	$C_{6,7}$	b_{45}	0	46	$C_{6,6}$	b_{45}	1
				45	$C_{6,8}$	b_{46}	0	46	$C_{6,7}$	b_{46}	0
								46	$C_{6,8}$	b_{47}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
47	$C_{1,2}$	b_8	3	48	$C_{1,2}$	b_8	3	49	$C_{1,2}$	b_8	3
47	$C_{1,3}$	b_9	2	48	$C_{1,3}$	b_9	2	49	$C_{1,3}$	b_9	2
47	$C_{1,4}$	b_{10}	2	48	$C_{1,4}$	b_{10}	2	49	$C_{1,4}$	b_{10}	2
47	$C_{1,5}$	b_{11}	1	48	$C_{1,5}$	b_{11}	1	49	$C_{1,5}$	b_{11}	1
47	$C_{1,6}$	b_{12}	1	48	$C_{1,6}$	b_{12}	1	49	$C_{1,6}$	b_{12}	1
47	$C_{1,7}$	b_{13}	1	48	$C_{1,7}$	b_{13}	1	49	$C_{1,7}$	b_{13}	1
47	$C_{2,2}$	b_{14}	3	48	$C_{1,8}$	b_{14}	1	49	$C_{1,8}$	b_{14}	1
47	$C_{2,3}$	b_{15}	2	48	$C_{2,2}$	b_{15}	3	49	$C_{2,2}$	b_{15}	3
47	$C_{2,4}$	b_{16}	2	48	$C_{2,3}$	b_{16}	2	49	$C_{2,3}$	b_{16}	2
47	$C_{2,5}$	b_{17}	1	48	$C_{2,4}$	b_{17}	2	49	$C_{2,4}$	b_{17}	2
47	$C_{2,6}$	b_{18}	1	48	$C_{2,5}$	b_{18}	1	49	$C_{2,5}$	b_{18}	1
47	$C_{2,7}$	b_{19}	1	48	$C_{2,6}$	b_{19}	1	49	$C_{2,6}$	b_{19}	1
47	$C_{2,8}$	b_{20}	1	48	$C_{2,7}$	b_{20}	1	49	$C_{2,7}$	b_{20}	1
47	$C_{3,2}$	b_{21}	2	48	$C_{2,8}$	b_{21}	1	49	$C_{2,8}$	b_{21}	1
47	$C_{3,3}$	b_{22}	2	48	$C_{3,2}$	b_{22}	2	49	$C_{3,2}$	b_{22}	2
47	$C_{3,4}$	b_{23}	1	48	$C_{3,3}$	b_{23}	2	49	$C_{3,3}$	b_{23}	2
47	$C_{3,5}$	b_{24}	1	48	$C_{3,4}$	b_{24}	1	49	$C_{3,4}$	b_{24}	1
47	$C_{3,6}$	b_{25}	1	48	$C_{3,5}$	b_{25}	1	49	$C_{3,5}$	b_{25}	1
47	$C_{3,7}$	b_{26}	1	48	$C_{3,6}$	b_{26}	1	49	$C_{3,6}$	b_{26}	1
47	$C_{3,8}$	b_{27}	1	48	$C_{3,7}$	b_{27}	1	49	$C_{3,7}$	b_{27}	1
47	$C_{4,2}$	b_{28}	2	48	$C_{3,8}$	b_{28}	1	49	$C_{3,8}$	b_{28}	0
47	$C_{4,3}$	b_{29}	2	48	$C_{4,2}$	b_{29}	2	49	$C_{4,2}$	b_{29}	2
47	$C_{4,4}$	b_{30}	1	48	$C_{4,3}$	b_{30}	2	49	$C_{4,3}$	b_{30}	2
47	$C_{4,5}$	b_{31}	1	48	$C_{4,4}$	b_{31}	1	49	$C_{4,4}$	b_{31}	1
47	$C_{4,6}$	b_{32}	1	48	$C_{4,5}$	b_{32}	1	49	$C_{4,5}$	b_{32}	1
47	$C_{4,7}$	b_{33}	0	48	$C_{4,6}$	b_{33}	1	49	$C_{4,6}$	b_{33}	1
47	$C_{4,8}$	b_{34}	0	48	$C_{4,7}$	b_{34}	0	49	$C_{4,7}$	b_{34}	0
47	$C_{5,2}$	b_{35}	2	48	$C_{4,8}$	b_{35}	0	49	$C_{4,8}$	b_{35}	0
47	$C_{5,3}$	b_{36}	1	48	$C_{5,2}$	b_{36}	2	49	$C_{5,2}$	b_{36}	2
47	$C_{5,4}$	b_{37}	1	48	$C_{5,3}$	b_{37}	1	49	$C_{5,3}$	b_{37}	1
47	$C_{5,5}$	b_{38}	1	48	$C_{5,4}$	b_{38}	1	49	$C_{5,4}$	b_{38}	1
47	$C_{5,6}$	b_{39}	1	48	$C_{5,5}$	b_{39}	1	49	$C_{5,5}$	b_{39}	1
47	$C_{5,7}$	b_{40}	0	48	$C_{5,6}$	b_{40}	0	49	$C_{5,6}$	b_{40}	0
47	$C_{5,8}$	b_{41}	0	48	$C_{5,7}$	b_{41}	0	49	$C_{5,7}$	b_{41}	0
47	$C_{6,2}$	b_{42}	2	48	$C_{5,8}$	b_{42}	0	49	$C_{5,8}$	b_{42}	0
47	$C_{6,3}$	b_{43}	1	48	$C_{6,2}$	b_{43}	2	49	$C_{6,2}$	b_{43}	2
47	$C_{6,4}$	b_{44}	1	48	$C_{6,3}$	b_{44}	1	49	$C_{6,3}$	b_{44}	1
47	$C_{6,5}$	b_{45}	1	48	$C_{6,4}$	b_{45}	1	49	$C_{6,4}$	b_{45}	1
47	$C_{6,6}$	b_{46}	0	48	$C_{6,5}$	b_{46}	1	49	$C_{6,5}$	b_{46}	1
47	$C_{6,7}$	b_{47}	0	48	$C_{6,6}$	b_{47}	0	49	$C_{6,6}$	b_{47}	1
47	$C_{6,8}$	b_{48}	0	48	$C_{6,7}$	b_{48}	0	49	$C_{6,7}$	b_{48}	0
				48	$C_{6,8}$	b_{49}	0	49	$C_{6,8}$	b_{49}	0
								49	$C_{6,9}$	b_{50}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
50	$C_{1,2}$	b_8	3	51	$C_{1,2}$	b_8	3	52	$C_{1,2}$	b_8	3
50	$C_{1,3}$	b_9	2	51	$C_{1,3}$	b_9	2	52	$C_{1,3}$	b_9	2
50	$C_{1,4}$	b_{10}	2	51	$C_{1,4}$	b_{10}	2	52	$C_{1,4}$	b_{10}	1
50	$C_{1,5}$	b_{11}	1	51	$C_{1,5}$	b_{11}	1	52	$C_{1,5}$	b_{11}	1
50	$C_{1,6}$	b_{12}	1	51	$C_{1,6}$	b_{12}	1	52	$C_{1,6}$	b_{12}	1
50	$C_{1,7}$	b_{13}	1	51	$C_{1,7}$	b_{13}	1	52	$C_{1,7}$	b_{13}	1
50	$C_{1,8}$	b_{14}	1	51	$C_{1,8}$	b_{14}	1	52	$C_{1,8}$	b_{14}	1
50	$C_{2,2}$	b_{15}	3	51	$C_{2,2}$	b_{15}	3	52	$C_{2,2}$	b_{15}	3
50	$C_{2,3}$	b_{16}	2	51	$C_{2,3}$	b_{16}	2	52	$C_{2,3}$	b_{16}	2
50	$C_{2,4}$	b_{17}	2	51	$C_{2,4}$	b_{17}	1	52	$C_{2,4}$	b_{17}	2
50	$C_{2,5}$	b_{18}	1	51	$C_{2,5}$	b_{18}	1	52	$C_{2,5}$	b_{18}	1
50	$C_{2,6}$	b_{19}	1	51	$C_{2,6}$	b_{19}	1	52	$C_{2,6}$	b_{19}	1
50	$C_{2,7}$	b_{20}	1	51	$C_{2,7}$	b_{20}	1	52	$C_{2,7}$	b_{20}	1
50	$C_{2,8}$	b_{21}	1	51	$C_{2,8}$	b_{21}	1	52	$C_{2,8}$	b_{21}	1
50	$C_{3,2}$	b_{22}	2	51	$C_{3,2}$	b_{22}	2	52	$C_{3,2}$	b_{22}	2
50	$C_{3,3}$	b_{23}	2	51	$C_{3,3}$	b_{23}	2	52	$C_{3,3}$	b_{23}	2
50	$C_{3,4}$	b_{24}	1	51	$C_{3,4}$	b_{24}	1	52	$C_{3,4}$	b_{24}	1
50	$C_{3,5}$	b_{25}	1	51	$C_{3,5}$	b_{25}	1	52	$C_{3,5}$	b_{25}	1
50	$C_{3,6}$	b_{26}	1	51	$C_{3,6}$	b_{26}	1	52	$C_{3,6}$	b_{26}	1
50	$C_{3,7}$	b_{27}	1	51	$C_{3,7}$	b_{27}	1	52	$C_{3,7}$	b_{27}	1
50	$C_{3,8}$	b_{28}	0	51	$C_{3,8}$	b_{28}	0	52	$C_{3,8}$	b_{28}	1
50	$C_{4,2}$	b_{29}	2	51	$C_{4,2}$	b_{29}	2	52	$C_{3,9}$	b_{29}	0
50	$C_{4,3}$	b_{30}	2	51	$C_{4,3}$	b_{30}	2	52	$C_{4,2}$	b_{30}	2
50	$C_{4,4}$	b_{31}	1	51	$C_{4,4}$	b_{31}	1	52	$C_{4,3}$	b_{31}	2
50	$C_{4,5}$	b_{32}	1	51	$C_{4,5}$	b_{32}	1	52	$C_{4,4}$	b_{32}	1
50	$C_{4,6}$	b_{33}	1	51	$C_{4,6}$	b_{33}	1	52	$C_{4,5}$	b_{33}	1
50	$C_{4,7}$	b_{34}	0	51	$C_{4,7}$	b_{34}	0	52	$C_{4,6}$	b_{34}	1
50	$C_{4,8}$	b_{35}	0	51	$C_{4,8}$	b_{35}	0	52	$C_{4,7}$	b_{35}	0
50	$C_{5,2}$	b_{36}	2	51	$C_{4,9}$	b_{36}	0	52	$C_{4,8}$	b_{36}	0
50	$C_{5,3}$	b_{37}	1	51	$C_{5,2}$	b_{37}	2	52	$C_{4,9}$	b_{37}	0
50	$C_{5,4}$	b_{38}	1	51	$C_{5,3}$	b_{38}	1	52	$C_{5,2}$	b_{38}	2
50	$C_{5,5}$	b_{39}	1	51	$C_{5,4}$	b_{39}	1	52	$C_{5,3}$	b_{39}	1
50	$C_{5,6}$	b_{40}	1	51	$C_{5,5}$	b_{40}	1	52	$C_{5,4}$	b_{40}	1
50	$C_{5,7}$	b_{41}	0	51	$C_{5,6}$	b_{41}	1	52	$C_{5,5}$	b_{41}	1
50	$C_{5,8}$	b_{42}	0	51	$C_{5,7}$	b_{42}	0	52	$C_{5,6}$	b_{42}	1
50	$C_{5,9}$	b_{43}	0	51	$C_{5,8}$	b_{43}	0	52	$C_{5,7}$	b_{43}	0
50	$C_{6,2}$	b_{44}	2	51	$C_{5,9}$	b_{44}	0	52	$C_{5,8}$	b_{44}	0
50	$C_{6,3}$	b_{45}	1	51	$C_{6,2}$	b_{45}	2	52	$C_{5,9}$	b_{45}	0
50	$C_{6,4}$	b_{46}	1	51	$C_{6,3}$	b_{46}	1	52	$C_{6,2}$	b_{46}	2
50	$C_{6,5}$	b_{47}	1	51	$C_{6,4}$	b_{47}	1	52	$C_{6,3}$	b_{47}	1
50	$C_{6,6}$	b_{48}	0	51	$C_{6,5}$	b_{48}	1	52	$C_{6,4}$	b_{48}	1
50	$C_{6,7}$	b_{49}	0	51	$C_{6,6}$	b_{49}	1	52	$C_{6,5}$	b_{49}	1
50	$C_{6,8}$	b_{50}	0	51	$C_{6,7}$	b_{50}	0	52	$C_{6,6}$	b_{50}	1
50	$C_{6,9}$	b_{51}	0	51	$C_{6,8}$	b_{51}	0	52	$C_{6,7}$	b_{51}	0
				51	$C_{6,9}$	b_{52}	0	52	$C_{6,8}$	b_{52}	0
								52	$C_{6,9}$	b_{53}	0

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
53	$C_{1,2}$	b_8	3	53	$C_{5,7}$	b_{44}	0	54	$C_{4,4}$	b_{34}	1
53	$C_{1,3}$	b_9	2	53	$C_{5,8}$	b_{45}	0	54	$C_{4,5}$	b_{35}	1
53	$C_{1,4}$	b_{10}	1	53	$C_{5,9}$	b_{46}	0	54	$C_{4,6}$	b_{36}	1
53	$C_{1,5}$	b_{11}	1	53	$C_{6,2}$	b_{47}	2	54	$C_{4,7}$	b_{37}	0
53	$C_{1,6}$	b_{12}	1	53	$C_{6,3}$	b_{48}	1	54	$C_{4,8}$	b_{38}	0
53	$C_{1,7}$	b_{13}	1	53	$C_{6,4}$	b_{49}	1	54	$C_{4,9}$	b_{39}	0
53	$C_{1,8}$	b_{14}	1	53	$C_{6,5}$	b_{50}	1	54	$C_{5,2}$	b_{40}	2
53	$C_{2,2}$	b_{15}	3	53	$C_{6,6}$	b_{51}	0	54	$C_{5,3}$	b_{41}	1
53	$C_{2,3}$	b_{16}	2	53	$C_{6,7}$	b_{52}	0	54	$C_{5,4}$	b_{42}	1
53	$C_{2,4}$	b_{17}	2	53	$C_{6,8}$	b_{53}	0	54	$C_{5,5}$	b_{43}	1
53	$C_{2,5}$	b_{18}	1	53	$C_{6,9}$	b_{54}	0	54	$C_{5,6}$	b_{44}	1
53	$C_{2,6}$	b_{19}	1					54	$C_{5,7}$	b_{45}	0
53	$C_{2,7}$	b_{20}	1	54	$C_{1,2}$	b_8	3	54	$C_{5,8}$	b_{46}	0
53	$C_{2,8}$	b_{21}	1	54	$C_{1,3}$	b_9	2	54	$C_{5,9}$	b_{47}	0
53	$C_{2,9}$	b_{22}	1	54	$C_{1,4}$	b_{10}	2	54	$C_{6,2}$	b_{48}	2
53	$C_{3,2}$	b_{23}	2	54	$C_{1,5}$	b_{11}	1	54	$C_{6,3}$	b_{49}	1
53	$C_{3,3}$	b_{24}	2	54	$C_{1,6}$	b_{12}	1	54	$C_{6,4}$	b_{50}	1
53	$C_{3,4}$	b_{25}	1	54	$C_{1,7}$	b_{13}	1	54	$C_{6,5}$	b_{51}	1
53	$C_{3,5}$	b_{26}	1	54	$C_{1,8}$	b_{14}	1	54	$C_{6,6}$	b_{52}	0
53	$C_{3,6}$	b_{27}	1	54	$C_{1,9}$	b_{15}	0	54	$C_{6,7}$	b_{53}	0
53	$C_{3,7}$	b_{28}	1	54	$C_{2,2}$	b_{16}	3	54	$C_{6,8}$	b_{54}	0
53	$C_{3,8}$	b_{29}	1	54	$C_{2,3}$	b_{17}	2	54	$C_{6,9}$	b_{55}	0
53	$C_{3,9}$	b_{30}	0	54	$C_{2,4}$	b_{18}	2				
53	$C_{4,2}$	b_{31}	2	54	$C_{2,5}$	b_{19}	1				
53	$C_{4,3}$	b_{32}	2	54	$C_{2,6}$	b_{20}	1				
53	$C_{4,4}$	b_{33}	1	54	$C_{2,7}$	b_{21}	1				
53	$C_{4,5}$	b_{34}	1	54	$C_{2,8}$	b_{22}	1				
53	$C_{4,6}$	b_{35}	1	54	$C_{2,9}$	b_{23}	0				
53	$C_{4,7}$	b_{36}	0	54	$C_{3,2}$	b_{24}	2				
53	$C_{4,8}$	b_{37}	0	54	$C_{3,3}$	b_{25}	2				
53	$C_{4,9}$	b_{38}	0	54	$C_{3,4}$	b_{26}	1				
53	$C_{5,2}$	b_{39}	2	54	$C_{3,5}$	b_{27}	1				
53	$C_{5,3}$	b_{40}	1	54	$C_{3,6}$	b_{28}	1				
53	$C_{5,4}$	b_{41}	1	54	$C_{3,7}$	b_{29}	1				
53	$C_{5,5}$	b_{42}	1	54	$C_{3,8}$	b_{30}	1				
53	$C_{5,6}$	b_{43}	1	54	$C_{3,9}$	b_{31}	0				
				54	$C_{4,2}$	b_{32}	2				
				54	$C_{4,3}$	b_{33}	2				

L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m	L	$C_{i,k}$	b_m	B_m
55	$C_{1,2}$	b_8	3	55	$C_{5,6}$	b_{44}	0	56	$C_{4,2}$	b_{32}	2
55	$C_{1,3}$	b_9	2	55	$C_{5,7}$	b_{45}	0	56	$C_{4,3}$	b_{33}	2
55	$C_{1,4}$	b_{10}	2	55	$C_{5,8}$	b_{46}	0	56	$C_{4,4}$	b_{34}	1
55	$C_{1,5}$	b_{11}	1	55	$C_{5,9}$	b_{47}	0	56	$C_{4,5}$	b_{35}	1
55	$C_{1,6}$	b_{12}	1	55	$C_{6,2}$	b_{48}	2	56	$C_{4,6}$	b_{36}	1
55	$C_{1,7}$	b_{13}	1	55	$C_{6,3}$	b_{49}	1	56	$C_{4,7}$	b_{37}	0
55	$C_{1,8}$	b_{14}	1	55	$C_{6,4}$	b_{50}	1	56	$C_{4,8}$	b_{38}	0
55	$C_{1,9}$	b_{15}	0	55	$C_{6,5}$	b_{51}	1	56	$C_{4,9}$	b_{39}	0
55	$C_{2,2}$	b_{16}	3	55	$C_{6,6}$	b_{52}	1	56	$C_{5,2}$	b_{40}	2
55	$C_{2,3}$	b_{17}	2	55	$C_{6,7}$	b_{53}	0	56	$C_{5,3}$	b_{41}	1
55	$C_{2,4}$	b_{18}	2	55	$C_{6,8}$	b_{54}	0	56	$C_{5,4}$	b_{42}	1
55	$C_{2,5}$	b_{19}	1	55	$C_{6,9}$	b_{55}	0	56	$C_{5,5}$	b_{43}	1
55	$C_{2,6}$	b_{20}	1	55	$C_{6,10}$	b_{56}	0	56	$C_{5,6}$	b_{44}	1
55	$C_{2,7}$	b_{21}	1					56	$C_{5,7}$	b_{45}	0
55	$C_{2,8}$	b_{22}	1	56	$C_{1,2}$	b_8	3	56	$C_{5,8}$	b_{46}	0
55	$C_{2,9}$	b_{23}	0	56	$C_{1,3}$	b_9	2	56	$C_{5,9}$	b_{47}	0
55	$C_{3,2}$	b_{24}	2	56	$C_{1,4}$	b_{10}	2	56	$C_{5,10}$	b_{48}	0
55	$C_{3,3}$	b_{25}	2	56	$C_{1,5}$	b_{11}	1	56	$C_{6,2}$	b_{49}	2
55	$C_{3,4}$	b_{26}	1	56	$C_{1,6}$	b_{12}	1	56	$C_{6,3}$	b_{50}	1
55	$C_{3,5}$	b_{27}	1	56	$C_{1,7}$	b_{13}	1	56	$C_{6,4}$	b_{51}	1
55	$C_{3,6}$	b_{28}	1	56	$C_{1,8}$	b_{14}	1	56	$C_{6,5}$	b_{52}	1
55	$C_{3,7}$	b_{29}	1	56	$C_{1,9}$	b_{15}	0	56	$C_{6,6}$	b_{53}	0
55	$C_{3,8}$	b_{30}	1	56	$C_{2,2}$	b_{16}	3	56	$C_{6,7}$	b_{54}	0
55	$C_{3,9}$	b_{31}	0	56	$C_{2,3}$	b_{17}	2	56	$C_{6,8}$	b_{55}	0
55	$C_{4,2}$	b_{32}	2	56	$C_{2,4}$	b_{18}	2	56	$C_{6,9}$	b_{56}	0
55	$C_{4,3}$	b_{33}	2	56	$C_{2,5}$	b_{19}	1	56	$C_{6,10}$	b_{57}	0
55	$C_{4,4}$	b_{34}	1	56	$C_{2,6}$	b_{20}	1				
55	$C_{4,5}$	b_{35}	1	56	$C_{2,7}$	b_{21}	1				
55	$C_{4,6}$	b_{36}	1	56	$C_{2,8}$	b_{22}	1				
55	$C_{4,7}$	b_{37}	0	56	$C_{2,9}$	b_{23}	0				
55	$C_{4,8}$	b_{38}	0	56	$C_{3,2}$	b_{24}	2				
55	$C_{4,9}$	b_{39}	0	56	$C_{3,3}$	b_{25}	2				
55	$C_{5,2}$	b_{40}	2	56	$C_{3,4}$	b_{26}	1				
55	$C_{5,3}$	b_{41}	1	56	$C_{3,5}$	b_{27}	1				
55	$C_{5,4}$	b_{42}	1	56	$C_{3,6}$	b_{28}	1				
55	$C_{5,5}$	b_{43}	1	56	$C_{3,7}$	b_{29}	1				
				56	$C_{3,8}$	b_{30}	1				
				56	$C_{3,9}$	b_{31}	0				

Annex H Bit Frame Format

<i>Symbol</i>	<i>Bit 1</i>	<i>Bit 0</i>	<i>Symbol</i>	<i>Bit 1</i>	<i>Bit 0</i>
0	$c_0(22)$	$c_1(21)$	36	$c_0(10)$	$c_1(9)$
1	$c_2(20)$	$c_3(19)$	37	$c_2(8)$	$c_3(7)$
2	$c_4(10)$	$c_5(1)$	38	$c_5(13)$	$c_6(4)$
3	$c_1(20)$	$c_0(21)$	39	$c_1(8)$	$c_0(9)$
4	$c_3(18)$	$c_2(19)$	40	$c_3(6)$	$c_2(7)$
5	$c_5(0)$	$c_4(9)$	41	$c_6(3)$	$c_5(12)$
6	$c_0(20)$	$c_1(19)$	42	$c_0(8)$	$c_1(7)$
7	$c_2(18)$	$c_3(17)$	43	$c_2(6)$	$c_3(5)$
8	$c_4(8)$	$c_6(14)$	44	$c_5(11)$	$c_6(2)$
9	$c_1(18)$	$c_0(19)$	45	$c_1(6)$	$c_0(7)$
10	$c_3(16)$	$c_2(17)$	46	$c_3(4)$	$c_2(5)$
11	$c_6(13)$	$c_4(7)$	47	$c_6(1)$	$c_5(10)$
12	$c_0(18)$	$c_1(17)$	48	$c_0(6)$	$c_1(5)$
13	$c_2(16)$	$c_3(15)$	49	$c_2(4)$	$c_3(3)$
14	$c_4(6)$	$c_6(12)$	50	$c_5(9)$	$c_6(0)$
15	$c_1(16)$	$c_0(17)$	51	$c_1(4)$	$c_9(5)$
16	$c_3(14)$	$c_2(15)$	52	$c_3(2)$	$c_2(3)$
17	$c_6(11)$	$c_4(5)$	53	$c_7(6)$	$c_5(8)$
18	$c_0(16)$	$c_1(15)$	54	$c_0(4)$	$c_1(3)$
19	$c_2(14)$	$c_3(13)$	55	$c_2(2)$	$c_3(1)$
20	$c_4(4)$	$c_6(10)$	56	$c_5(7)$	$c_7(5)$
21	$c_1(14)$	$c_0(15)$	57	$c_1(2)$	$c_0(3)$
22	$c_3(12)$	$c_2(13)$	58	$c_3(0)$	$c_2(1)$
23	$c_6(9)$	$c_4(3)$	59	$c_7(4)$	$c_5(6)$
24	$c_0(14)$	$c_1(13)$	60	$c_0(2)$	$c_1(1)$
25	$c_2(12)$	$c_3(11)$	61	$c_2(0)$	$c_4(14)$
26	$c_4(2)$	$c_6(8)$	62	$c_5(5)$	$c_7(3)$
27	$c_1(12)$	$c_0(13)$	63	$c_1(0)$	$c_0(1)$
28	$c_3(10)$	$c_2(11)$	64	$c_4(13)$	$c_3(22)$
29	$c_6(7)$	$c_4(1)$	65	$c_7(2)$	$c_5(4)$
30	$c_0(12)$	$c_1(11)$	66	$c_0(0)$	$c_2(22)$
31	$c_2(10)$	$c_3(9)$	67	$c_3(21)$	$c_4(12)$
32	$c_4(0)$	$c_6(6)$	68	$c_5(3)$	$c_7(1)$
33	$c_1(10)$	$c_0(11)$	69	$c_2(21)$	$c_1(22)$
34	$c_3(8)$	$c_2(9)$	70	$c_4(11)$	$c_3(20)$
35	$c_6(5)$	$c_5(14)$	71	$c_7(0)$	$c_5(2)$

Annex I Speech Synthesis Window

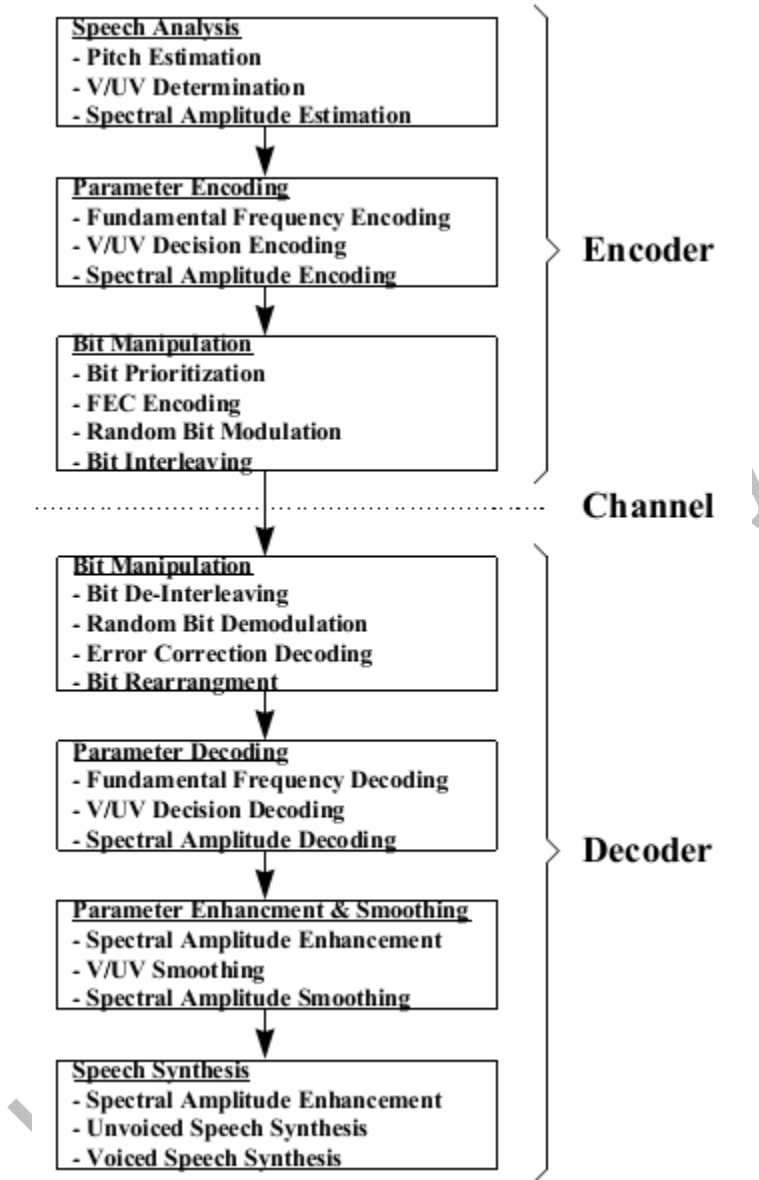
n	$w_s(n)$	n	$w_s(n)$	n	$w_s(n)$	n	$w_s(n)$
-105	0.00	-74	0.62	-43	1.00	-12	1.00
-104	0.02	-73	0.64	-42	1.00	-11	1.00
-103	0.04	-72	0.66	-41	1.00	-10	1.00
-102	0.06	-71	0.68	-40	1.00	-9	1.00
-101	0.08	-70	0.70	-39	1.00	-8	1.00
-100	0.10	-69	0.72	-38	1.00	-7	1.00
-99	0.12	-68	0.74	-37	1.00	-6	1.00
-98	0.14	-67	0.76	-36	1.00	-5	1.00
-97	0.16	-66	0.78	-35	1.00	-4	1.00
-96	0.18	-65	0.80	-34	1.00	-3	1.00
-95	0.20	-64	0.82	-33	1.00	-2	1.00
-94	0.22	-63	0.84	-32	1.00	-1	1.00
-93	0.24	-62	0.86	-31	1.00	0	1.00
-92	0.26	-61	0.88	-30	1.00	1	1.00
-91	0.28	-60	0.90	-29	1.00	2	1.00
-90	0.30	-59	0.92	-28	1.00	3	1.00
-89	0.32	-58	0.94	-27	1.00	4	1.00
-88	0.34	-57	0.96	-26	1.00	5	1.00
-87	0.36	-56	0.98	-25	1.00	6	1.00
-86	0.38	-55	1.00	-24	1.00	7	1.00
-85	0.40	-54	1.00	-23	1.00	8	1.00
-84	0.42	-53	1.00	-22	1.00	9	1.00
-83	0.44	-52	1.00	-21	1.00	10	1.00
-82	0.46	-51	1.00	-20	1.00	11	1.00
-81	0.48	-50	1.00	-19	1.00	12	1.00
-80	0.50	-49	1.00	-18	1.00	13	1.00
-79	0.52	-48	1.00	-17	1.00	14	1.00
-78	0.54	-47	1.00	-16	1.00	15	1.00
-77	0.56	-46	1.00	-15	1.00	16	1.00
-76	0.58	-45	1.00	-14	1.00	17	1.00
-75	0.60	-44	1.00	-13	1.00	18	1.00

n	$w_s(n)$	n	$w_s(n)$	n	$w_s(n)$
19	1.00	50	1.00	81	0.48
20	1.00	51	1.00	82	0.46
21	1.00	52	1.00	83	0.44
22	1.00	53	1.00	84	0.42
23	1.00	54	1.00	85	0.40
24	1.00	55	1.00	86	0.38
25	1.00	56	0.98	87	0.36
26	1.00	57	0.96	88	0.34
27	1.00	58	0.94	89	0.32
28	1.00	59	0.92	90	0.30
29	1.00	60	0.90	91	0.28
30	1.00	61	0.88	92	0.26
31	1.00	62	0.86	93	0.24
32	1.00	63	0.84	94	0.22
33	1.00	64	0.82	95	0.20
34	1.00	65	0.80	96	0.18
35	1.00	66	0.78	97	0.16
36	1.00	67	0.76	98	0.14
37	1.00	68	0.74	99	0.12
38	1.00	69	0.72	100	0.10
39	1.00	70	0.70	101	0.08
40	1.00	71	0.68	102	0.06
41	1.00	72	0.66	103	0.04
42	1.00	73	0.64	104	0.02
43	1.00	74	0.62	105	0.00
44	1.00	75	0.60		
45	1.00	76	0.58		
46	1.00	77	0.56		
47	1.00	78	0.54		
48	1.00	79	0.52		
49	1.00	80	0.50		

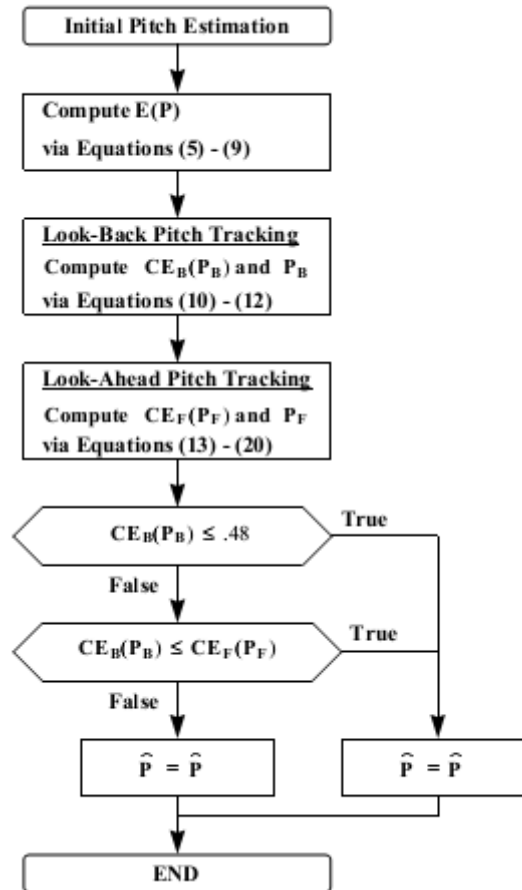
Annex J Log Magnitude Prediction Residual Block Lengths

$\hat{L}$	$\hat{f}_1$	$\hat{f}_2$	$\hat{f}_3$	$\hat{f}_4$	$\hat{f}_5$	$\hat{f}_6$	$\hat{L}$	$\hat{f}_1$	$\hat{f}_2$	$\hat{f}_3$	$\hat{f}_4$	$\hat{f}_5$	$\hat{f}_6$
9	1	1	1	2	2	2	33	5	5	5	6	6	6
10	1	1	2	2	2	2	34	5	5	6	6	6	6
11	1	2	2	2	2	2	35	5	6	6	6	6	6
12	2	2	2	2	2	2	36	6	6	6	6	6	6
13	2	2	2	2	2	3	37	6	6	6	6	6	7
14	2	2	2	2	3	3	38	6	6	6	6	7	7
15	2	2	2	3	3	3	39	6	6	6	7	7	7
16	2	2	3	3	3	3	40	6	6	7	7	7	7
17	2	3	3	3	3	3	41	6	7	7	7	7	7
18	3	3	3	3	3	3	42	7	7	7	7	7	7
19	3	3	3	3	3	4	43	7	7	7	7	7	8
20	3	3	3	3	4	4	44	7	7	7	7	8	8
21	3	3	3	4	4	4	45	7	7	7	8	8	8
22	3	3	4	4	4	4	46	7	7	8	8	8	8
23	3	4	4	4	4	4	47	7	8	8	8	8	8
24	4	4	4	4	4	4	48	8	8	8	8	8	8
25	4	4	4	4	4	5	49	8	8	8	8	8	9
26	4	4	4	4	5	5	50	8	8	8	8	9	9
27	4	4	4	5	5	5	51	8	8	8	9	9	9
28	4	4	5	5	5	5	52	8	8	9	9	9	9
29	4	5	5	5	5	5	53	8	9	9	9	9	9
30	5	5	5	5	5	5	54	9	9	9	9	9	9
31	5	5	5	5	5	6	55	9	9	9	9	9	10
32	5	5	5	5	6	6	56	9	9	9	9	10	10

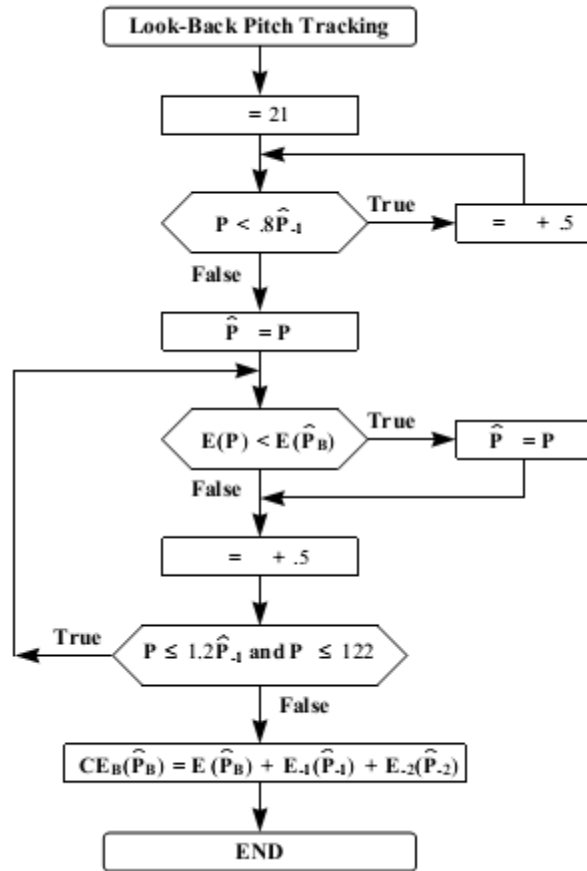
Annex K Flow Charts



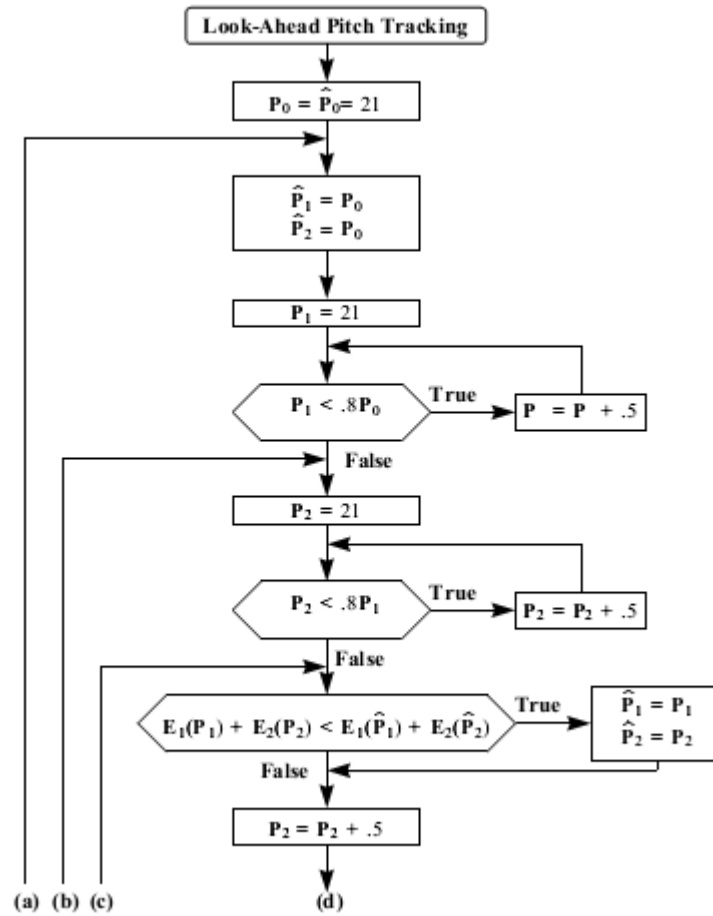
Flow Chart 1: IMBE Voice Coder



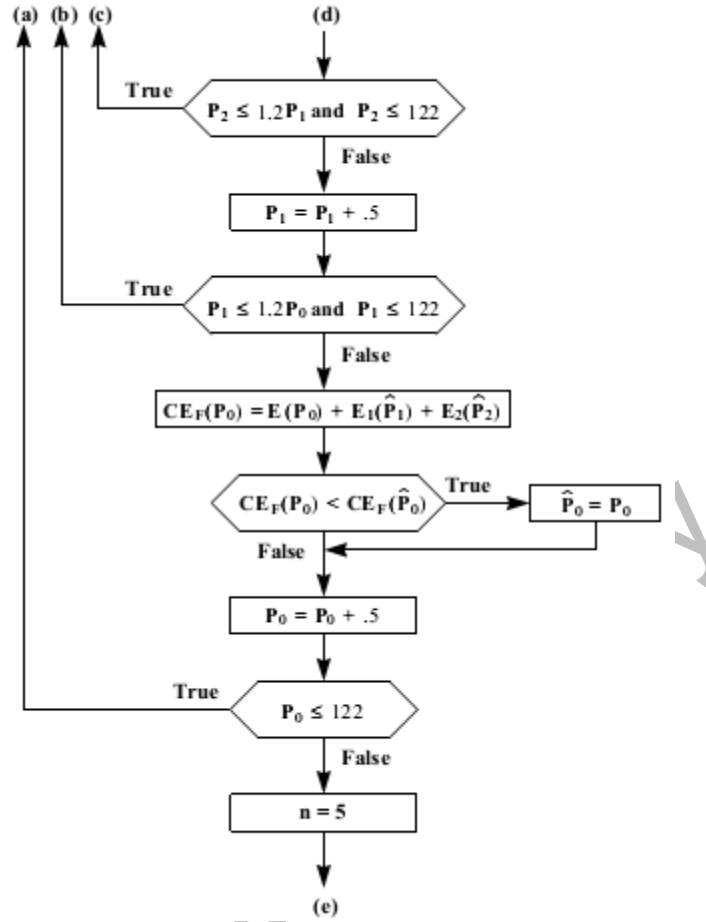
Flow Chart 2: Initial Pitch Estimation



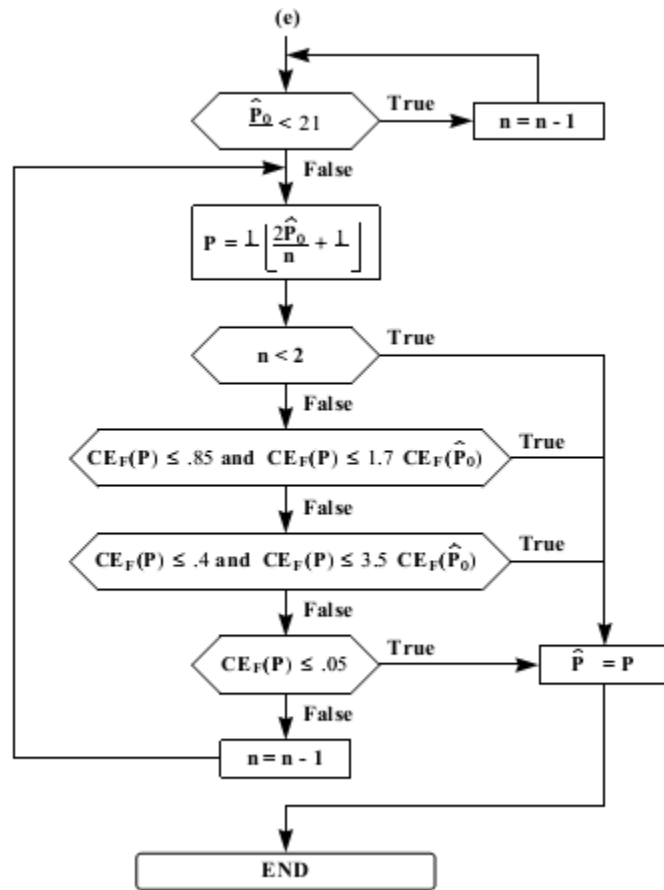
Flow Chart 3: Look-Back Pitch Tracking



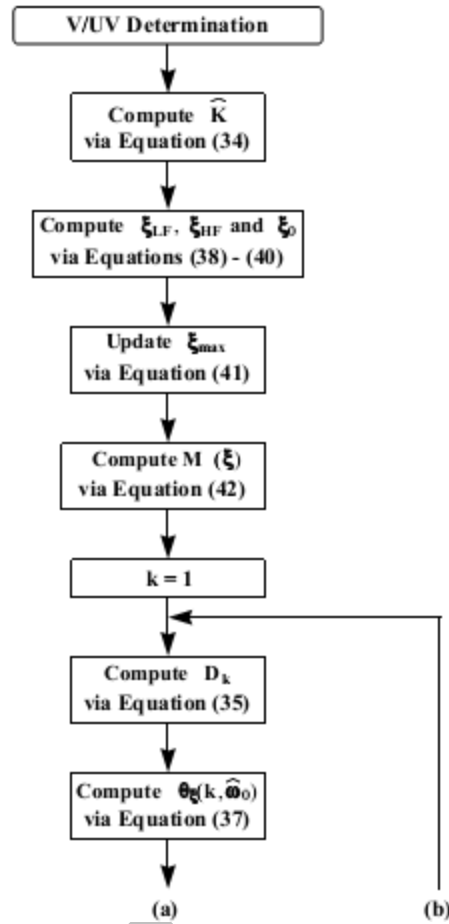
Flow Chart 4: Look-Ahead Pitch Tracking (1 of 3)



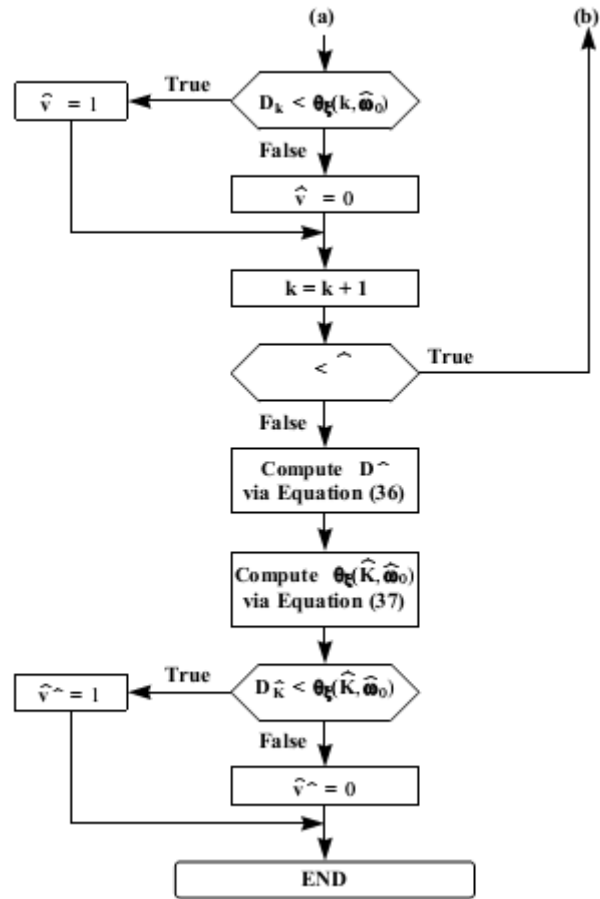
Flow Chart 4: Look-Ahead Pitch Tracking (2 of 3)



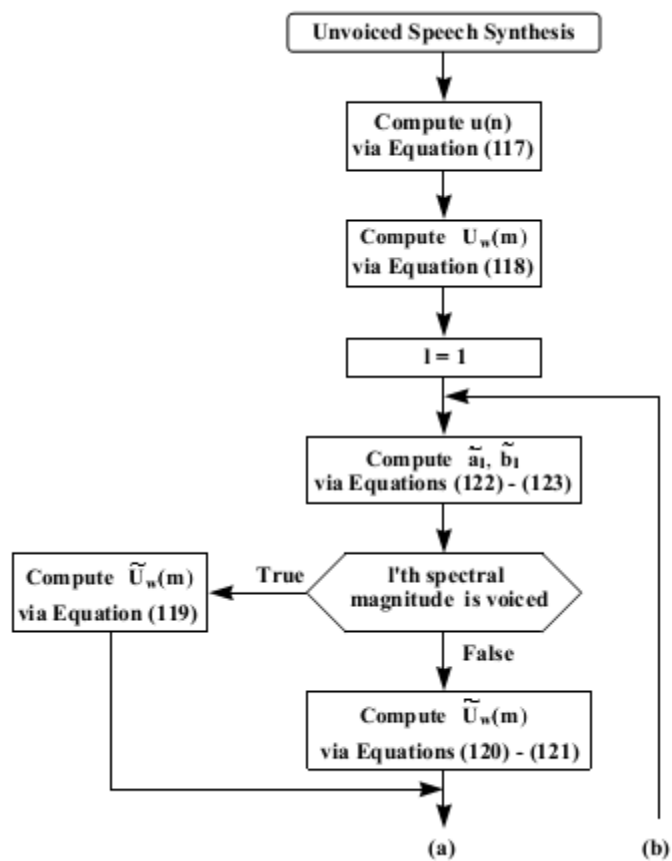
Flow Chart 4: Look-Ahead Pitch Tracking (3 of 3)



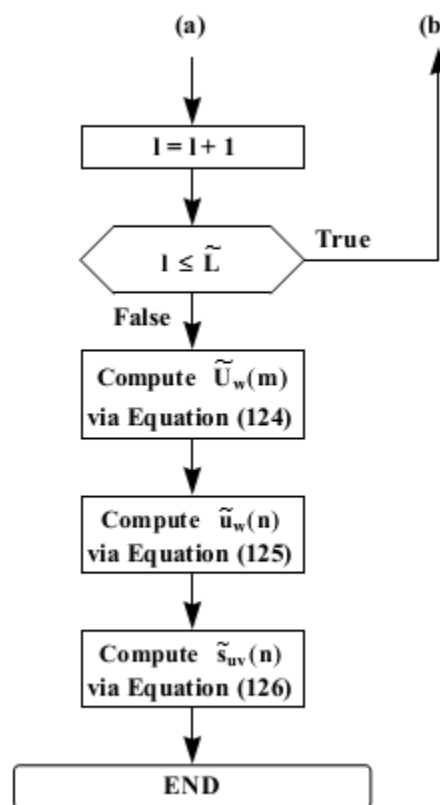
Flow Chart 5: V/UV Determination (1 of 2)



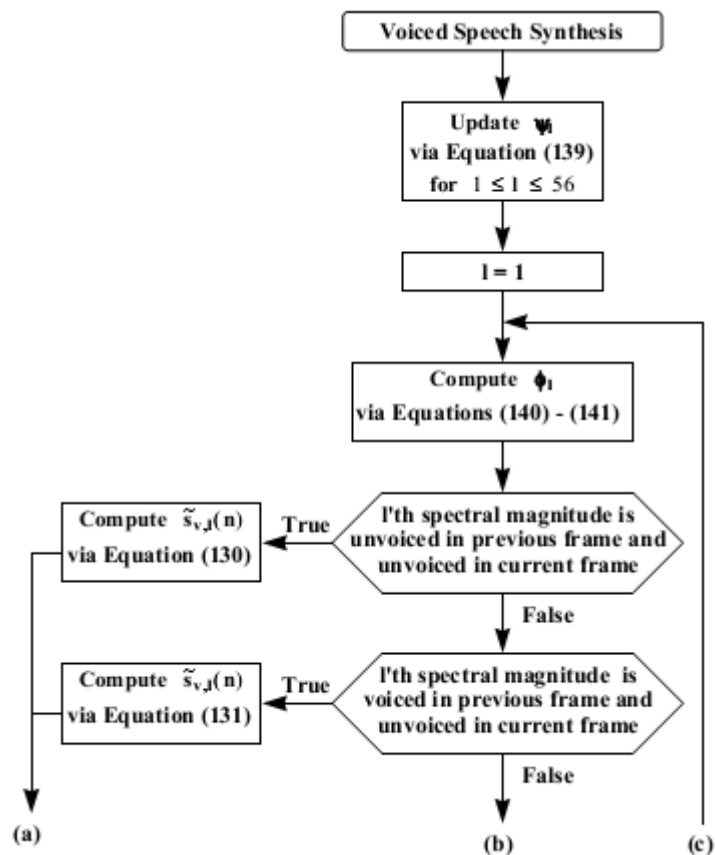
Flow Chart 5: V/UV Determination (2 of 2)



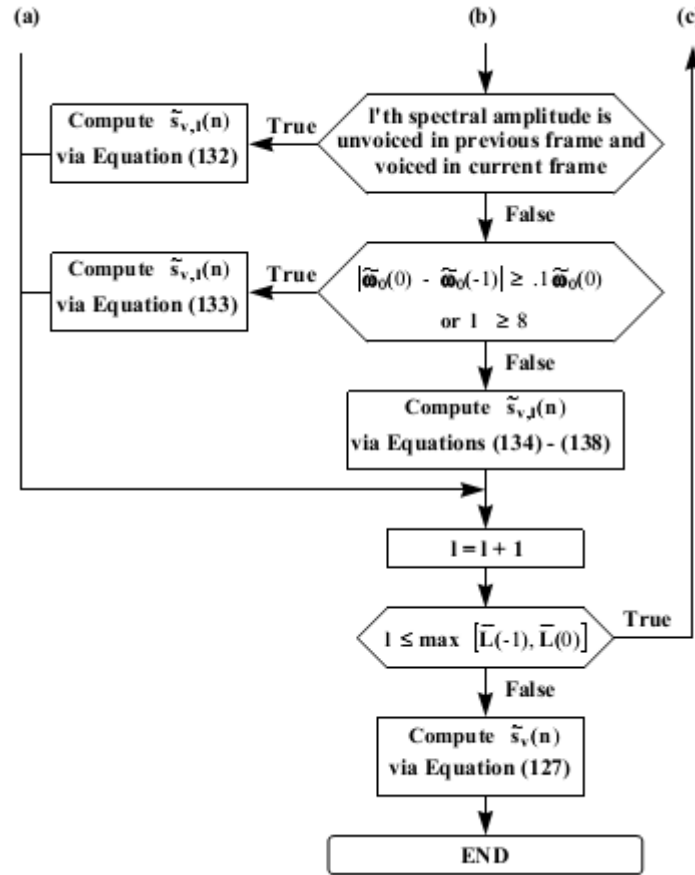
Flow Chart 6: Unvoiced Speech Synthesis (1 of 2)



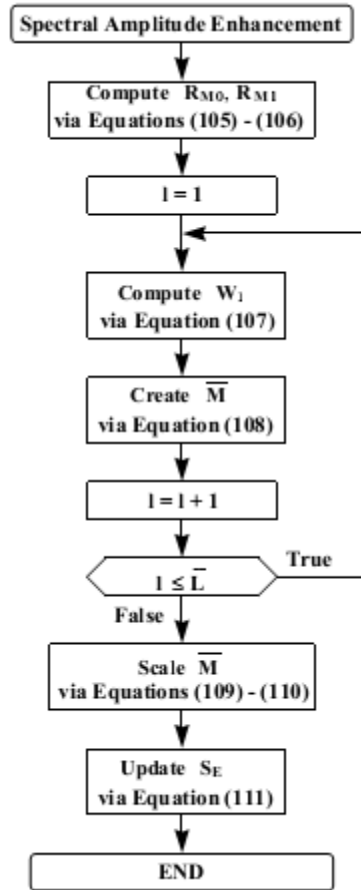
Flow Chart 6: Unvoiced Speech Synthesis (2 of 2)



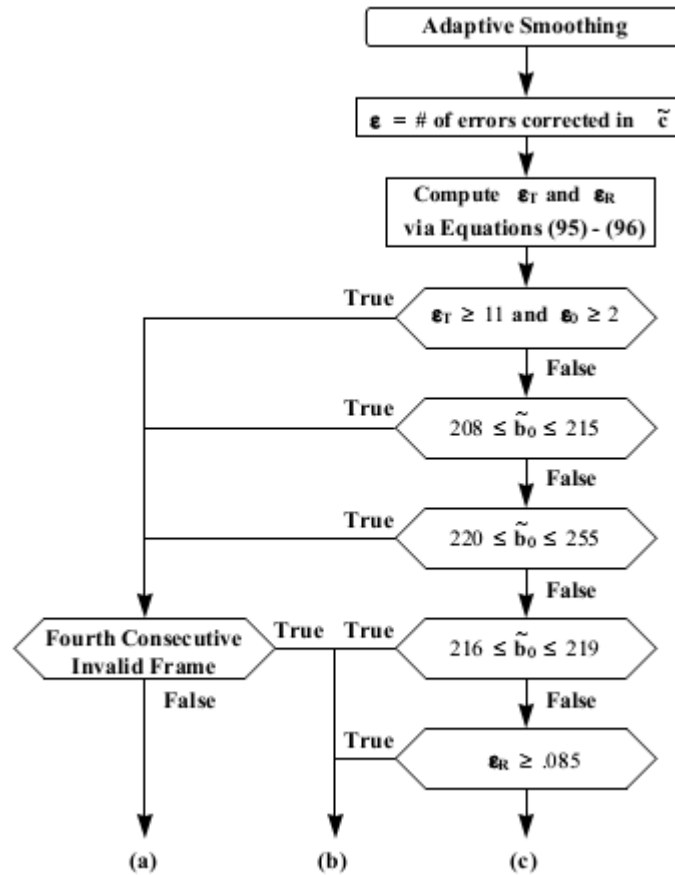
Flow Chart 7: Voiced Speech Synthesis (1 of 2)



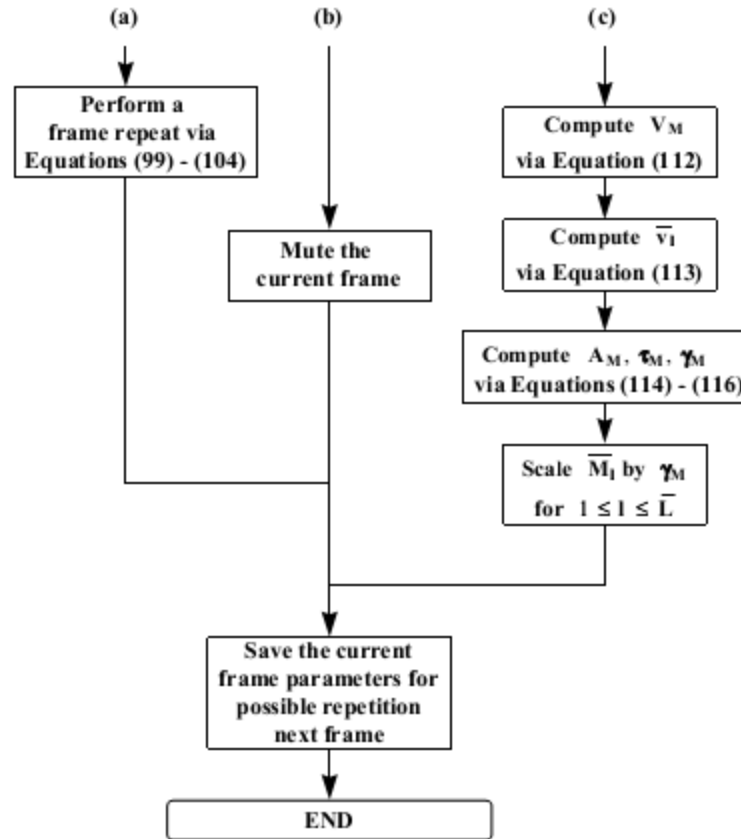
Flow Chart 7: Voiced Speech Synthesis (2 of 2)



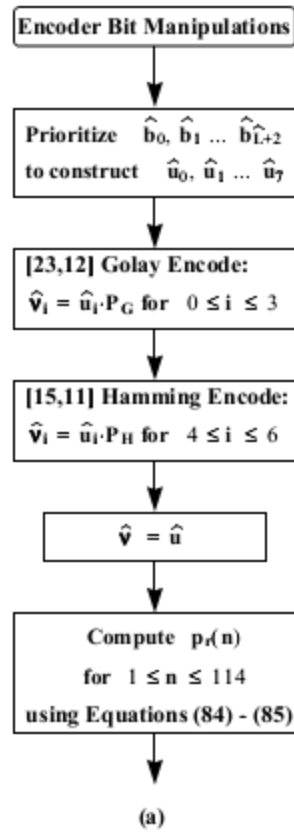
Flow Chart 8: Spectral Amplitude Enhancement



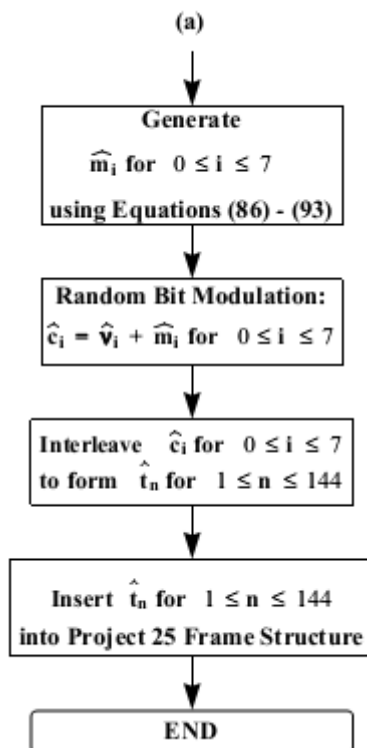
Flow Chart 9: Adaptive Smoothing (1 of 2)



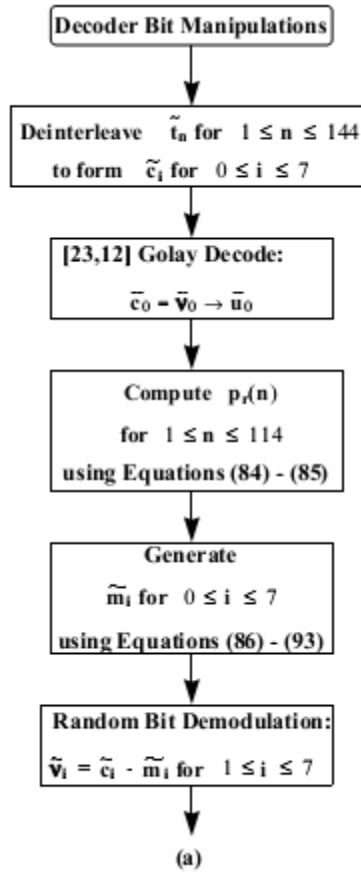
Flow Chart 9: Adaptive Smoothing (2 of 2)



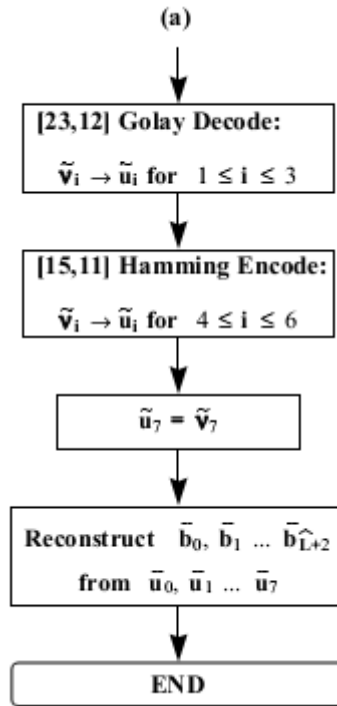
Flow Chart 10: Encoder Bit Manipulations (1 of 2)



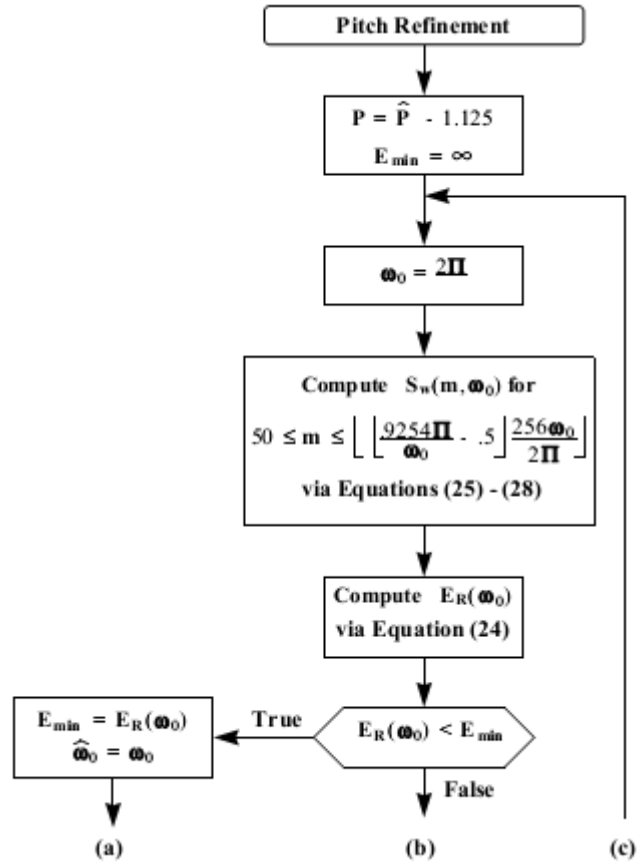
Flow Chart 10: Encoder Bit Manipulations (2 of 2)



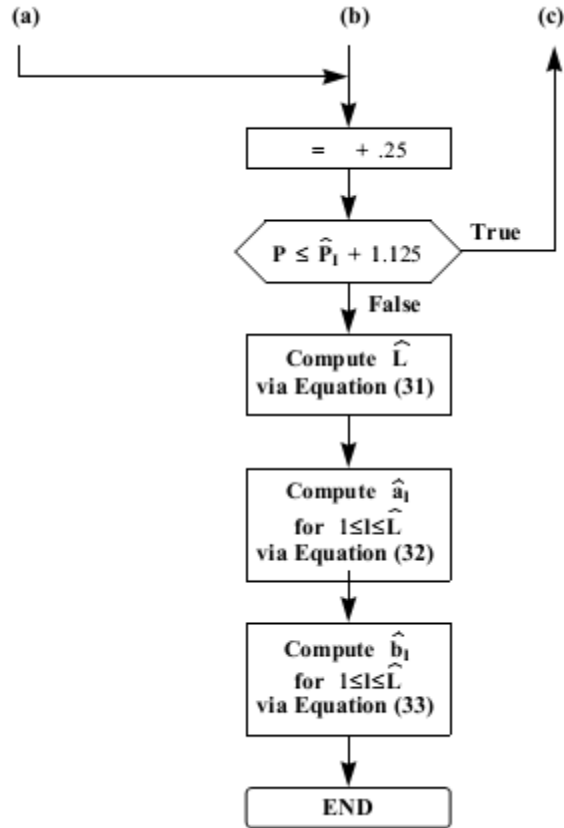
Flow Chart 11: Decoder Bit Manipulations (1 of 2)



Flow Chart 11: Decoder Bit Manipulations (2 of 2)



Flow Chart 12: Pitch Refinement (1 of 2)



Flow Chart 12: Pitch Refinement (2 of 2)

Annex L (Normative) Fundamental Frequency Quantization Table

b_0	L	ω_0	b_0	L	ω_0	b_0	L	ω_0
0	9	0.049971	40	17	0.027122	80	31	0.014721
1	9	0.049215	41	17	0.026712	81	31	0.014496
2	9	0.048471	42	17	0.026304	82	32	0.014277
3	9	0.047739	43	17	0.025906	83	32	0.014061
4	9	0.047010	44	18	0.025515	84	33	0.013847
5	9	0.046299	45	18	0.025129	85	33	0.013636
6	10	0.045601	46	18	0.024746	86	34	0.013430
7	10	0.044905	47	18	0.024372	87	34	0.013227
8	10	0.044226	48	19	0.024002	88	35	0.013025
9	10	0.043558	49	19	0.023636	89	36	0.012829
10	10	0.042900	50	19	0.023279	90	36	0.012634
11	10	0.042246	51	20	0.022926	91	37	0.012444
12	11	0.041609	52	20	0.022581	92	37	0.012253
13	11	0.040979	53	20	0.022236	93	38	0.012068
14	11	0.040356	54	21	0.021900	94	38	0.011887
15	11	0.039747	55	21	0.021570	95	39	0.011703
16	11	0.039148	56	21	0.021240	96	40	0.011528
17	11	0.038559	57	22	0.020920	97	40	0.011353
18	12	0.037971	58	22	0.020605	98	41	0.011183
19	12	0.037399	59	22	0.020294	99	42	0.011011
20	12	0.036839	60	23	0.019983	100	42	0.010845
21	12	0.036278	61	23	0.019684	101	43	0.010681
22	12	0.035732	62	23	0.019386	102	43	0.010517
23	13	0.035198	63	24	0.019094	103	44	0.010359
24	13	0.034672	64	24	0.018805	104	45	0.010202
25	13	0.034145	65	24	0.018520	105	46	0.010050
26	13	0.033636	66	25	0.018242	106	46	0.009895
27	13	0.033133	67	25	0.017965	107	47	0.009747
28	14	0.032635	68	26	0.017696	108	48	0.009600
29	14	0.032148	69	26	0.017431	109	48	0.009453
30	14	0.031670	70	26	0.017170	110	49	0.009312
31	14	0.031122	71	27	0.016911	111	50	0.009172
32	15	0.030647	72	27	0.016657	112	51	0.009033
33	15	0.030184	73	28	0.016409	113	52	0.008896
34	15	0.029728	74	28	0.016163	114	52	0.008762
35	15	0.029272	75	29	0.015923	115	53	0.008633
36	16	0.028831	76	29	0.015686	116	54	0.008501
37	16	0.028395	77	30	0.015411	117	55	0.008375
38	16	0.027966	78	30	0.015177	118	56	0.008249
39	16	0.027538	79	30	0.014946	119	56	0.008125

Annex M (Normative) V/UV Quantization Vectors

b_1	v_0	v_1	v_2	v_3	v_4	v_5	v_6	v_7
0	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	0
3	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	0	0
5	1	1	0	1	1	1	1	1
6	1	1	1	0	1	1	1	1
7	1	1	1	1	1	0	1	1
8	1	1	1	1	0	0	0	0
9	1	1	1	1	1	0	0	0
10	1	1	1	0	0	0	0	0
11	1	1	1	0	0	0	0	1
12	1	1	0	0	0	0	0	0
13	1	1	1	0	0	0	0	0
14	1	0	0	0	0	0	0	0
15	1	1	1	0	0	0	0	0
16	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0
19	0	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0	0
21	0	0	0	0	0	0	0	0
22	0	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0
25	0	0	0	0	0	0	0	0
26	0	0	0	0	0	0	0	0
27	0	0	0	0	0	0	0	0
28	0	0	0	0	0	0	0	0
29	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0
31	0	0	0	0	0	0	0	0

Annex N (Normative) Log Magnitude Prediction Residual Block Lengths

L	J_1	J_2	J_3	J_4
9	2	2	2	3
10	2	2	3	3
11	2	3	3	3
12	2	3	3	4
13	3	3	3	4
14	3	3	4	4
15	3	3	4	5
16	3	4	4	5
17	3	4	5	5
18	4	4	5	5
19	4	4	5	6
20	4	4	6	6
21	4	5	6	6
22	4	5	6	7
23	5	5	6	7
24	5	5	7	7
25	5	6	7	7
26	5	6	7	8
27	5	6	8	8
28	6	6	8	8
29	6	6	8	9
30	6	7	8	9
31	6	7	9	9
32	6	7	9	10
33	7	7	9	10
34	7	8	9	10
35	7	8	10	10
36	7	8	10	11
37	8	8	10	11
38	8	9	10	11
39	8	9	11	11
40	8	9	11	12
41	8	9	11	13
42	8	9	12	13
43	8	10	12	13
44	9	10	12	13
45	9	10	12	14
46	9	10	13	14
47	9	11	13	14
48	10	11	13	14
49	10	11	13	15
50	10	11	14	15
51	10	12	14	15
52	10	12	14	16
53	11	12	14	16
54	11	12	15	16
55	11	12	15	17
56	11	13	15	17

Annex O (Normative) Gain Quantizer Levels

b_2	Δ_r
0	-2.000000
1	-0.670000
2	0.297941
3	0.663728
4	1.036829
5	1.438136
6	1.890077
7	2.227970
8	2.478289
9	2.667544
10	2.793619
11	2.893261
12	3.020630
13	3.138586
14	3.237579
15	3.322570
16	3.432367
17	3.571863
18	3.696650
19	3.814917
20	3.920932
21	4.022503
22	4.123569
23	4.228291
24	4.370569
25	4.543700
26	4.707695
27	4.848879
28	5.056757
29	5.326468
30	5.777581
31	6.874496

Annex P (Normative) PRBA24 Vector Quantizer Levels

b_3	G_2	G_3	G_4	b_3	G_2	G_3	G_4
0	0.526055	-0.328567	-0.304727	50	0.286428	-0.210542	-0.029587
1	0.441044	-0.303127	-0.201114	51	0.257656	-0.261837	-0.056566
2	1.030896	-0.324730	-0.397204	52	-0.235852	-0.310760	-0.165147
3	0.839696	-0.351933	-0.224909	53	-0.334949	-0.385870	-0.197362
4	0.272958	-0.176118	-0.098893	54	0.094870	-0.241144	0.059122
5	0.221466	-0.160045	-0.061026	55	0.060177	-0.225884	0.031140
6	0.496555	-0.211499	0.047305	56	-0.301184	-0.306545	-0.446189
7	0.424376	-0.223752	0.069911	57	-0.293528	-0.504146	-0.429844
8	0.264531	-0.353355	-0.330505	58	-0.055084	-0.379015	-0.125887
9	0.273650	-0.253004	-0.250241	59	-0.115434	-0.375008	-0.059939
10	0.484531	-0.297627	-0.071051	60	-0.777425	-0.592163	-0.107585
11	0.410814	-0.224961	-0.084998	61	-0.950500	-0.893847	-0.181762
12	0.039519	-0.252904	-0.115128	62	-0.259402	-0.396726	0.010357
13	0.017423	-0.296519	-0.045921	63	-0.368905	-0.449026	0.038299
14	0.225113	-0.224371	0.037882	64	0.279719	-0.063196	-0.184628
15	0.183424	-0.260492	0.050491	65	0.255265	-0.067248	-0.121124
16	0.308704	-0.073205	-0.405880	66	0.458433	-0.103777	0.010074
17	0.213125	-0.101632	-0.333208	67	0.437231	-0.092496	-0.031028
18	0.617735	-0.137299	-0.213670	68	0.082265	-0.028050	-0.041262
19	0.514382	-0.126485	-0.170204	69	0.045920	-0.051719	-0.030155
20	0.130009	-0.076955	-0.229303	70	0.271149	-0.043613	0.112085
21	0.061740	-0.108259	-0.203887	71	0.246881	-0.065274	0.105436
22	0.244473	-0.110094	-0.051689	72	0.056590	-0.117773	-0.142283
23	0.230452	-0.076147	-0.028190	73	0.058824	-0.104418	-0.099608
24	0.059837	-0.254595	-0.562704	74	0.213781	-0.111974	0.031269
25	0.011630	-0.135223	-0.432791	75	0.187554	-0.070340	0.011834
26	0.207077	-0.152248	-0.148391	76	-0.185701	-0.081106	-0.073803
27	0.158078	-0.128800	-0.122150	77	-0.266112	-0.074133	-0.085370
28	-0.265982	-0.144742	-0.199894	78	-0.029368	-0.046490	0.124679
29	-0.356479	-0.204740	-0.156465	79	-0.017378	-0.102882	0.140482
30	0.000324	-0.139549	-0.066471	80	0.114700	0.092738	-0.244271
31	0.001888	-0.170557	-0.025025	81	0.072922	0.007863	-0.231476
32	0.402913	-0.581478	-0.274626	82	0.270022	0.031819	-0.094208
33	0.191289	-0.540335	-0.193040	83	0.254403	0.024805	-0.050389
34	0.632914	-0.401410	-0.006636	84	-0.182905	0.021629	-0.168481
35	0.471086	-0.463144	0.061489	85	-0.225864	-0.010109	-0.130374
36	0.044829	-0.438487	0.033433	86	0.040089	0.013969	0.016028
37	0.015513	-0.539475	-0.006719	87	0.001442	0.010551	0.032942
38	0.336218	-0.351311	0.214087	88	-0.287472	-0.036130	-0.296798
39	0.239967	-0.380836	0.157681	89	-0.332344	-0.108862	-0.342196
40	0.347609	-0.901619	-0.688432	90	0.012700	0.022917	-0.052501
41	0.064067	-0.826753	-0.492089	91	-0.040681	-0.001805	-0.050548
42	0.303089	-0.396757	-0.108446	92	-0.718522	-0.061234	-0.278820
43	0.235590	-0.446122	0.006437	93	-0.879205	-0.213588	-0.303508
44	-0.236964	-0.652532	-0.135520	94	-0.234102	-0.065407	0.013686
45	-0.418285	-0.793014	-0.034730	95	-0.281223	-0.076139	0.046830
46	-0.038262	-0.516984	0.273681	96	0.141967	-0.193679	-0.055697
47	-0.037419	-0.958198	0.214749	97	0.100318	-0.161222	-0.063062
48	0.061624	-0.238233	-0.237184	98	0.265859	-0.132747	0.078209
49	-0.013944	-0.235704	-0.204811	99	0.244805	-0.139776	0.122123

b_3	G_2	G_3	G_4	b_3	G_2	G_3	G_4
100	-0.121802	-0.179976	0.031732	150	0.359915	0.101273	-0.052997
101	-0.185318	-0.214011	0.018117	151	0.318117	0.125888	-0.003486
102	0.047014	-0.153961	0.218068	152	0.150452	0.050219	-0.409155
103	0.047305	-0.187402	0.282114	153	0.188753	0.091894	-0.325733
104	-0.027533	-0.415868	-0.333841	154	0.334922	0.029098	-0.098587
105	-0.125886	-0.334492	-0.290317	155	0.324508	0.015809	-0.135408
106	-0.030602	-0.190918	0.097454	156	-0.042506	0.038667	-0.208535
107	-0.054936	-0.209948	0.158977	157	-0.083003	0.094758	-0.174054
108	-0.507223	-0.295876	-0.217183	158	0.094773	0.102653	-0.025701
109	-0.581733	-0.403194	-0.208936	159	0.063284	0.118703	-0.000071
110	-0.299719	-0.289679	0.297101	160	0.355965	-0.139239	-0.191705
111	-0.363169	-0.362718	0.436529	161	0.392742	-0.105496	-0.132103
112	-0.124627	-0.042100	-0.157011	162	0.663678	-0.204627	-0.031242
113	-0.161571	-0.092846	-0.183636	163	0.609381	-0.146914	0.079610
114	0.084520	-0.100217	-0.000901	164	0.151855	-0.132843	-0.007125
115	0.055655	-0.136381	0.032764	165	0.146404	-0.161917	0.024842
116	-0.545087	-0.197713	-0.026888	166	0.400524	-0.135221	0.232289
117	-0.662772	-0.179815	0.026419	167	0.324931	-0.116605	0.253458
118	-0.165583	-0.148913	0.090382	168	0.169066	-0.215132	-0.185604
119	-0.240772	-0.182830	0.105474	169	0.128681	-0.189394	-0.160279
120	-0.576315	-0.359473	-0.456844	170	0.356194	-0.116992	-0.038381
121	-0.713430	-0.554156	-0.476739	171	0.342866	-0.144687	0.020265
122	-0.275628	-0.223640	-0.051584	172	-0.065545	-0.202593	-0.043688
123	-0.359501	-0.230758	-0.027006	173	-0.124296	-0.260225	-0.035370
124	-1.282559	-0.284807	-0.233743	174	0.083224	-0.235149	0.153301
125	-1.060476	-0.399911	-0.562698	175	0.046256	-0.309608	0.190944
126	-0.871952	-0.272197	0.016126	176	0.187385	-0.008168	-0.198575
127	-0.747922	-0.329404	0.276696	177	0.190401	-0.018699	-0.136858
128	0.643086	0.046175	-0.660078	178	0.398009	-0.025700	-0.007458
129	0.738204	-0.127844	-0.433708	179	0.346948	-0.022258	-0.020905
130	1.158072	0.025571	-0.177856	180	-0.047064	-0.085629	-0.080677
131	0.974840	-0.009417	-0.112337	181	-0.067523	-0.128972	-0.119538
132	0.418014	0.032741	-0.124545	182	0.186086	-0.016828	0.070014
133	0.381422	-0.001557	-0.085504	183	0.187364	0.017133	0.075949
134	0.768280	0.056085	0.095375	184	-0.112669	-0.037433	-0.298944
135	0.680004	0.052035	0.152318	185	-0.068276	-0.114504	-0.265795
136	0.473182	0.012560	-0.264221	186	0.147510	-0.040616	-0.013687
137	0.345153	0.036627	-0.248756	187	0.133084	-0.062849	-0.032637
138	0.746238	-0.025880	-0.106050	188	-0.416571	-0.041544	-0.125088
139	0.644319	-0.058256	-0.095133	189	-0.505337	-0.044193	-0.157651
140	0.185924	-0.022230	-0.070540	190	-0.154132	-0.075106	0.050466
141	0.146068	-0.009550	-0.057871	191	-0.148036	-0.059719	0.121516
142	0.338488	0.013022	0.069961	192	0.490555	0.157659	-0.222208
143	0.298969	0.047403	0.052598	193	0.436700	0.120500	-0.205869
144	0.346002	0.256253	-0.380261	194	0.754525	0.269323	0.045810
145	0.313092	0.163821	-0.314004	195	0.645077	0.271923	0.013942
146	0.719154	0.103108	-0.252648	196	0.237023	0.115337	-0.026429
147	0.621429	0.172423	-0.265180	197	0.204895	0.121020	-0.008541
148	0.240461	0.104684	-0.202582	198	0.383999	0.153963	0.171763
149	0.206946	0.139642	-0.138016	199	0.385026	0.222074	0.239731

b_3	G_2	G_3	G_4	b_3	G_2	G_3	G_4
200	0.198232	0.072972	-0.108179	250	-0.061004	0.107744	0.037257
201	0.147882	0.074743	-0.123341	251	-0.100991	0.080302	0.062701
202	0.390929	0.075205	0.081828	252	-0.927022	0.285660	-0.240549
203	0.341623	0.089405	0.069389	253	-1.153224	0.277232	-0.322538
204	-0.003381	0.159694	-0.016026	254	-0.569012	0.108135	0.172634
205	-0.043653	0.206860	-0.040729	255	-0.555273	0.131461	0.325930
206	0.135515	0.107824	0.179310	256	0.518847	0.065683	-0.132877
207	0.081086	0.119673	0.174282	257	0.501324	-0.006585	-0.094884
208	0.192637	0.400335	-0.341906	258	1.066190	-0.150380	0.201791
209	0.171196	0.284921	-0.221516	259	0.858377	-0.166415	0.081686
210	0.377807	0.359087	-0.151523	260	0.320584	-0.031499	0.039534
211	0.411052	0.297925	-0.099774	261	0.311442	-0.075120	0.026013
212	-0.010060	0.261887	-0.149567	262	0.625829	-0.019856	0.346041
213	-0.107877	0.287756	-0.116982	263	0.525271	-0.003948	0.284868
214	0.158003	0.209727	0.077988	264	0.312594	-0.075673	-0.066642
215	0.109710	0.232272	0.088135	265	0.295732	-0.057895	-0.042207
216	0.000698	0.209353	-0.395208	266	0.550446	-0.029110	0.046850
217	-0.094015	0.230322	-0.279928	267	0.465467	-0.068987	0.096167
218	0.137355	0.230881	-0.124115	268	0.122669	-0.051786	0.044283
219	0.103058	0.166855	-0.100386	269	0.079669	-0.044145	0.045805
220	-0.305058	0.305422	-0.176026	270	0.238778	-0.031835	0.171694
221	-0.422049	0.337137	-0.293297	271	0.200734	-0.072619	0.178726
222	-0.121744	0.185124	0.048115	272	0.342512	0.131270	-0.163021
223	-0.171052	0.200312	0.052812	273	0.294028	0.111759	-0.125793
224	0.224091	-0.010673	-0.019727	274	0.589523	0.121808	-0.049372
225	0.200266	-0.020167	0.001798	275	0.550506	0.132318	0.017485
226	0.382742	0.032362	0.161665	276	0.164280	0.047560	-0.058383
227	0.345631	-0.019705	0.164451	277	0.120110	0.049242	-0.052403
228	0.029431	0.045010	0.071518	278	0.269181	0.035000	0.103494
229	0.031940	0.010876	0.087037	279	0.297466	0.038517	0.139289
230	0.181935	0.039112	0.202316	280	0.094549	-0.030880	-0.153376
231	0.181810	0.033189	0.253435	281	0.080363	0.024359	-0.127578
232	-0.008677	-0.066679	-0.144737	282	0.281351	0.055178	0.000155
233	-0.021768	-0.021288	-0.125903	283	0.234900	0.039477	0.013957
234	0.136766	0.000100	0.059449	284	-0.118161	0.011976	-0.034270
235	0.135405	-0.020446	0.103793	285	-0.157654	0.027765	-0.005010
236	-0.289115	0.039747	-0.012256	286	0.102631	0.027283	0.099723
237	-0.338683	0.025909	-0.034058	287	0.077285	0.052532	0.115583
238	-0.016515	0.048584	0.197981	288	0.329398	-0.278552	0.016316
239	-0.046790	0.011816	0.199964	289	0.305993	-0.267896	0.094952
240	0.094214	0.127422	-0.169936	290	0.775270	-0.394995	0.290748
241	0.048279	0.096189	-0.148153	291	0.583180	-0.252159	0.285391
242	0.217391	0.081732	0.013677	292	0.192226	-0.182242	0.126859
243	0.179656	0.084671	0.031434	293	0.185908	-0.245779	0.159940
244	-0.227367	0.118176	-0.039803	294	0.346293	-0.250404	0.355682
245	-0.327096	0.159747	-0.018931	295	0.354160	-0.364521	0.472337
246	0.000834	0.113118	0.125325	296	0.134942	-0.313666	-0.115181
247	-0.014617	0.128924	0.163776	297	0.126077	-0.286568	-0.039927
248	-0.254570	0.154329	-0.232018	298	0.405618	-0.211792	0.199095
249	-0.353068	0.124341	-0.174409	299	0.312099	-0.213642	0.190972

b_3	G_2	G_3	G_4	b_3	G_2	G_3	G_4
300	-0.071392	-0.297366	0.081426	350	-0.094447	0.159393	0.164848
301	-0.165839	-0.301986	0.160640	351	-0.113612	0.120702	0.221656
302	0.147808	-0.290712	0.298198	352	0.204918	-0.078894	0.075524
303	0.063302	-0.310149	0.396302	353	0.161232	-0.090256	0.088701
304	0.141444	-0.081377	-0.076621	354	0.378460	-0.033687	0.309964
305	0.115936	-0.104440	-0.039885	355	0.311701	-0.049984	0.316881
306	0.367023	-0.087281	0.096390	356	0.019311	-0.050048	0.212387
307	0.330038	-0.117958	0.127050	357	0.002473	-0.062855	0.278462
308	0.002897	-0.062454	0.025151	358	0.151448	-0.090652	0.410031
309	-0.052404	-0.082200	0.041975	359	0.162778	-0.071291	0.531252
310	0.181553	-0.137004	0.230489	360	-0.083704	-0.076839	-0.020798
311	0.140768	-0.094604	0.265928	361	-0.092832	-0.043492	0.029202
312	-0.101763	-0.209566	-0.135964	362	0.136844	-0.077791	0.186493
313	-0.159056	-0.191005	-0.095509	363	0.089536	-0.086826	0.184711
314	0.045016	-0.081562	0.075942	364	-0.270255	-0.058858	0.173048
315	0.016808	-0.112482	0.068593	365	-0.350416	-0.009219	0.273260
316	-0.408578	-0.132377	0.079163	366	-0.105248	-0.205534	0.425159
317	-0.431534	-0.214646	0.157714	367	-0.135030	-0.197464	0.623550
318	-0.096931	-0.101938	0.200304	368	-0.051717	0.069756	-0.043829
319	-0.167867	-0.114851	0.262964	369	-0.081050	0.056947	-0.000205
320	0.393882	0.086002	0.008961	370	0.190388	0.016366	0.145922
321	0.338747	0.048405	-0.004187	371	0.142662	0.002575	0.159182
322	0.877844	0.374373	0.171008	372	-0.352890	0.011117	0.091040
323	0.740790	0.324525	0.242248	373	-0.367374	0.056547	0.147209
324	0.200218	0.070150	0.085891	374	-0.003179	0.026570	0.282541
325	0.171760	0.090531	0.102579	375	-0.069934	-0.005171	0.337678
326	0.314263	0.126417	0.322833	376	-0.496181	0.026464	0.019432
327	0.313523	0.065445	0.403855	377	-0.690384	0.069313	-0.004175
328	0.164261	0.057745	-0.005490	378	-0.146138	0.046372	0.161839
329	0.122141	0.024122	0.009190	379	-0.197581	0.034093	0.241003
330	0.308248	0.078401	0.180577	380	-0.989567	0.040993	0.049384
331	0.251222	0.073868	0.160457	381	-1.151075	0.210556	0.237374
332	-0.047526	0.023725	0.086336	382	-0.335366	-0.058208	0.480168
333	-0.091643	0.005539	0.093179	383	-0.502419	-0.093761	0.675240
334	0.079339	0.044135	0.206697	384	0.862548	0.264137	-0.294905
335	0.104213	0.011277	0.240060	385	0.782668	0.251324	-0.122108
336	0.226607	0.186234	-0.056881	386	1.597797	0.463818	-0.133153
337	0.173281	0.158131	-0.059413	387	1.615756	0.060653	0.084764
338	0.339400	0.214501	0.052905	388	0.435588	0.209832	0.095050
339	0.309166	0.188181	0.058028	389	0.431013	0.165328	0.047909
340	0.014442	0.194715	0.048945	390	1.248164	0.265923	0.488086
341	-0.028793	0.194766	0.089078	391	1.009933	0.345440	0.473702
342	0.069564	0.206743	0.193568	392	0.477017	0.194237	-0.058012
343	0.091532	0.202786	0.269680	393	0.401362	0.186915	-0.054137
344	-0.071196	0.135604	-0.103744	394	1.202158	0.284782	-0.066531
345	-0.118288	0.152837	-0.060151	395	1.064907	0.203766	0.046383
346	0.146856	0.143174	0.061789	396	0.255848	0.133398	0.046049
347	0.104379	0.143672	0.056797	397	0.218680	0.128833	0.065326
348	-0.541832	0.250034	-0.017602	398	0.490817	0.182041	0.286583
349	-0.641583	0.278411	-0.111909	399	0.440714	0.106576	0.301120

b_3	G_2	G_3	G_4	b_3	G_2	G_3	G_4
400	0.604263	0.522925	-0.238629	450	1.259194	0.901494	0.256085
401	0.526329	0.377577	-0.198100	451	1.296139	0.607949	0.302184
402	1.038632	0.606242	-0.121253	452	0.319619	0.307231	0.099647
403	0.995283	0.552202	0.110700	453	0.287232	0.359355	0.186844
404	0.262232	0.313664	-0.086909	454	0.751306	0.676688	0.499386
405	0.230835	0.273385	-0.054268	455	0.479609	0.553030	0.560447
406	0.548466	0.490721	0.278201	456	0.276377	0.214032	-0.003661
407	0.466984	0.355859	0.289160	457	0.238146	0.223595	0.028806
408	0.367137	0.236160	-0.228114	458	0.542688	0.266205	0.171393
409	0.309359	0.233843	-0.171325	459	0.460188	0.283979	0.158288
410	0.465268	0.276569	0.010951	460	0.057385	0.309853	0.144517
411	0.378124	0.250237	0.011131	461	-0.006881	0.348152	0.097310
412	0.061885	0.296810	-0.011420	462	0.244434	0.247298	0.322601
413	0.000125	0.350029	-0.011277	463	0.253992	0.335420	0.402241
414	0.163815	0.261191	0.175863	464	0.354006	0.579776	-0.130176
415	0.165132	0.308797	0.227800	465	0.267043	0.461976	-0.058178
416	0.461418	0.052075	-0.016543	466	0.534049	0.626549	0.046747
417	0.472372	0.046962	0.045746	467	0.441835	0.468260	0.057556
418	0.856406	0.136415	0.245074	468	0.110477	0.628795	0.102950
419	0.834616	0.003254	0.372643	469	0.031409	0.489068	0.090605
420	0.337869	0.036994	0.232513	470	0.229564	0.525640	0.325454
421	0.267414	0.027593	0.252779	471	0.105570	0.582151	0.509738
422	0.584983	0.113046	0.583119	472	0.005690	0.521474	-0.157885
423	0.475406	-0.024234	0.655070	473	0.104463	0.424022	-0.080647
424	0.264823	-0.029292	0.004270	474	0.223784	0.389860	0.060904
425	0.246071	-0.019109	0.030048	475	0.159806	0.340571	0.062061
426	0.477401	0.021039	0.155448	476	-0.173976	0.573425	0.027383
427	0.458453	-0.043959	0.187850	477	-0.376008	0.587868	0.133042
428	0.067059	-0.061227	0.126904	478	-0.051773	0.348339	0.231923
429	0.044608	-0.034575	0.150205	479	-0.122571	0.473049	0.251159
430	0.191304	-0.003810	0.316776	480	0.324321	0.148510	0.116006
431	0.153078	0.029915	0.361303	481	0.282263	0.121730	0.114016
432	0.320704	0.178950	-0.088835	482	0.690108	0.256346	0.418128
433	0.300866	0.137645	-0.056893	483	0.542523	0.294427	0.461973
434	0.553442	0.162339	0.131987	484	0.056944	0.107667	0.281797
435	0.490083	0.123682	0.146163	485	0.027844	0.106858	0.355071
436	0.118950	0.083109	0.034052	486	0.160456	0.177656	0.528819
437	0.099344	0.066212	0.054329	487	0.227537	0.177976	0.689465
438	0.228325	0.122445	0.309219	488	0.111585	0.097896	0.109244
439	0.172093	0.135754	0.323361	489	0.083994	0.133245	0.115789
440	0.064213	0.063405	-0.058243	490	0.208740	0.142084	0.208953
441	0.011906	0.088795	-0.069678	491	0.156072	0.143303	0.231368
442	0.194232	0.129185	0.125708	492	-0.185830	0.214347	0.309774
443	0.155182	0.174013	0.144099	493	-0.311053	0.240517	0.328512
444	-0.217068	0.112731	0.093497	494	-0.041749	0.090901	0.511373
445	-0.307590	0.171146	0.110735	495	-0.156164	0.098486	0.478020
446	-0.014897	0.138094	0.232455	496	0.151543	0.263073	-0.033471
447	-0.036936	0.170135	0.279166	497	0.126322	0.213004	-0.007014
448	0.681886	0.437121	0.078458	498	0.245313	0.217564	0.120210
449	0.548559	0.376914	0.092485	499	0.259136	0.225542	0.176601

b_3	G_2	G_3	G_4
500	-0.190632	0.260214	0.141755
501	-0.189271	0.331768	0.170606
502	0.054763	0.294766	0.357775
503	-0.033724	0.257645	0.365069
504	-0.184971	0.396532	0.057728
505	-0.293313	0.400259	0.001123
506	-0.015219	0.232287	0.177913
507	-0.022524	0.244724	0.240753
508	-0.520342	0.347950	0.249265
509	-0.671997	0.410782	0.153434
510	-0.253089	0.412356	0.489854
511	-0.410922	0.562454	0.543891

Limited Use Only

Annex Q (Normative) PRBA58 Vector Quantizer Levels

Quantization Vectors for b_4

b_4	G_5	G_6	G_7	G_8
0	-0.103660	0.094597	-0.013149	0.081501
1	-0.170709	0.129958	-0.057316	0.112324
2	-0.095113	0.080892	-0.027554	0.003371
3	-0.154153	0.113437	-0.074522	0.003446
4	-0.109553	0.153519	0.006858	0.040930
5	-0.181931	0.217882	-0.019042	0.040049
6	-0.096246	0.144191	-0.024147	-0.035120
7	-0.174811	0.193357	-0.054261	-0.071700
8	-0.183241	-0.052840	0.117923	0.030960
9	-0.242634	0.009075	0.098007	0.091643
10	-0.143847	-0.028529	0.040171	-0.002812
11	-0.198809	0.006990	0.020668	0.026641
12	-0.233172	-0.028793	0.140130	-0.071927
13	-0.309313	0.056873	0.108262	-0.018930
14	-0.172782	-0.002037	0.048755	-0.087065
15	-0.242901	0.036076	0.015064	-0.064366
16	0.077107	0.172685	0.159939	0.097456
17	0.024820	0.209676	0.087347	0.105204
18	0.085113	0.151639	0.084272	0.022747
19	0.047975	0.196695	0.038770	0.029953
20	0.113925	0.236813	0.176121	0.016635
21	0.009708	0.267969	0.127660	0.015872
22	0.114044	0.202311	0.096892	-0.043071
23	0.047219	0.260395	0.050952	-0.046996
24	-0.055095	0.034041	0.200464	0.039050
25	-0.061582	0.069566	0.113048	0.027511
26	-0.025469	0.040440	0.132777	-0.039098
27	-0.031388	0.064010	0.067559	-0.017117
28	-0.074386	0.086579	0.228232	-0.055461
29	-0.107352	0.120874	0.137364	-0.030252
30	-0.036897	0.089972	0.155831	-0.128475
31	-0.059070	0.097879	0.084489	-0.075821
32	-0.050865	-0.025167	-0.086636	0.011256
33	-0.051426	0.013301	-0.144665	0.038541
34	-0.073831	-0.028917	-0.142416	-0.025268
35	-0.083910	0.015004	-0.227113	-0.002808
36	-0.030840	-0.009326	-0.070517	-0.041304
37	-0.022018	0.029381	-0.124961	-0.031624
38	-0.064222	-0.014640	-0.108798	-0.092342
39	-0.038801	0.038133	-0.188992	-0.094221
40	-0.154059	-0.183932	-0.019894	0.082105
41	-0.188022	-0.113072	-0.117380	0.090911
42	-0.243301	-0.207086	-0.053735	-0.001975
43	-0.275931	-0.121035	-0.161261	0.004231
44	-0.118142	-0.157537	-0.036594	-0.008679
45	-0.153627	-0.111372	-0.103095	-0.009460
46	-0.173458	-0.180158	-0.057130	-0.103198
47	-0.208509	-0.127679	-0.149336	-0.109289
48	0.096310	0.047927	-0.024094	-0.057018
49	0.044289	0.075486	-0.008505	-0.067635

b_4	G_5	G_6	G_7	G_8
50	0.076751	0.025560	-0.066428	-0.102991
51	0.025215	0.090417	-0.058616	-0.114284
52	0.125980	0.070078	0.016282	-0.112355
53	0.070859	0.118988	0.001180	-0.116359
54	0.097520	0.059219	-0.026821	-0.172850
55	0.048226	0.145459	-0.050093	-0.188853
56	0.007242	-0.135796	0.147832	-0.034080
57	0.012843	-0.069616	0.077139	-0.047909
58	-0.050911	-0.116323	0.082521	-0.056362
59	-0.039630	-0.055678	0.036066	-0.067992
60	0.042694	-0.091527	0.150940	-0.124225
61	0.029225	-0.039401	0.071664	-0.113665
62	-0.025085	-0.099013	0.074622	-0.138674
63	-0.031220	-0.035717	0.020870	-0.143376
64	0.040638	0.087903	-0.049500	0.094607
65	0.026860	0.125924	-0.103449	0.140882
66	0.075166	0.110186	-0.115173	0.067330
67	0.036642	0.163193	-0.188762	0.103724
68	0.028179	0.095124	-0.053258	0.028900
69	0.002307	0.148211	-0.096037	0.046189
70	0.072227	0.137595	-0.095629	0.001339
71	0.033308	0.221480	-0.152201	0.012125
72	0.003458	-0.085112	0.041850	0.113836
73	-0.040610	-0.044880	0.029732	0.177011
74	0.011404	-0.054324	-0.012426	0.077815
75	-0.042413	-0.030930	-0.034844	0.122946
76	-0.002206	-0.045698	0.050651	0.054886
77	-0.041729	-0.016110	0.048005	0.102125
78	0.013963	-0.022204	0.001613	0.028997
79	-0.030218	-0.002052	-0.004365	0.065343
80	0.299049	0.046260	0.076320	0.070784
81	0.250160	0.098440	0.012590	0.137479
82	0.254170	0.095310	0.018749	0.004288
83	0.218892	0.145554	-0.035161	0.069784
84	0.303486	0.101424	0.135996	-0.013096
85	0.262919	0.165133	0.077237	0.071721
86	0.319358	0.170283	0.054554	-0.072210
87	0.272983	0.231181	-0.014471	0.011689
88	0.134116	-0.026693	0.161400	0.110292
89	0.100379	0.026517	0.086236	0.130478
90	0.144718	-0.000895	0.093767	0.044514
91	0.114943	0.022145	0.035871	0.069193
92	0.122051	0.011043	0.192803	0.022796
93	0.079482	0.026156	0.117725	0.056565
94	0.124641	0.027387	0.122956	-0.025369
95	0.090708	0.027357	0.064450	0.013058
96	0.159781	-0.055202	-0.090597	0.151598
97	0.084577	-0.037203	-0.126698	0.119739
98	0.192484	-0.100195	-0.162066	0.104148
99	0.114579	-0.046270	-0.219547	0.100067

b_4	G_5	G_6	G_7	G_8
100	0.153083	-0.010127	-0.086266	0.068648
101	0.088202	-0.010515	-0.102196	0.046281
102	0.164494	-0.057325	-0.132860	0.024093
103	0.109419	-0.013999	-0.169596	0.020412
104	0.039180	-0.209168	-0.035872	0.087949
105	0.012790	-0.177723	-0.129986	0.073364
106	0.045261	-0.256694	-0.088186	0.004212
107	-0.005314	-0.231202	-0.191671	-0.002628
108	0.037963	-0.153227	-0.045364	0.003322
109	0.030800	-0.126452	-0.114266	-0.010414
110	0.044125	-0.184146	-0.081400	-0.077341
111	0.029204	-0.157393	-0.172017	-0.089814
112	0.393519	-0.043228	-0.111365	-0.000740
113	0.289581	0.018928	-0.123140	0.000713
114	0.311229	-0.059735	-0.198982	-0.081664
115	0.258659	0.052505	-0.211913	-0.034928
116	0.300693	0.011381	-0.083545	-0.086683
117	0.214523	0.053878	-0.101199	-0.061018
118	0.253422	0.028496	-0.156752	-0.163342
119	0.199123	0.113877	-0.166220	-0.102584
120	0.249134	-0.165135	0.028917	0.051838
121	0.156434	-0.123708	0.017053	0.043043
122	0.214763	-0.101243	-0.005581	-0.020703
123	0.140554	-0.072067	-0.015063	-0.011165
124	0.241791	-0.152048	0.106403	-0.046857
125	0.142316	-0.131899	0.054076	-0.026485
126	0.206535	-0.086116	0.046640	-0.097615
127	0.129759	-0.081874	0.004693	-0.073169

Annex R (Normative) Quantization Tables for Higher Order Coefficients

Quantization table for b_5

b_5	$H_{1,1}$	$H_{1,2}$	$H_{1,3}$	$H_{1,4}$
0	0.264108	0.045976	-0.200999	-0.122344
1	0.479006	0.227924	-0.016114	-0.006835
2	0.077297	0.080775	-0.068936	0.041733
3	0.185486	0.231840	0.182410	0.101613
4	-0.012442	0.223718	-0.277803	-0.034370
5	-0.059507	0.139621	-0.024708	-0.104205
6	-0.248676	0.255502	-0.134894	-0.058338
7	-0.055122	0.427253	0.025059	-0.045051
8	-0.058898	-0.061945	0.028030	-0.022242
9	0.084153	0.025327	0.066780	-0.180839
10	-0.193125	-0.082632	0.140899	-0.089559
11	0.000000	0.033758	0.276623	0.002493
12	-0.396582	-0.049543	-0.118100	-0.208305
13	-0.287112	0.096620	0.049650	-0.079312
14	-0.543760	0.171107	-0.062173	-0.010483
15	-0.353572	0.227440	0.230128	-0.032089
16	0.248579	-0.279824	-0.209589	0.070903
17	0.377604	-0.119639	0.008463	-0.005589
18	0.102127	-0.093666	-0.061325	0.052082
19	0.154134	-0.105724	0.099317	0.187972
20	-0.139232	-0.091146	-0.275479	-0.038435
21	-0.144169	0.034314	-0.030840	0.022207
22	-0.143985	0.079414	-0.194701	0.175312
23	-0.195329	0.087467	0.067711	0.186783
24	-0.123515	-0.377873	-0.209929	-0.212677
25	0.068698	-0.255933	0.120463	-0.095629
26	-0.106810	-0.319964	-0.089322	0.106947
27	-0.158605	-0.309606	0.190900	0.089340
28	-0.489162	-0.432784	-0.151215	-0.005786
29	-0.370883	-0.154342	-0.022545	0.114054
30	-0.742866	-0.204364	-0.123865	-0.038888
31	-0.573077	-0.115287	0.208879	-0.027698

Quantization table for b_6

b_6	$H_{2,1}$	$H_{2,2}$	$H_{2,3}$	$H_{2,4}$
0	-0.143886	0.235528	-0.116707	0.025541
1	-0.170182	-0.063822	-0.096934	0.109704
2	0.232915	0.269793	0.047064	-0.032761
3	0.153458	0.068130	-0.033513	0.126553
4	-0.440712	0.132952	0.081378	-0.013210
5	-0.480433	-0.249687	-0.012280	0.007112
6	-0.088001	0.167609	0.148323	-0.119892
7	-0.104628	0.102639	0.183560	0.121674
8	0.047408	-0.000908	-0.214196	-0.109372
9	0.113418	-0.240340	-0.121420	0.041117
10	0.385609	0.042913	-0.184584	-0.017851
11	0.453830	-0.180745	0.050455	0.030984
12	-0.155984	-0.144212	0.018226	-0.146356
13	-0.104028	-0.260377	0.146472	0.101389
14	0.012376	-0.000267	0.006657	-0.013941
15	0.165852	-0.103467	0.119713	-0.075455

Quantization table for b_7

b_7	$H_{3,1}$	$H_{3,2}$	$H_{3,3}$	$H_{3,4}$
0	0.182478	0.271794	-0.057639	0.026115
1	0.110795	0.092854	0.078125	-0.082726
2	0.057964	0.000833	0.176048	0.135404
3	-0.027315	0.098668	-0.065801	0.116421
4	-0.222796	0.062967	0.201740	-0.089975
5	-0.193571	0.309225	-0.014101	-0.034574
6	-0.389053	-0.181476	0.107682	0.050169
7	-0.345604	0.064900	-0.065014	0.065642
8	0.319393	-0.055491	-0.220727	-0.067499
9	0.460572	0.084686	0.048453	-0.011050
10	0.201623	-0.068994	-0.067101	0.108320
11	0.227528	-0.173900	0.092417	-0.066515
12	-0.016927	0.047757	-0.177686	-0.102163
13	-0.052553	-0.065689	0.019328	-0.033060
14	-0.144910	-0.238617	-0.195206	-0.063917
15	-0.024159	-0.338822	0.003581	0.060995

Quantization table for b_8

b_8	$H_{4,1}$	$H_{4,2}$	$H_{4,3}$	$H_{4,4}$
0	0.323968	0.008964	-0.063117	0.027909
1	0.010900	-0.004030	-0.125016	-0.080818
2	0.109969	0.256272	0.042470	0.000749
3	-0.135446	0.201769	-0.083426	0.093888
4	-0.441995	0.038159	0.022784	0.003943
5	-0.155951	0.032467	0.145309	-0.041725
6	-0.149182	-0.223356	-0.065793	0.075016
7	0.096949	-0.096400	0.083194	0.049306

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Annex S (Normative) Bit Frame Format

<i>Symbol</i>	<i>Bit 1</i>	<i>Bit 0</i>
0	$c_0(23)$	$c_0(5)$
1	$c_1(10)$	$c_2(3)$
2	$c_0(22)$	$c_0(4)$
3	$c_1(9)$	$c_2(2)$
4	$c_0(21)$	$c_0(3)$
5	$c_1(8)$	$c_2(1)$
6	$c_0(20)$	$c_0(2)$
7	$c_1(7)$	$c_2(0)$
8	$c_0(19)$	$c_0(1)$
9	$c_1(6)$	$c_3(13)$
10	$c_0(18)$	$c_0(0)$
11	$c_1(5)$	$c_3(12)$
12	$c_0(17)$	$c_1(22)$
13	$c_1(4)$	$c_3(11)$
14	$c_0(16)$	$c_1(21)$
15	$c_1(3)$	$c_3(10)$
16	$c_0(15)$	$c_1(20)$
17	$c_1(2)$	$c_3(9)$
18	$c_0(14)$	$c_1(19)$
19	$c_1(1)$	$c_3(8)$
20	$c_0(13)$	$c_1(18)$
21	$c_1(0)$	$c_3(7)$
22	$c_0(12)$	$c_1(17)$
23	$c_2(10)$	$c_3(6)$
24	$c_0(11)$	$c_1(16)$
25	$c_2(9)$	$c_3(5)$
26	$c_0(10)$	$c_1(15)$
27	$c_2(8)$	$c_3(4)$
28	$c_0(9)$	$c_1(14)$
29	$c_2(7)$	$c_3(3)$
30	$c_0(8)$	$c_1(13)$
31	$c_2(6)$	$c_3(2)$
32	$c_0(7)$	$c_1(12)$
33	$c_2(5)$	$c_3(1)$
34	$c_0(5)$	$c_1(11)$
35	$c_2(4)$	$c_3(0)$

Annex T (Normative) Tone Frame Parameters

I_D	f_0	l_1	l_2
0 - 4	N/A	N/A	N/A
5 - 12	31.250 I_D	1	1
13 - 25	15.625 I_D	2	2
26 - 38	10.417 I_D	3	3
39 - 51	7.8125 I_D	4	4
52 - 64	6.2500 I_D	5	5
65 - 76	5.2803 I_D	6	6
77 - 89	4.4643 I_D	7	7
90 - 102	3.9063 I_D	8	8
103 - 115	3.4722 I_D	9	9
116 - 122	3.1250 I_D	10	10
123 - 127	N/A	N/A	N/A
128	78.5	12	17
129	173.48	4	7
130	70	10	19
131	87	8	17
132	109.95	7	11
133	191.68	4	7
134	70.17	11	21
135	71.06	12	17
136	121.58	7	11
137	212	4	7
138	116.41	6	14
139	96.15	8	17
140	71	12	23
141	234.26	4	7
142	134.38	7	9
143	134.35	7	11
144	68.33	12	17
145	150.89	4	7
146	67.82	9	17
147	86.5	7	15
148	95.79	7	11
149	166.92	4	7
150	67.7	10	19
151	74.74	10	14
152	105.9	7	11
153	92.78	8	14
154	101.55	6	14
155	84.02	8	17
156	67.83	11	21
157	102.3	8	14
158	117	7	9
159	117.49	7	11
160	87.78	4	5
161	70.83	6	7
162	122	4	5
163	70	5	7
164 - 254	N/A	N/A	N/A
255	250	0	0

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